T · · Systems · · ·

rvsEVO

Version 7.00

User Manual

rvsEVO
Version 7.00
User Manual

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Berlin 10781

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1 Introduction

In this chapter you will find a short description of rvs® and rvsEVO as well as an explanation of typographic conventions used in the present manual.

1.1 Short Description of the System

What is rvs®

rvs® = Rechner-Verbund-System

The abbreviation rvs® stands for the German word **Rechner-Verbund-System**. The rvs® computer communication system is a well-established base service for electronic data interchange (EDI).

rvs® serves to ensure transmission of electronic data between heterogeneous computer platforms using different network protocols.

To do so, rvs® implements a universal network model, which you can configure in each network node.

rvs® provides an efficient and reliable transport service for both standardized EDI message types and files of any format or contents. You can receive only such files that are explicitly destined for rvs®. This means that rvs® does not allow any unauthorized access to remote or to own data files.

The system was originally developed by Volkswagen AG and has been used in the German and European automobile industries for several years and by banks, insurances and industries worldwide.

rvs® uses the OFTP protocol.

What rvs® is not

rvs® is not an online system. It neither supports direct terminal-like access to other sites, nor provides a communication pipe from application to application on a data record level. You cannot directly execute transfers in your own application. You rather can place send orders from within your application to rvs® which will be handled asynchronously.

rvs® is not a job scheduling system.

rvs® does not care about the contents of the files it transfers. It only acts as a transparent transport medium and performs no semantic interpretation of the data it carries.

rvs® is not an EDI converter. Please contact your sales partner if you need to convert among specific message formats (e.g. VDA, ODETTE, EDIFACT, XML):

Phone: +49 375 606 19 902

E-Mail: rvs-service@t-systems.com

rvs® is not a network control or monitoring tool.

What is rvsEVO

rvsEVO is a communication software with a graphical user interface based, like rvs®, on the OFTP protocol.

Available features

The following features are available in rvsEVO 6.05:

- graphical user interface (GUI)
- sending files to neighbour or to routed stations
- receiving files from neighbour or from routed stations
- receiving files on virtual stations; sending files from virtual stations
- activating neighbour stations
- support of the OFTP version 1.3, 1.4 and 2.0
- viewing information about receive, send, failed and ended jobs
- displaying monitor messages
- log files for tracing the monitor activities and for troubleshooting
- deleting or releasing EERPs (End-to-End-Response) if necessary
- archiving information about the send or receive jobs in the revision log
- code conversion with various code conversion tables
- format conversion
- defining job filters and actions when sending or receiving files
- compression and encryption
- comfortable certificate management
- Automatic Certificate Exchange (ACX)
- backup and recovery
- No limit for size of transmitted files
- Support of SNMP Monitoring (for more information see the rvsSNMP-Agent User Manual)
- remote access to rvsEVO Server via Remote GUI
- remote access to rvsEVO Server via internet
- Oracle database connection
- Microsoft SQL database connection
- Postgres database connection
- PKI connection
- Application clustering
- Automatic update installation (Distributed Update)
- Operation of rvsEVO in container environment

– Semi-automatic establishment of connections to partner stations and update of partner stations (Odette Partner Data eXchange)

rvsEVO is implemented in Java. There exist four options of rvsEVO: rvsEVO Enterprise Edition, rvsEVO Standard Edition, rvsEVO Light Edition, rvsEVO Tiny Edition. Please see chapter 1.2 to 1.5 for description.

rvsEVO uses a batch interface and file system to communicate with the application. If capable of doing so, the linked application can indicate successful processing and indicate successful dispatch.

At present, rvsEVO supports the following networks: TCP/IP, TLS, Proxy TCP/IP and Proxy TLS.

Attention:

From version 6.03.00 on TLSv1 and TLSv1.1 are disabled. For further information, please see also rvsEVO Operator Manual, chapter „Error Situations of rvsEVO“ / „Connection Failure“ / „Examples with outgoing call, TLS Connection“.

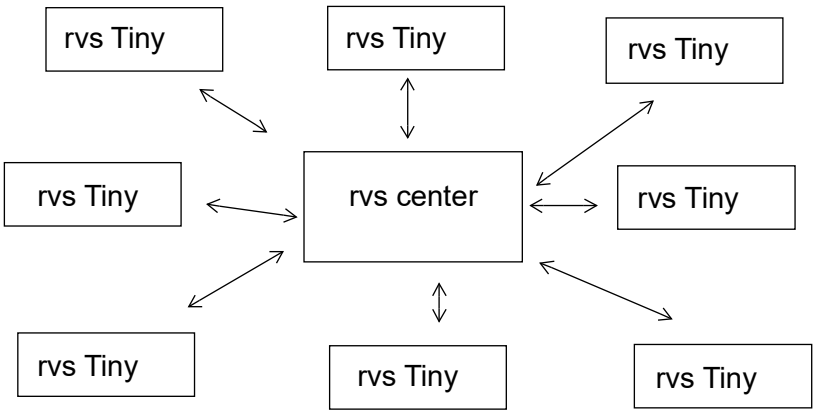
Hint:

For more information on supported platforms please read the `$RVS_HOME/docs/readme.txt` release notes. You can find the explanation of **$RVS_HOME** in chapter 1.6.Representation Formats

Please see the rvsEVO Product Datasheet for binding information of the technical data. You can download the datasheet from the following address: `https://system-communication-products.t-systems-service.com/de/en/data-communication-suite/rvs/rvs-pd/download-742440`.

1.2 rvsEVO Tiny Edition

The Tiny Edition supports the communication with only one neighbour station: rvs® center. The number of partner station is restricted to 1. Other stations can be reached via routing. Therefore, rvsEVO Tiny Edition is particularly suitable for star topology.



Features of rvsEVO Tiny Edition (subject to modifications):

– 5 partner station (neighbour and routing stations)
– 1 neighbour station (rvs® center)
– 1 session
– max. 4 routing partner
– communication components TCP/IP and TLS
– encryption and compression
– SNMP Agent
– code conversion (PC-Mainframe)

Additionally, the following components can be linked:

– File Service Module
– Distributed Update
– rvsEVO in container environment

1.3 rvsEVO Light Edition

Features of rvsEVO Light Edition (subject to modifications):

– max. 5 partner stations (direct or routing partner)
– 2 parallel connections (sessions)
– routing functionality
– communication components TCP/IP and TLS
– encryption and compression
– Derby database embedded
– SNMP Agent
– code conversion (PC-Mainframe)

Additionally, the following components can be linked:

– File Service Module
– Distributed Update
– rvsEVO in container environment

1.4 rvsEVO Standard Edition

Features of rvsEVO Standard Edition (subject to modifications):

– 999 partner stations (direct or routing partner)
– 20 parallel connections (sessions)
– routing functionality
– 1 virtual station

– communication components TCP/IP and TLS
– encryption and compression
– Derby database embedded
– SNMP Agent
– code conversion (PC-Mainframe)
– remote GUI

Additionally, the following components can be linked:

– File Service Module
– further virtual stations
– Distributed Update
– rvsEVO in container environment

1.5 rvsEVO Enterprise Edition

rvsEVO Enterprise Edition provides all the basic requirements for secure data transfer with a simultaneous performance even at high loads. rvsEVO Enterprise Edition includes an automatic adjustment of the processing capacity and uses a professional external database.

Features of rvsEVO Enterprise Edition (subject to modifications):

– 9.999 partner stations (direct or routing partner)
– 999 parallel connections (sessions)
– routing functionality
– 1 virtual station
– all communication components
– encryption and compression
– Oracle database connection
– Microsoft SQL database connection
– Postgres database connection
– SNMP Agent
– code conversion (PC-Mainframe)
– PKI connection
– File Service Module
– remote GUI (10 users)
– Application clustering

Additionally, the following components can be linked:

– rvs® OFTP Proxy for OFTP2
– further virtual stations

– Distributed Update

– rvsEVO in container environment

Hint: The available features are limited by the license key.

For more Information on rvsEVO Editions please contact your sales partner (Phone: +49 375 606 19 902; E-Mail: rvs-service@t-systems.com)

1.6 Representation Formats

This chapter describes the typographic conventions used in this manual and explains the meaning of specially highlighted expressions.

Typographic conventions

• Instructions begin with a bullet.

– Other lists begin with the en dash.

Character styles:

`Courier`	Commands, menu commands, file names, path names, programs, examples, scripts, options, qualifiers, data sets, fields, modes, window names, dialog boxes and statuses
BOLD and **UPPER CASE**	Parameters, environment variables, variables
"Inverted comma"	Links to other manuals, sections and chapters, literature
Bold	Important terms, names of operating systems, proper names, buttons, function keys

$RVS_HOME

As user directories are found on different locations for the different operating systems, we use the variable **$RVS_HOME** in this manual. Default values are:

– `C:\Programs\rvsEVO` for **Windows**.

Substitute the variable with your correct path.

1.7 Target Group

This manual is meant for regular users of rvsEVO as well as administrators. It provides an overview of the basic rvsEVO functions.

Skills

The following skills are required to be able to use rvs EVO:

– good knowledge of the current operating system

– knowledge of the communication techniques in use TCP/IP, TLS, Proxy TCP/IP or Proxy TLS.

Before starting to work with rvsEVO it is advisable to have read this book.

2 Installation of rvsEVO

This chapter describes the system requirements as well as the rvsEVO installation procedure.

Hint: From version 6.02.00 upward the installer comprises the installation of rvsEVO as well as the installation of rvsRuntime (please see chapter 21 "Multitenancy Functionality of rvsEVO" and the rvsEVO Operator Manual, for more information about rvsRuntime).

2.1 System Requirement

To successfully operate rvsEVO you need the following software:

Software

– Operating system: **Windows 10 / Windows 11 / Windows Server 2016 / Windows Server 2019 / Windows Server 2022 / Windows Server 2025** or **UNIX (AIX)** as well as **Linux** and **zLinux**.

– Java runtime environment (JRE) version 21.

Hints:

– With Windows Linux and AIX systems the installer distributes the required Java runtime environment version into the rvsEVO installation folder.

– With UNIX systems you can select an installed JRE version 21 during the installation.

– Initially, you need at least 500 MB free space on your hard disk. Depending on the amount of usage, the retention period for old entries, and the time between database cleanups, the space requirement may be considerably larger.

Hotfix

Please download the hotfixes as well, if available. Please unzip the zip file and copy the.jar files into `$RVS_HOME/lib` directory after the installation is complete.

You can download rvsEVO from the following web page:

https://system-communication-products.t-systems.net/en-DE/rvsevo

If that is not possible for you, please contact your sales partner:

Phone from Germany: 0800 664 77 45

Phone from other countries: +49 375 606 19 902

Email: rvs-service@t-systems.com

Networks

Communication line based on TCP/IP, TLS, Proxy TCP/IP and Proxy TLS.

2.2 Obtaining a License

You need a license key to work with rvsEVO.

rvs® after sales service

Please contact the rvs® Service Support Center to receive a license key.

from Germany for free, telephone: 0800 664 77 45

from other countries, telephone: +49 375 606 19 902

e-mail: rvs-service@t-systems.com

To receive a license key:

- Please send the hostname and the **Odette ID** to the rvs® Service Support Center. You get the hostname as typing hostname in the command prompt window (`Run -> cmd`). You can request the Odette ID with the Odette organisation (`https://www.odette.org/odette-id`).
- You will receive your licensekey file by email.
- Save the license key file in the `$RVS_HOME/conf/` directory as `license.properties`.
  If the directory which contains the installer also contains a `license.properties` file this file is copied into `$RVS_HOME/conf` directory during installation.

Note: For more information about the ODETTE-ID see chapter 6.2.2 "Parameter Groups of Local Station".

Renew a License

Use the tool `importLicense` to import a new license key at runtime. This could be required if your license key expires or if you need a license for added rvsEVO components. Please see chapter 25.7.18 "importLicense: Import of rvsEVO License Key" for more information.

Test License

For receiving a test license, please contact the rvs® Service Support Center too. In this case you must not inform the rvs® Service Support Center about your hostname and the Odette ID.

2.3 New Installation of rvsEVO and rvsRuntime

This chapter describes the installation of rvsEVO. Please read the chapter 2.1 "System Requirement" before installing.

Attention:

For successful installation the installing user needs execute rights for `TEMP` directory (defined via **TEMP** environment variable). If he has no execute rights, he needs to create a `TEMP` directory with execute rights before the installation procedure is started and to define this directory via **$IATEMPDIR** environment variable.

Please see Chapter 25 "rvsEVO Database" for installation of rvsEVO Enterprise Edition with Oracle or MS-SQL database.

Hints:

rvsEVO can be installed without input of any data. You only have to press the return button. In this case the configuration will done with the default values. Exception: this procedure cannot be executed in German language on console.

Please read the rvsEVO Operator Manual, Chapter „Installation of an rvsEVO Server Farm“ for Information about the installation of a server farm.

2.3.1 Installation on Windows Systems

- Start Windows and log in as a Windows user with administrator rights.

  Hints:

  To set up the Windows services, it is necessary tu run the installer with administrator privileges.

  Unlock the firewall before the installation procedure for the ports of the listeners that should be used for receiving data from your partner station (if the Windows firewall is enabled).

- Start the installer `rvsEVO-setup-[version]-Windows64.exe` by double-clicking or in the console. In the second case the installation mode can be indicated via `-i` parameter.

  – `-i awt`: graphical mode

  – `-i console`: console mode

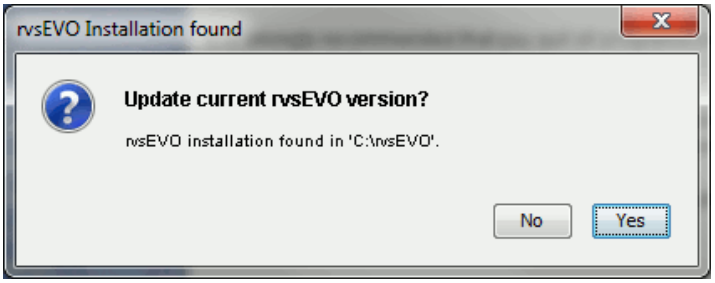
  Example:

  ```
  rvsEVO-setup-7.00.00-Windows64.exe -i console
  ```

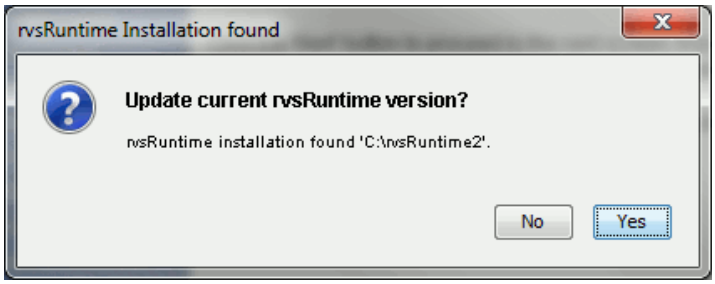
- With the first window you get information on installation of rvsEVO.
- Thereafter you can choose the language of the installation wizard.

With the following dialogs you indicate the kind of installation: update or new installation of rvsEVO or rvsRuntime.

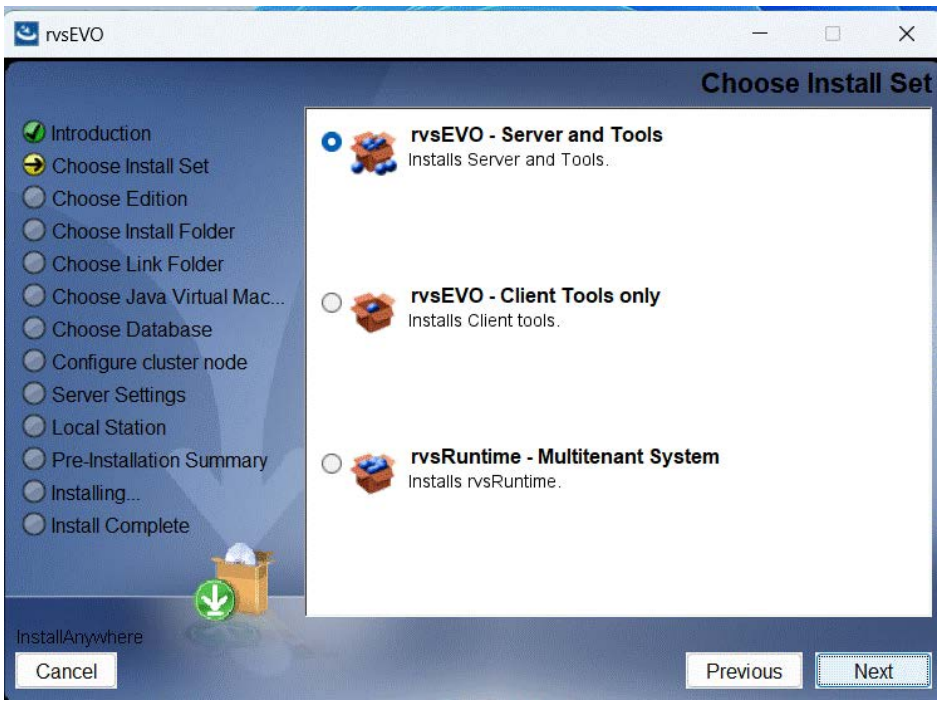
- If there is already an rvsEVO installation on your machine, first you were asked whether you would like to update your current rvsEVO. If you decide for upgrading your existing rvsEVO please continue as described in chapter 2.4 "rvsEVO Update Installation", else click on **No** button for going on.



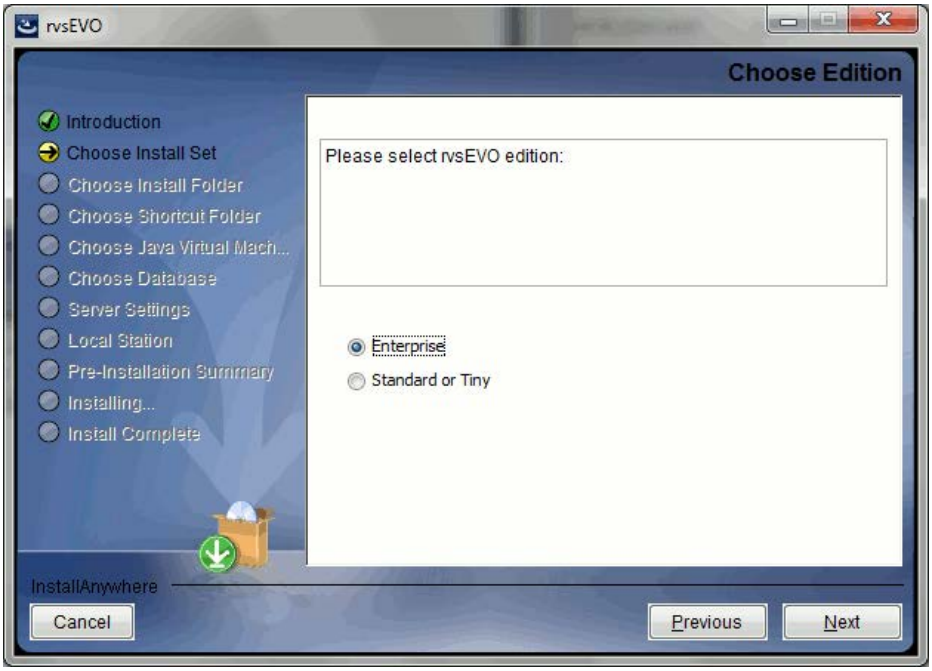
- If there is already an rvsRuntime installation on your machine, you have to decide whether you would like to update your current rvsRuntime. If you decide for upgrading your existing rvsRuntime please see the rvsEVO Operator Manual, chapter „Multitenancy Functionality of rvsEVO“ / „Installation of rvsRuntime“ / „Update Installation of rvsRuntime“, else click on **No** button for going on.



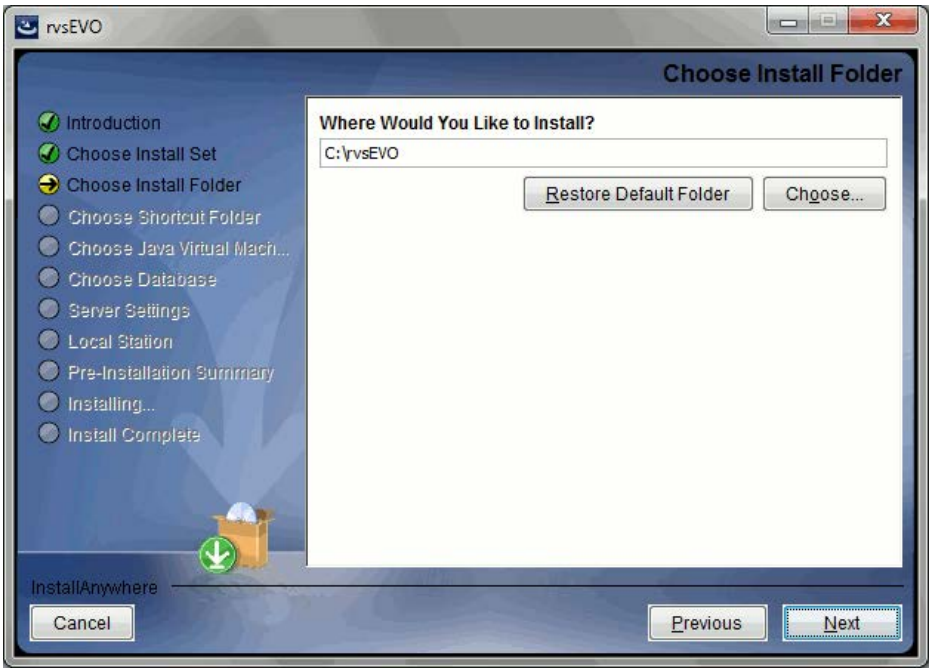
- Next you have to choose the install set. For Information about installation of rvsRuntime see the rvsEVO Operator Manual, chapter „Multitenancy Functionality of rvsEVO“ / „Installation of rvsRuntime“. Click on **Next** button for going on.



- In the following window you can decide which rvsEVO edition is to be installed.

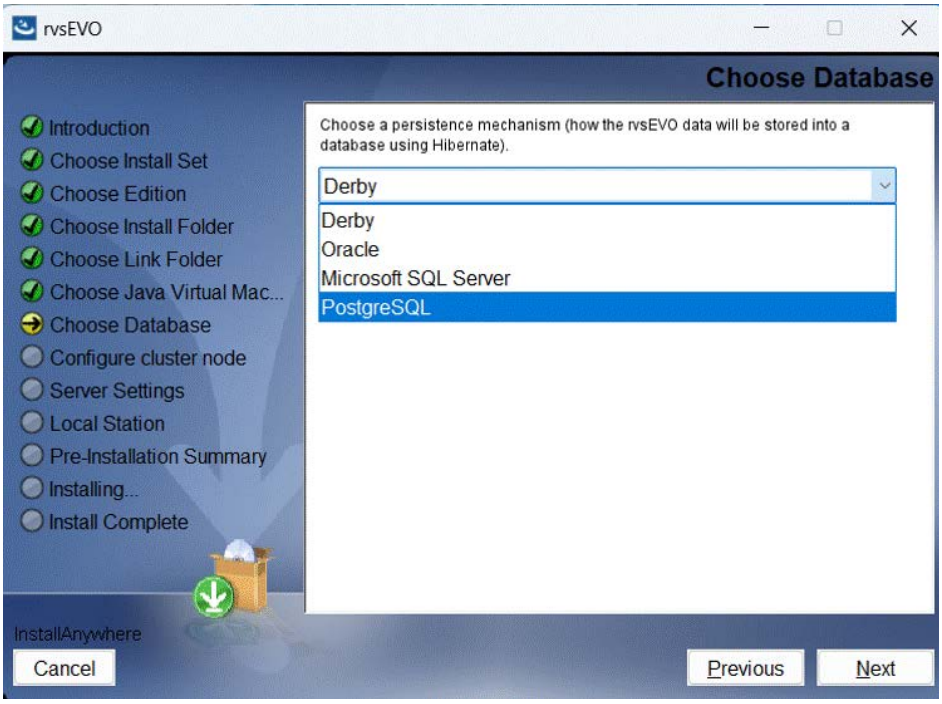


- Via the dialog below you can indicate the rvsEVO destination directory. Alphanumeric characters without special characters, underscores, hyphens and spaces are permitted for the directory name. If you choose an existing installation folder of rvsEVO or rvsRuntime a warning is displayed and via a mouse-click on **OK** button you can indicate another directory name.



The following dialogs refer to installation of Enterprise Edition only

- With installation of rvsEVO Enterprise Edition you can choose the database in which your data should be saved. Oracle, Postgres and MS SQL database are available.



- Set your Oracle connection parameters if you decided to use an Oracle database:
  - **jdbc.url** has the following syntax:

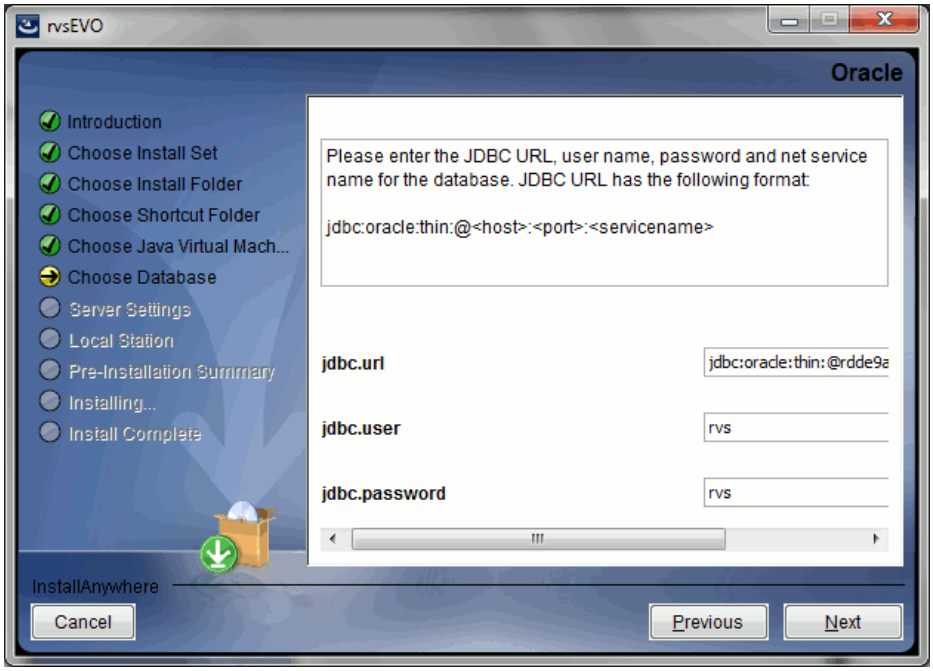
    `jdbc:oracle:thin:@<server>:<port>/<service_name>`

    `server` is the name or IP address of the machine where Oracle server is installed.

    Default port for Oracle is 1521.

    `service_name` is Oracle service name.
  - **jdbc.user** is the user setting up on Oracle database
  - **jdbc.passsword** is his password.

    For further information see the chapter 25.3 "Oracle"

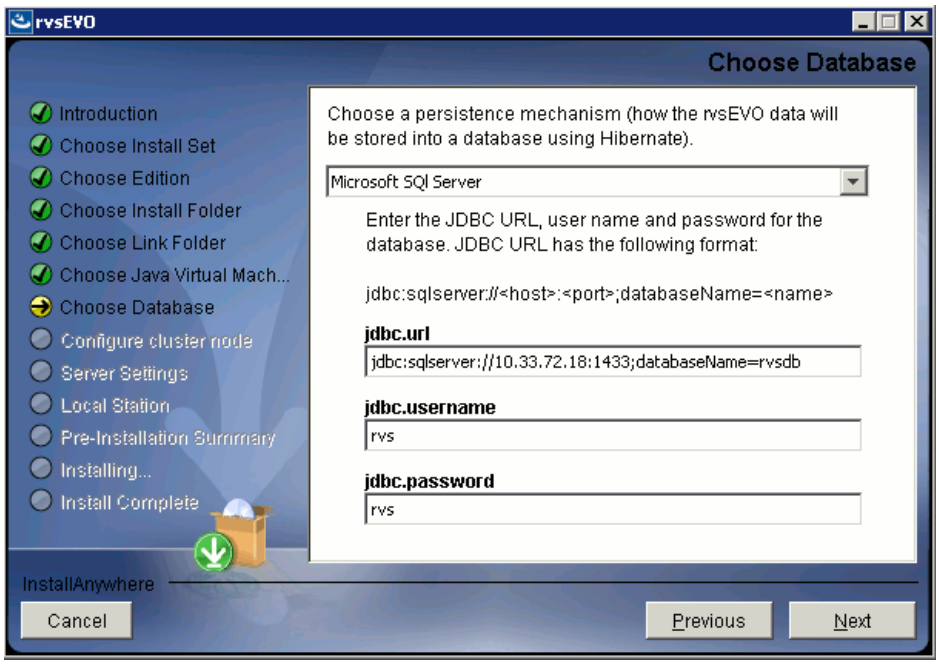


Analogous to Oracle database, set the MS SQL connection parameters if you decided to use an Oracle database:

– **jdbc.url** has the following syntax:

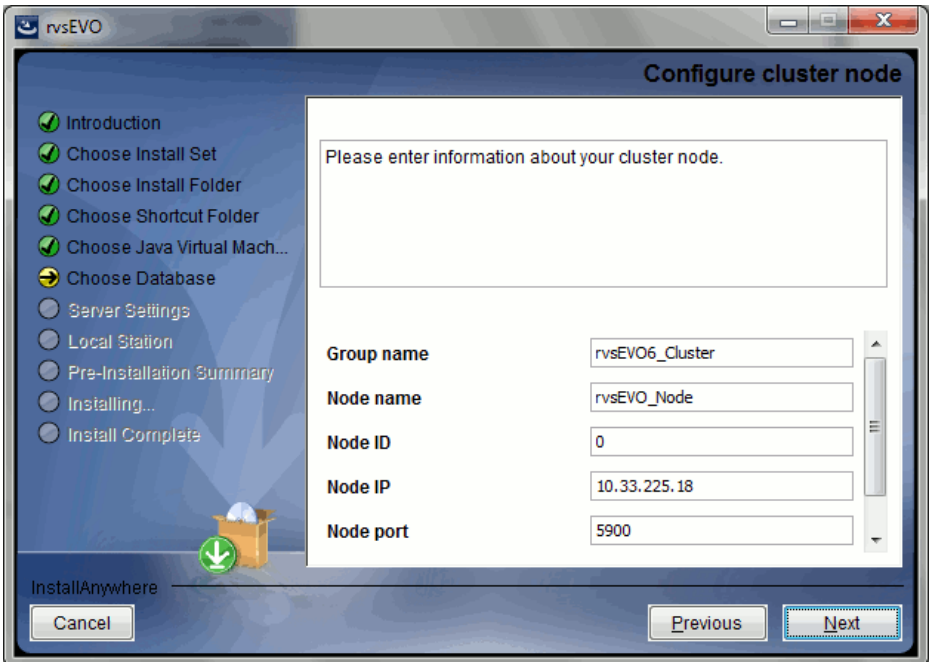
  `jdbc:sqlserver://<host>:<port>;databaseName=<name>`

  Default port for MS SQL is 1433.

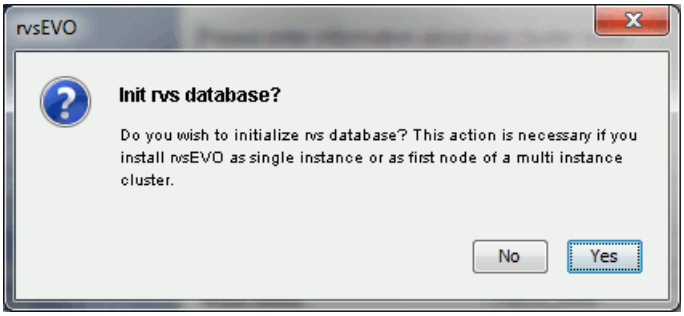


- The following window contains the parameters for setting up the cluster node. **Group name** is the name of your server farm. The Group name must be the same on all cluster nodes. **Node name** is an unambiguous designation of the node to be installed. Furthermore,you were asked for the **Node IP,** the **Node port** and the IP addresses or hostnames of the other nodes of the server farm (**Nodes**).

  In case of installing a single instance of rvsEVO you can take over the given values. Otherwise please follow the instruction given by the rvsEVO Operator Manual, chapter „Installation of an rvsEVO Server Farm“.



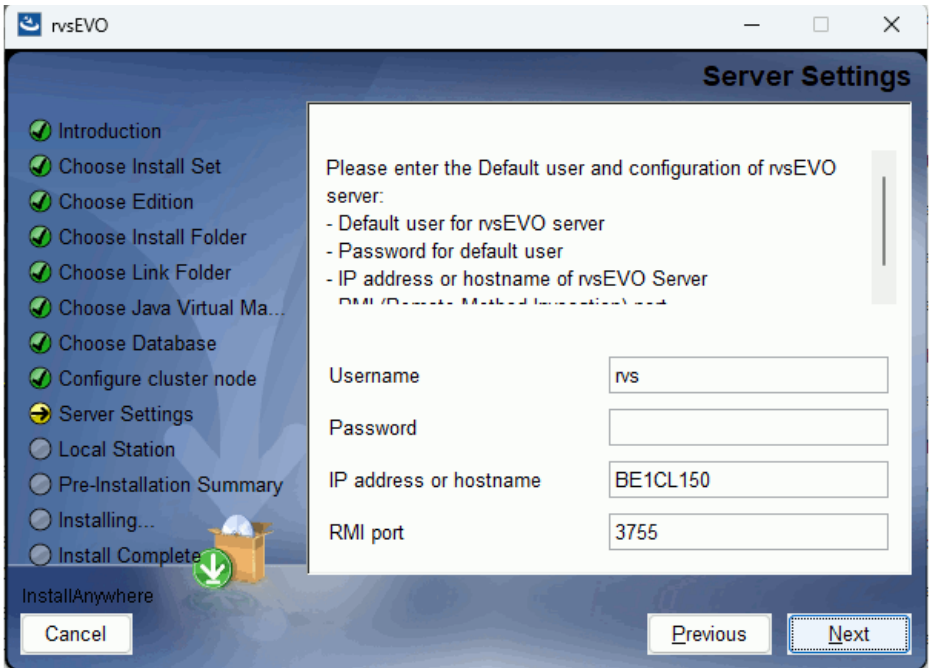
- The dialog below offers to initialize the database. If you are installing a server farm, you have to initialize the data base only during installing the first node. Please see chapter 19 "Server Farm of rvsEVO" for more information about cluster functionality.



Continuance for all editions

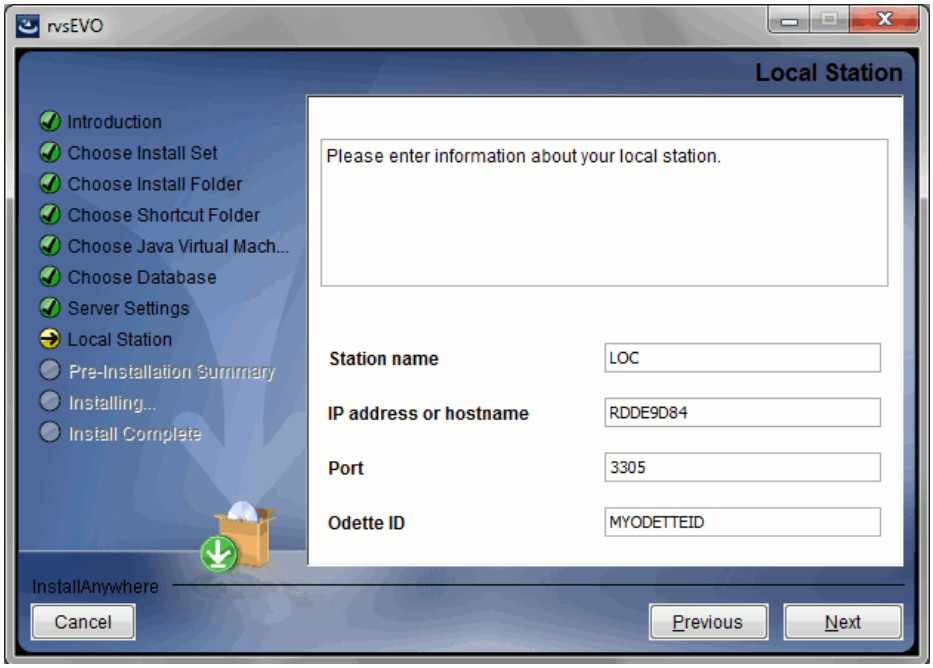
- Subsequently, please define the username and the password of the default user with administrator rights as well as the IP address or

hostname (default: hostname of your machine) and the RMI port (default: 3755). Username is a string of alphanumeric characters, hyphen, dot and underline, password is restricted to 7-bit ASCII characters.



- If you have already installed rvs® portable on your system, you can migrate the settings of the local station from rvs® portable. Please read the rvsEVO Operator Manuel chapter „Migration from rvs® portable to rvsEVO“ for further information about migration procedure.

  If you install just the GUI you can also ignore the dialog below. It deals with the settings of the local station. Please read the chapter 6.2.2 "Parameter Groups of Local Station" for more information about setting the station parameters (e.g. how to obtain the ODETTE ID). The neighbour station will be configured after installation procedure via GUI. (See chapter 6.3.1 "Add a Neighbour Station").



- In the next dialog you are given a brief overview of selections you have made. The required and the currently available disk space is also displayed. Press the **Next** button to start installation and to copy the installation files into the directories you specified.
- The last dialogs inform you about the successful installation of rvsEVO.

2.3.2 Installation on Linux and AIX Systems

The Installation on Linux and AIX systems is similar to installation on Windows systems.

For Linux, please use the following installer:

```
rvsEVO-setup-[version]-Linux64.bin.gz
```

For AIX, please use the following installer:

```
rvsEVO-setup-[version]-AIX.bin.gz
```

The installers for Linux and AIX systems are provided in a packaged format and must be unpacked before installing rvsEVO:

```
gunzip rvsEVO-setup-[version]-[platform].bin.gz
```

Then, execution permissions must be granted:

```
chmod 744 rvsEVO-setup-[version]-[platform].bin
```

After that, the rvsEVO installer can be started:

```
./rvsEVO-setup-[version]-[platform].bin
```

Note: No spaces are allowed in the installation path.

2.3.3 Installation on UNIX Systems

For an installation on UNIX systems, please use the following installer:

`rvsEVO-setup-[version]-GenericUnix.bin.gz`

The following exception applies when installing on UNIX systems:

– The installer uses JRE version 21. Add the JRE to the environment variable $PATH:

Example:

```
PATH="/home/rvsuser/JRE/temurin/jdk-21.0.8+9-jre/
bin":$PATH
```

The installer for UNIX systems is provided in a packaged format and must be unpacked before installing rvsEVO:

`gunzip rvsEVO-setup-[version]-GenericUnix.bin.gz`

Execution permissions must be granted:

`chmod 744 rvsEVO-setup-[version]-GenericUnix.bin`

After that, the installer can be run. The rvsEVO installer can be started:

`./rvsEVO-setup-[version]-GenericUnix.bin`

Additional notes:

– No spaces are allowed in the installation path.

– You will also be prompted to select the Java runtime environment

that will be used to run rvsEVO.

rvsEVO is approved for use with Java version 21.

2.4 rvsEVO Update Installation

Requirement

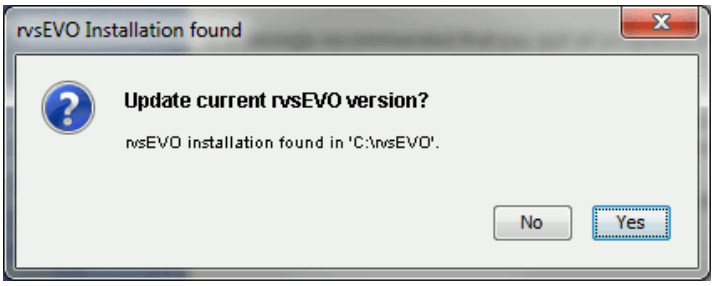
The system variable **RVS_HOME** is to be set for the logged on user (the value of **RVS_HOME** is the installation directory of rvsEVO). Set **RVS_HOME** by choosing `Start -> Control Panel -> System -> Advanced -> Environment Variables`.

Important: Save your rvsEVO installation before the update installation is executed like described in the rvsEVO Operator Manual chapter „Fall Back after Update Installation“.

- Start the `rvsEVO-setup-[version]-.exe` installer by double-clicking or in the console.

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- Click the **Update** button if the installer asks you whether you would like to install a new version or to update your current rvsEVO.



- In the next dialog you are given a brief overview of selections you have made (installation directory, link directory). The required and the currently available disk space is also displayed. Press the **Install** button to start the update installation.
- The last window informs you about the successfully installation of rvsEVO. Finish the installation routine by pressing the button **Done**.

Hints:

From version 6.05.00 rvsEVO provides automatical updating of multiple defined partner stations (targets) via rvsEVO controller station. Please read the rvsEVO Operator Manual, chapter 'Distributed Update' for further information.

`$RVS_HOME/bin` directory:

The directory is overwritten during update installation. Please save the files which were customized before the update installation is started.

Exception: the `setcp.bat` and `rvsEVOService.ini` files (on Linux systems `setcp.sh`) are saved with the extension `.backup`. Your additional settings are adopted into the new files.

`$RVS_HOME/conf directory:`

During the update installation a new `$RVS_HOME/conf/rvs-system.properties.new` file is added. Please take over your added configurations of your old `rvs-system.properties` file to the new one and rename `rvs-system.properties.new` as `rvs-system.properties`.

A new `$RVS_HOME/conf/rvsEvoClient.prefs` is added as well. Your old file is saved as `rvsEvoClient.prefs.backup`. Please take over your settings to the new file after update installation is complete.

If the directory which contains the installer also contains a `license.properties` file this file is copied into `$RVS_HOME/conf` directory. In case the `conf` directory already contains a `license.properties` file a timestamp is added to the old filename.

2.5 Migration from rvs® portable to rvsEVO

Via migration functionality, rvsEVO offers the import of settings from rvs® portable.

Please read the rvsEVO Operator Manuel chapter „Migration from rvs® portable to rvsEVO“for information about migration procedure.

2.6 rvsEVO Version

You can see which rvsEVO version you have installed on your computer in the rvsWeb-GUI at the bottom left under **RVSEVO**/Version. As an alternative you can execute the script `$RVS_HOME/bin/rvsver.bat`.

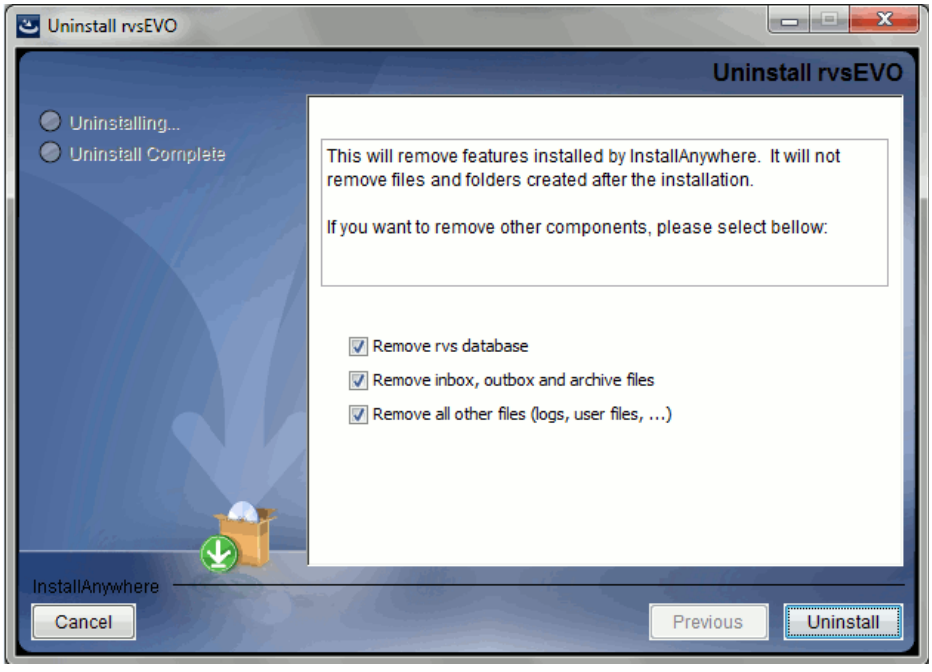
2.7 Uninstall rvsEVO

In order to uninstall rvsEVO, please follow the procedure given below:

Windows:

- Start the program `$RVS_HOME/UninstallerData/Uninstall.exe` and follow the dialog.
- In the first dialog you can indicate the components to be deleted. Those are available:
  - `Remove rvs database`
  - `Remove inbox, outbox and archive files`
  - `Remove all other files (logs, user files ...)`

  Click on the **Uninstall** button to uninstall rvsEVO.



- The next window displays the components which have not been deleted. Click on **<done>** in order to exit the uninstalling program.

Hint: Depending on files which could not be deleted a restart of your system is necessary. In this case a further window informs you about the needed restart.

UNIX:

Use the shell-script `$RVS_HOME/UninstallerData/Uninstall rvsEVO_StandardEdition.sh`.

Uninstall procedure on UNIX-systems can be executed analogous to rvsEVO installation.

3 Start and Stop of rvsEVO

This chapter provides information concerning the starting and stopping of rvsEVO Server and rvsWeb-GUI as well as informations about the installation as a Windows service.

3.1 Starting of rvsEVO Server

The server must be started before the graphical user interface can be started.

3.1.1 Server Start on Windows Systems

Start the rvsEVO Server via the command line tool `$RVS_HOME\bin\startServer.bat`.

Windows Service

By default, rvsEVO is installed as a Windows service.

The term service means a program that can be started from the operating system and works in the background.

rvsEVO as Windows service can also be started via the batch script `rvsEVOservice.exe`.

Syntax:

```
rvsEVOService <-install [ini]> <-configure [ini]>
<-remove [ini]> <-debug [ini]> <-run [ini]>
```

Optional rvsEVO Service Parameters:

Parameter	Description
-install [ini]	Installs rvsEVO as service. `$RVS_HOME\bin\rvsEVOService.ini` is used by default.
-configure [ini]	For new configuration of an installed rvsEVO Windows service. `$RVS_HOME\bin\rvsEVOService.ini` is used by default.
-remove [ini]	Removes rvsEVO as service. `$RVS_HOME\bin\rvsEVOService.ini` is used by default.
-debug [ini]	For starting a console application for debugging. `$RVS_HOME\bin\rvsEVOService.ini` is used by default.
-run [ini]	Starts rvsEVO as service on console. `$RVS_HOME\bin\rvsEVOService.ini` is used by default.

Hint: After the installation of rvsEVO as a service, you will find rvsEVO in the list of system services. Via double-click on „`rvs Server`“ in the list, the properties will be opened and offers you the options to start, stop, pause or resume the service or to indicate the startup type (Automatic, Manual, Disabled)

3.1.2 Server Start on Linux und UNIX Systems

Start the rvsEVO Server via shell script `$RVS_HOME/bin/startServer.sh`.

rvsEVOService

Alternatively, the rvsEVO Server can be started as nohup process in the background detached from the user session via `$RVS_HOME/bin/startService.sh` script. This means that the rvsEVO Server is not stopped when the user session is terminated. The server messages are written into `$RVS_HOME/log/server.out` file.

Syntax:

```
sh ./rvsEVOService.sh [--no-wait]
```

Optional Parameter:

Parameter	Description
--no-wait	The shell will be released immediately.

Hint: You get the following error message if you try to boot an already started rvsEVO Server:

```
* rvsEVO Server is already started.
Please check server process.
(to enforce startup of server please delete /home/
evotest/rvsEVO/system/status)
Status: STARTED 26070
```

3.1.3 Automatic Server Start on Linux Systems

On Linux systems the rvsEVO Server is started via `$RVS_HOME/bin/startServer.sh` or `$RVS_HOME/bin/rvsEVOService.sh` script by default.

The `$RVS_HOME/tools/scripts/Linux` directory contains a script for enabling the automatic start and stop of `rvsEVO` together with the operation system. In the same directory you can find the `readme-Linux.txt` file, in which the installation steps are described.

3.2 Stopping of rvsEVO Server

rvsEVO can be stopped via `$RVS_HOME/bin/stopServer` script.

Syntax:

```
stopServer [-m <mode>] [-async] [-h] [--verbose]
```

Optional Parameters:

Parameter	Description
-async	terminates `stopServer` without waiting
-m <mode>	sets the shutdown mode (integer value, default=0)
-h, --help, -?	Requests help information (usage).
--verbose	verbose output

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4 Start and Display the rvsWeb-GUI

4.1 Introduction

The rvsWeb-GUI (**G**raphical **U**ser **I**nterface) allows you to work with rvsEVO without much practice. The most important functionalities can be performed using the rvsWeb-GUI.

For logged-in users, similar functionality is available as in the RCP GUI, depending on the logged-in user's role.

For further information, please see Chapter 13 "User Management"

4.2 Configure and Start of rvsWeb-GUI

After an update from rvsEVO 6.xx or a new installation, the rvsWEB-GUI is automatically activated. The file r**vs-rest.properties** might get adjusted to activate the rvsWEB-GUI.

(*rvs.web.server.activate=true* and *rvs.web.server.port=8433*)

You can enable and configure the rvsWeb GUI in the `$RVS_HOME/conf/rvsrest.properties` file. Please refer to the rvsEVO Operator Manual, chapter "Customizing rvsEVO" / "Customizing rvsWeb and REST API" for further information.

The following browsers are supported by the rvsWeb-GUI:

– Edge

– Firefox

– Chrome

The connection can be established directly to the rvs instance or to a homepage, through which the instances can be reached.

You can configure access via a homepage using the `$RVS_HOME/conf/rvs-rest.properties` file.

Further information can be found in the rvsEVO Operator Manual, in the chapter "Customizing rvsEVO" / "Customizing rvsWeb".

Please start the rvsWeb-GUI in the following format:

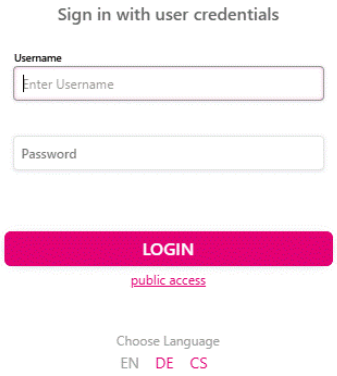
`https://<hostname>:<port>/rvs/web`

Note:

The hostname and port must match the settings in the `rvsrest.properties` file.

You can also start the rvsWeb-GUI via the script `$RVS_HOME/bin/startGUI.bat`.

After starting the following dialog window will open and you can login to the server:



Required specifications:

– `User Name`: Login to rvsEVO (to the server) with the user set up in the user management (user ID)

– `Password`: Password of the user

Hint:

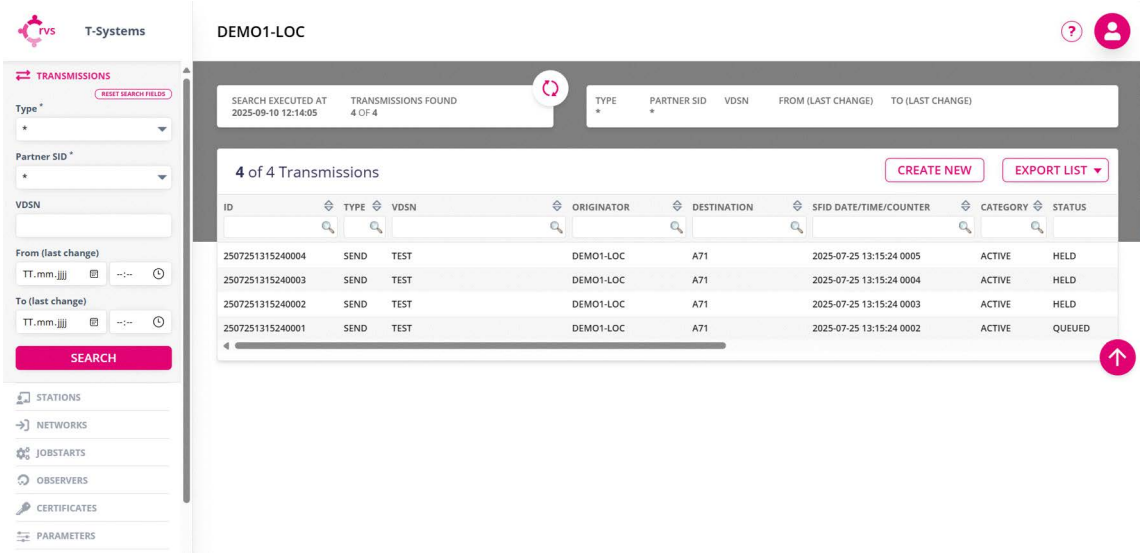
The rvsWeb-GUI can be accessed with limited functionality through the "Public Access" link without logging in.

You can change the language directly here by clicking on the available versions: English (EN), German (DE), and Czech (CS).

4.3 Structure of the rvsWeb GUI in list view

After logging into the rvsWeb-GUI, the main application window will be displayed: **TRANSMISSIONS**

Enter search criteria **Type** and **Partner SID** in the side bar and click the **SEARCH** button to view the Transmissions.



The rvsWeb-GUI is divided into distinct areas that present different informations:

Title bar:

– SID of the local station

– Help button

– User icon with options Change language settings (not available for the rvsEVO standard user)

– Log out of the rvsWeb-GUI

Side bar:

– Filter areas

– Navigation bar

Header bar:

– Filter criteria and execution time of the displayed list

– Refresh button

– Search criteria entered in the sidebar

List area:

– Number of filtered items and the total number

Buttons:

– CREATE NEW: Add a new item (depending on user role)

– TOOLS: Executable operations (depending on user role)

– EXPORT LIST: Export lists in different formats
– Column heading with sorting function. By clicking on the arrow icon next to the column heading, you can sort the list entries in ascending or descending order.
– Filter row with live filter function. Acts as a "second-level filter." Entering "25" in the ID search field displays all transfers that contain "25" in the ID.
– Items in list view

Select one or more rows, then right-click to access several functions:

– Copy cell - Copy to the clipboard, from which the cell can be inserted, for example, into a text file.
– Copy 1 selected row
– Copy all rows
– Delete 1 selected row

4.4 Detail view in the rvsWeb-GUI

The detail view shows you detailed information about an object and can be opened by double-clicking a list item.

In the detail view, you can edit parameters or access additional functionalities via the **ACTIONS** button.

Clicking the **EDIT** button unlocks the editable fields.

Changes are saved using the **SAVE** button. This button is only enabled after an allowed change has been made.

You can close the form using the **CLOSE** button.Clicking the **CANCEL** button will discard any changes.

Hint:

This function is only available to users with the **Administrator** or **Operator** role. Users with the **User** role cannot make any changes.

The detail view contains all information about the item. Using the COPY TO CLIPBOARD button, the text can be copied and, for example, pasted into a text file.

5 Transmission with rvsEVO

This chapter provides information concerning file transmssion and output of transmission data via Transfer perspective.

5.1 Search Transmissions

After logging into the rvsWeb-GUI, the main application window, **TRANSMISSIONS**, is displayed.

In the **Type** search field in the sidebar, the following options for the list output are provided by rvsEVO by default:

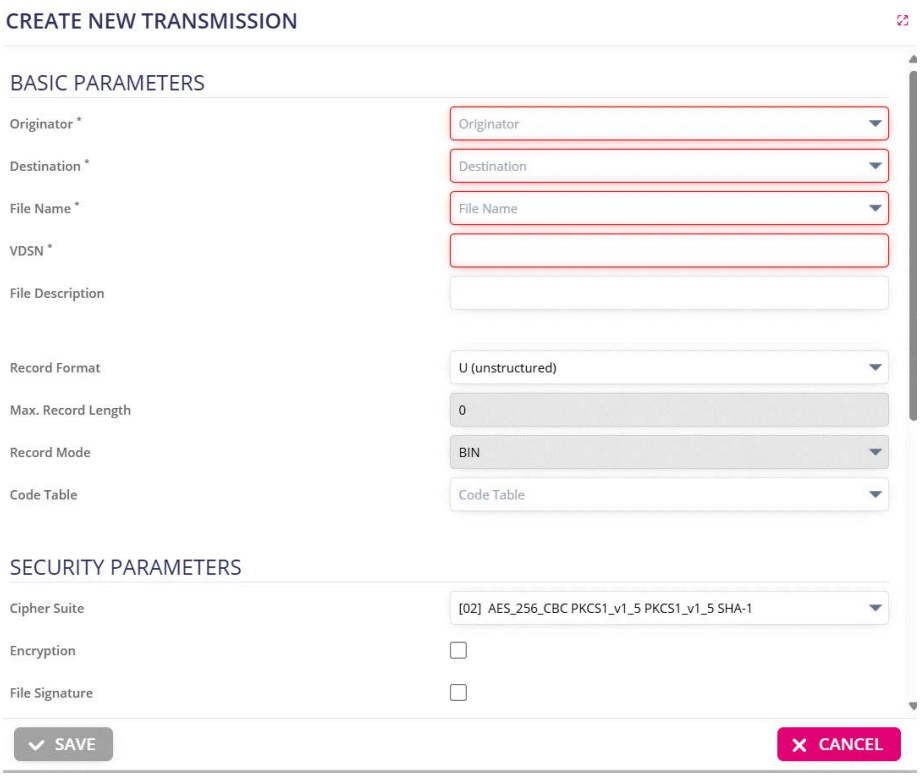
– `SEND`: output of send transmissions
– `RECV`: output of received transmissions
– `EERP`: output of successful ERP jobs
– `NERP`: output of unsuccessful ERP jobs

Additional search criteria can be entered, and clicking the **SEARCH** button will display the list of transmissions.

5.2 Create a new Transmission

If you create a new transmission you can only select files from the `outbox` directory.

You can create a new transmission using the button **CREATE NEW**.



This window prompts for the basic, security and additional parameters.

The following tables provide a description of the transmission parameters.

Basic parameters:

Parameter	Description
Originator	Station ID of originator. Possible values: SID of local station (default) SID of virtual Station **Hint:** Depending on parameter **Adressing Mode** input of **SID Originator** or **OID Originator** is required.
Destination	Station ID of destination. **Hint:** Depending on parameter **Adressing Mode** input of **SID Destination** or **OID Destination** is required.
File Name	Name of the file to be send.
VDSN	Virtual file name; the length of the file name used for ODETTE transfer must not exceed 26 characters. Default: Name of the File to be transmitted. **Hint:** Via defintion `rvs_evo.controller.underscore_in_vdsn_allowed=true` in $`RVS_HOME/conf/rvs-system.properties` file you can configure that underscores are allowed in VDSN.
File Description	Comment, free text, available only in connection with OFTP version 2. If your partner does not support OFTP version 2, this field will be ignored.
Record Format	Format of the file to be sent: - `T`=text file; a stream of ASCII characters - `U`=unstructured (binary); byte stream - `V`=variable; variable record length - `F`=fixed; fixed record length Please see also the parameter **Record Mode**.
Max. Record Length	Maximal record length for the files in **Format** `F` or `V`, please see also the parameter **Record Mode**

Basic parameters:

Parameter	Description
Record Mode	In this parameter you indicate the mode of files in format 'F' or 'V'. Possible values: - TXT: text file - BIN: binary file Text File in format 'F': If the record length deviate from the specified one (Parameter **MaxRecl**) the line will be filled with blanks or will be cut up to defined value. Text File in format 'V': longer records will be cut up to defined value. Binary files: The length of all records is the max. record size except the length of the last one. Binary files in format 'F': If the size of the last record is shorter than the specified one the line will be filled to the defined value.
Code Table	For ASCII - EBCDIC conversion the following conversion tables are available: ASCII-IBM037, ASCII-IBM273, ASCII-IBM500, ANSI-IBM073, ANSI-IBM273, ANSI-IBM500. For EBCDIC - ASCII conversion the following conversion tables are available: IBM037-ASCII, IBM273-ASCII, IBM500-ASCII, IBM037-ANSI, IBM273-ANSI, IBM500-ANSI. For using your own conversion table, please see the rvsEVO Operator Manual, chapter „Customizing rvsEVO“ / „Conversion Tables“ . **Hint**: Not available for files in format 'T'

Security parameters:

Parameter	Description
Cipher Suite	File security Algorithms: (CipherSuite: Symmetric Encryption, Asymmetric Encryption, Asymmetric Signature, Hash Function) - **02:** AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-1 - **04**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-256 - **06**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-512 - **07**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA3-512

Security parameters:

Parameter	Description
Encryption	Active only in connection with setting of parameter **Security Feature Set**=`4`. If the check box is activated file encryption takes place.
File Signature	Active only in connection with setting of parameter **Security Feature Set**=`4`. If the check box is activated file signature takes place
Request signed EERP/ NERP	Active only in connection with setting of parameter **Security Feature Set**=`4`. With OFTP 2.0 you have the possibility to request the signature of EERP and NERP. If the check box is activated signed EERP or NERP is requested. The send job ends in `FAILED` status if your partner sends no signed EERP or NERP.
Compression	Active only in connection with setting of parameter **Security Feature Set**=`4`. If the check box is activated Odette compression takes place.

Additional Parameters:

Parameter	Description
Serialization	If the check box is activated the files will be sent in the same order, as the send jobs were created. The next job will only be sent, if the previous is completely finished. All send jobs for the serialization must have the same label (see parameter Label).
Label	Group name of serialized send jobs. User specified (descriptive) label for this job. (See parameter **Serialization**)
Disposition	Disposition of local file after successful send attempt. Possible values: - `K` (Keep): file will not be deleted after sending - `D` (Delete): file will be deleted after sending.
No Copy	With this parameter you can decide whether the file to be transmitted is to copy into Outbox directory or not. **Attention**: With a server farm, this parameter must not be activated because the transmitted files need to be copied to outbox directory on the NFS drive, so that each node is able to access the file. (See also chapter 19 "Server Farm of rvsEVO").

Additional Parameters:

Parameter	Description
Hold	If the check box is activated the send job is created in `HOLD` status.
Schedule	With this parameter you can define what time the send job should be launched. Press the icon on the right to select date and time. Default: at once.
Priority	Priority for send jobs. Possible values: - `1` bis `5` `1` is highest priority, `5` is lowest priority, default: `3`
External Job ID	Parameter for an external JobID, which can refer to several rvsEVO JobIDs. The external JobID is a string of alphanumeric characters. By default, **External Job ID** is an optional parameter. With `<entry key="client.transfer.sendJobWizard.externalJobId" value="mandatory"/>` definition in `$RVSHOME/conf/rvsEvoClient.prefs` file you can set up **External Job ID** to mandatory parameter. This setting is only valid for send jobs which are created via rvsEVO GUI.

5.2.1 Local Transmission Delivery

There are two ways for sending files to the own local or virtual station.

Looptest: The file is transmitted via a neighbour station back to the own station. For this procedure **LooptestNeighbourSID** parameter has to be set. (see the chapter "ODETTE Parameters" im operator manual).

Local delivery without network connection. For this procedure **LooptestNeighbourSID** parameter must not be set (see the chapter "ODETTE Parameters" im operator manual).

5.2.2 Role of Routing Station with Send Jobs for Routed Stations

In case of sending data via routing station to the destination, first the routing station acts as receiver and subsequently the routing station creates a new send job for forwarding the file to the destination. In the first OFTP session the routing station functions as responder in the second one as initiator.

5.3 Delete Transmissions

Hint: This functionality is available to users with the role of Administrator or Operator.

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Deleting the entries of processed send and receive jobs is possible by right-clicking on the transmission and selecting **Forced delete.**

The transmission status will be changed to **Deleted** and can be removed from the database by selecting **Cleanup**.

See also cmd tool `cleanup` (chapter 16.7 "cleanup: Deletion of Jobs and Monlog Data from the Database")

5.4 Hold Transmission and Send Next

Hint: This functionality is available to users with the role of Administrator or Operator.

By right-clicking on a transmission, additional functionalities are displayed:

– HOLD transmission - the selected transmission will be hold

– Send Next - the selected transmission will be sent

5.5 Detail View of a Transmission

By double-clicking on a transmission, the detail view opens, where you can review the data and perform various actions:

– hold transmission

– send next

– forced delete

Hint: This functionality is available to users with the role of Administrator or Operator.

The following tables describe the transmission attributes.

Basic parameters:

Parameter	Description
ID	rvsEVO job ID
VDSN	Virtual file name for Odette transfer.
File Description	File description. Text comment, possible with OFTP version 2. If your destination device does not support OFTP 2.0 this field is ignored.
Originator	SID of Originator station
Destination	SID of recipient
Filename	Path and name of the received or sent file.
Source File Name	The entire path of the original file; this file is copied into rvsEVO outbox directory.
Category	category

Basic parameters:

Parameter	Description
Status	Job status. For possible status values please see the chapter Kapitel "Status Values for Send Jobs and Receive Jobs"
Last Change	Time of the last change in job status

Transmission

Parameter	Description
SFID Date/Time	Date and time of creation of transmission job
Transmission Start Time	Start of transfer, format „YYYY-MMDD HH:mm:ss“
Transmission End Time	End of transfer, format „YYYY-MMDD HH:mm:ss“
Transmitted Bytes	Number of transmitted bytes
Original File Size	Size of original file before processing through the service provider
Retry Count	Number of failed send attempts.
Priority	Priority for send jobs. Possible values: - `1` bis `5` `1` is highest priority, `5` is lowest priority, default: `3`
Serialization	If this parameter is activated your files are sent in serial order. All files with the same ID (**Label**) are sent in the same group.
Label	Label for serialization. All files which have been sent in the same group should have the same ID (label). See also parameter **Serialization**
Send Attempts	Number of failed send attempts. After successful transmission the parameter is set to zero.
Schedule Time	Fixed date and time for transmission
Restart Position	Restart position in case of restart of the transmission

Format parameter

Parameter	Description
Record Format	Format of the file to be sent: - `T`=text file; a stream of ASCII characters - `U`=unstructured (binary); byte stream - `V` (variable); variable record length - `F` (fixed); fixed record length. Please see also the parameter **Record Mode**.
Record Length	Maximum record length
Record Mode	In this parameter you indicate the mode of files in format 'F' or 'V'. Possible values: - `TXT`: text file - `BIN`: binary file
Code Table	Name and path to conversion table: For ASCII(ANSI) - EBCDIC conversion: `ASCII-IBM037, ASCII-IBM273, ASCII-IBM500, ANSI-IBM037, ANSI-IBM273`. For EBCDIC - ASCII(ANSI) conversion: `IBM037-ASCII, IBM273-ASCII, IBM500-ASCII, IBM037-ANSI, IBM273-ANSI`

Security Features and NERP

Parameter	Description
Cypher Suite	Encoding algorithm
Encryption	File encryption if the check box is activated
File Signature	If the check box is activated, the file is signed
EERP/NERP Signature	If the check box is activated, signed EERP/NERP is requested
Compression	Odette compression if the check box is activated.
NERP Reason Code	Code of NERP
NERP Reason Text	Description of NERP Code
NERP Creator OdetteID	Odette ID of NERP creater
Error Number	Error code
Error Description	Description of error code

5.5.1 Status Values for Send Jobs and Receive Jobs

Status Values for Send Jobs and Receive Jobs:

- DELETED (Job got deleted)
- DELETED_WITH_NERP (Routing job got deleted and NERP was sent to originator)
- ENDED (Job has successfully ended)
- ENDED_WITH_JS_ERROR (Job run into error during call of jobstart)
- FAILED_WITH_NERP (NERP received / sent)
- FATAL_ERROR (Job run into fatal error)
- RESTART (wait after interruption of transmission to send file again)
- SP_ENDED (Service provider has processed job successfully)
- SP_FAILED (Service provider failed during processing of job)
- SP_PROCESSING (Service provider is processing job)

Status for Send Jobs:

- CREATED (Job was created, but not sent)
- FAILED_WITH_INVALID_EERP_SIGNATUR (Received EERP with invalid signature)
- FAILED_WITH_INVALID_NERP_SIGNATUR (Received NERP with invalid signature)
- FAILED_WITH_INVALID_PROTOCOL_VERSION (Transmissions failed based on invalid protocol version)
- FAILED_WITH_SFNA (Receiving of SFNA with setting retry = No)
- HELD (Send job got holded)
- RELEASED (Send job got released)
- SCHEDULED (Send job created - execution deadline not reached)
- SYNCHRONIZE_ERROR (Send job run into synchronize error)
- WF_CDT (Send job waits for credit renewal)
- WF_EERP (Send job waits for EERP)
- WF_EERP_ROUTING (Wait for EERP during Routing)
- WF_EFID_ANSWER (Send job waits for answer to already sent EFID)
- WF_SFID_ANSWER (Send job waits for answer to already sent SFID)

Status for Receive Jobs:

- EERP_HELD (File has completely been received; EERP is on hold and needs to get released)
- EERP_RELEASED (User released EERP, but it is still not send)
- FAILED_WITHOUT_NERP (transmission failed; no NERP is sent)

– NERP_RELEASED_JS (NERP released by JS, but it is still not sent)
– POSTPROCESSING_FAILED (File has completely been received but post-processing failed)
– RECEIVING (Receiving file data after EFPA)
– RESTART_AFTER_EFNA (Transmission with error; EFNA send; waiting for retransmission)
– RESTART_AFTER_EFPA_FAILURE (File already delivered but EFPA failed; assuming partner will send file again)
– ROUTING_HELD_DURING_SHUTDOWN (Couldn't create routing send job, because of shutdown
– PDX_HELD (Received Odette Partner Data needs processing)

Hint: Please see „rvsEVO Operator Manual“ chapter „ODETTE Protocol“ for ODETTE protocol sequences (such as EFNA, SFID, EFID, EERP, ...).

5.6 Operations on Jobs

You have the possibility to execute operations for a job via right-click on the job in the list or via buttons at the bottom in the detailed view.

The available functions depends on the job status.

Active Send Transmissions

– Hold Transmission
– Release Transmission
– Restart
– Send Next
– Delete Transmission

Hint: In case of deleting a routing send job the routing receive job is deleted as well.

– Forced Delete (allows to delete an active job without pausing it before)

Active Receive Transmission

– Restart SP (after bug fixing, to activate a job in SP_FAILED status)
– Delete EERP
– Forced Delete (allows to delete an active job without pausing it before)
– Release EERP
– Release NERP

Ended Transmissions

– Recreate (only for ended send jobs)
– Cleanup (deletion of transmission data from the database)

Failed Transmissions

– Recreate (only for send jobs)

– Cleanup (deletion of transmission data from the database)

5.7 Display Transmission queries

To display transmission queries, switch to the Query tab under **Transmissions** in the main screen.

The following transmission queries for list output are already provided by rvsEVO and displayed in the list under **Query***:

ENDED JOBS: Output of completed transmissions

FAILED JOBS: Output of failed transmissions

ACTIVE SEND JOBS: Output of active send transmissions

ACTIVE RECEIVE JOBS: Output of active receive transmissions

ALL JOBS: Output of all transmissions

NOTE:

If these standard queries are not present, you can import them using the tool **createDefaultQueries**. For more information see the chapter 25.7.2 "createDefaultQueries".

Custom-created transmission queries are additionally displayed in this table.

5.8 Execute Transmission queries

You can execute a transmission query by selecting the desired transmission query in the **Query*** table.

The result will be displayed on the right in the **Transmissions** table view.

5.9 Change Transmission queries

To change a transmission query, click the **Manage Queries** button. The list of the transmission queries will appear.

Double-click the transmission query you want to modify to open the details view.

Click the **EDIT** button to make changes.

Click the **SAVE** button to save the changes.

5.10 Create a Transmission query

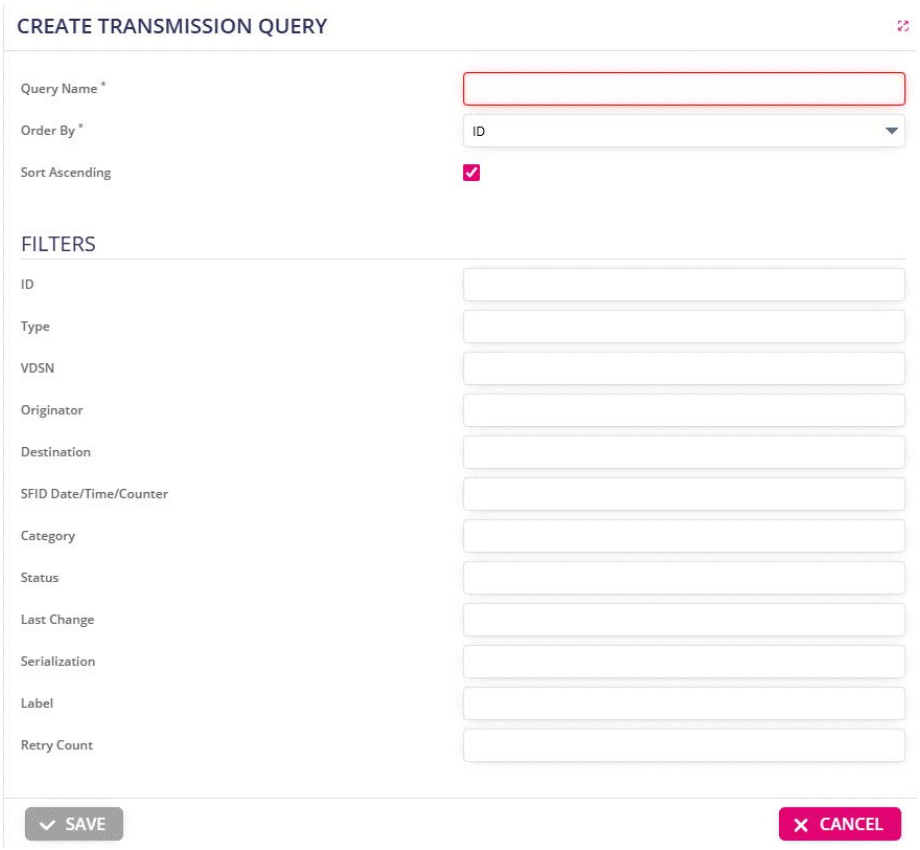
To create your own transmission query, please proceed as follows:

– Switch to the **Query** tab

– Click the **Manage Queries** button

– Click the **CREATE NEW** button

– Enter the name under which the query should be saved in the **Query Name** field.
– In the Order by* field, select the parameter by wich the output should be sorted.The default value is **"ID"**
– The **Sort Ascending** checkbox is selected by default. Change the selection if you do not want the output to be sorted in ascending order.
– In the **FILTERS** section, you can define the filter criteria by which the list should be created (e.g., the filter **Type** SEND for send jobs, RECV for receive jobs, or *ERP for receipts (for routing stations) can be added to the Type parameter).
– Once you have defined all the criteria for the transmission list, you can save your query using the **SAVE** button. It will be stored in the list of transmission queries.



Note:

The **SFID Date/Time** and **Last Change** filter criteria can be used with wildcards (“?” as a placeholder for one character and “*” for any number of characters) as well as with “<”and ">” characters.

The date must be entered in the format `YYYY-MMDD hh:mm:ss`.

Examples:

`< 2023-04-25 10:00:17`: All transmissions older than the specified date will be filtered out.

`< 2023-04*`: All transmissions modified before April 2023 will be filtered out.

Relative date filters can also be set.

The query operators "older than" and "younger than" are permitted, along with the time units "minute", "minutes", "hour", "hours", "day", "days", "month","months", "year", and "years".

Examples:

```
older than 3 days
```

```
younger than 1 month
```

Using "from" ... "until", you can filter transmissions within a specific time period.

Example:

```
from 2015-10-01 until 2015-10-31
```

Additionally, the filter criterion "Last Change" "today" is available with the query operators "older than", "<", and "from ... until":

Examples:

```
<today
```

```
from 2017-04-01 until today
```

```
older than today
```

```
today
```

Negation is also permitted for the following filter criteria:

Filter criteria	Description
Type	`!SEND,!RECV,!EERP,!NERP`
VDSN	`!<VDSN>`
Originator	`!<SID>`
Destination	`!<SID>`
SFID Date/Time	In combination with query operators: - `!<from ... until>` - `!today` - `!<YYYY-MM-DD hh:mm:ss>`
Category	`!<category>`
Status	`!<Status>`

Filter criteria	Description
Last Change	In combination with query operators: - !<from ... until> - !today - !<YYYY-MM-DD hh:mm:ss>
Serialization	!yes, !no
Label	!<label>
Retry Count	!<Number of failed transmission attempts>

5.11 Delete Transmission Query

You can delete a transmission query by right-clicking on the corresponding transmission query in the **Transmission Queries** view (after clicking the **Manage Queries** button) and then selecting `Delete 1 selected row` from the drop-down list.

5.12 Query Parameters

The following table describes the transmission query attributes.

Parameter	Description
ID	Job ID automatically assigned by rvsEVO
Type	Direction:: - SEND=send - RECV=receive - EERP=output of successful ERP jobs - NERP=output of insuccessful ERP jobs
VDSN	Virtual filename used for the Odette transfer
Originator	Stations-ID from the originator
Destination	Stations-ID from then destination
SFID Date/Time/Counter	Date and time the transmission order was created
Category	Category of the transmission: `ACTIVE`, `FAILED` oder `ENDED`
Status	Transfer status. For possible values, see chapter 5.5.1 "Status Values for Send Jobs and Receive Jobs"
Last Change	Date and time of the last change of the transmission

Parameter	Description
Serialization	Serialization ensures an orderly transmission sequence
Label	A label can be used to create a group for serialization
Retry Count	Number of failed transmission attempts

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6 Management of Stations

To be able to exchange data with rvsEVO, you need at least one neighbour station, routed or virtual station in addition to your local station with which you want to exchange files.

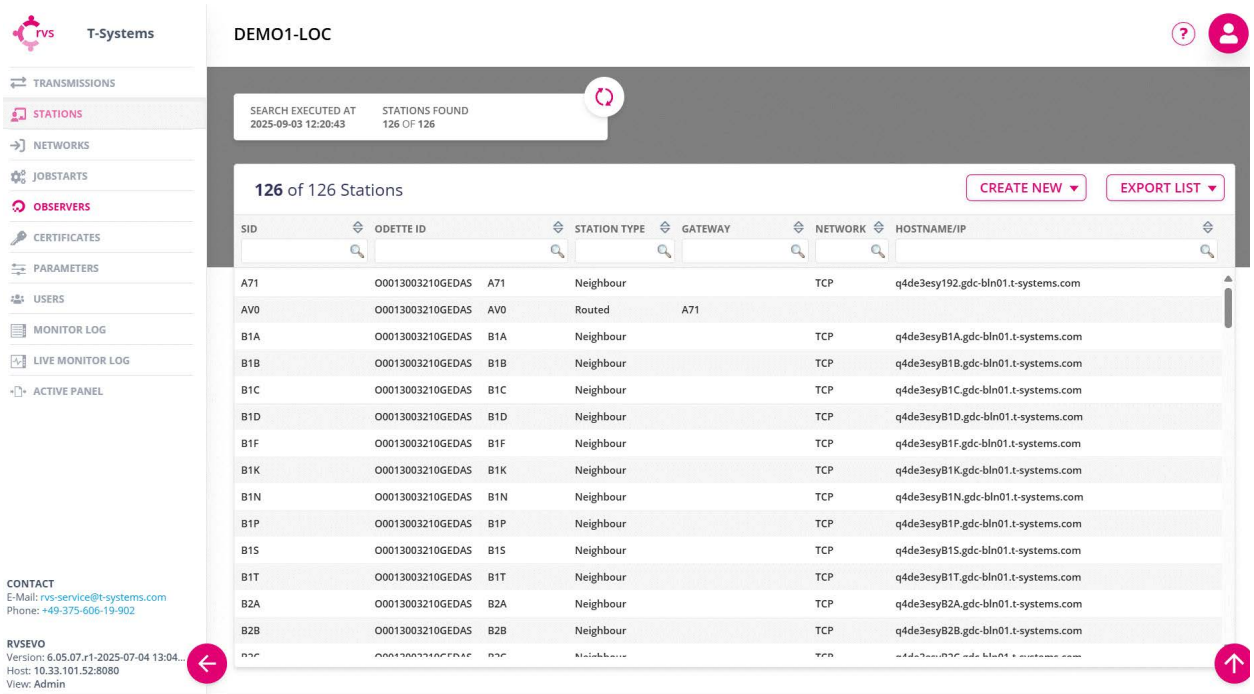
The obligatory parameters for the local station have already been polled during the installation.

Hint: The management of stations is available to users with the role of Administrator or Operator. User with the role „User“ kann only open these data in read-only mode.

6.1 User Interface of Stations

The stations management is reachable via sidebar item `Stations`.

The illustration below shows the default view of the stations:



The following functions can be executed using the button **CREATE NEW**:

– Neighbour Station

– Routed Station

– Virtual Station

The view `Stations` contain the list area, which shows all stations (local station, Neighbour Stations, Routed Stations and Virtual Stations) stored in the rvsEVO database.

After neighbour or virtual stations have been set up, they appear in the list view with their own station type.

Depending on the user‘s role, you can start the following operations with a right mouse click on a station in the list view:

Local Station:

– Change Station ID

Virtual Station:

– Duplicate Station

– Change Station ID

– Delete 1 selected station

Neighbour station and routed Station:

– Activate Station (only partner station)

– Activate Station without Transfer (only partner station)

– Certificate Request to Station

– Certificate Deliver to Station

– Add Routed Station (only partner station)

– change station Type or to Routed

– Duplicate Station

– Change Station ID

– Delete 1 selected station

Detailed view of a station

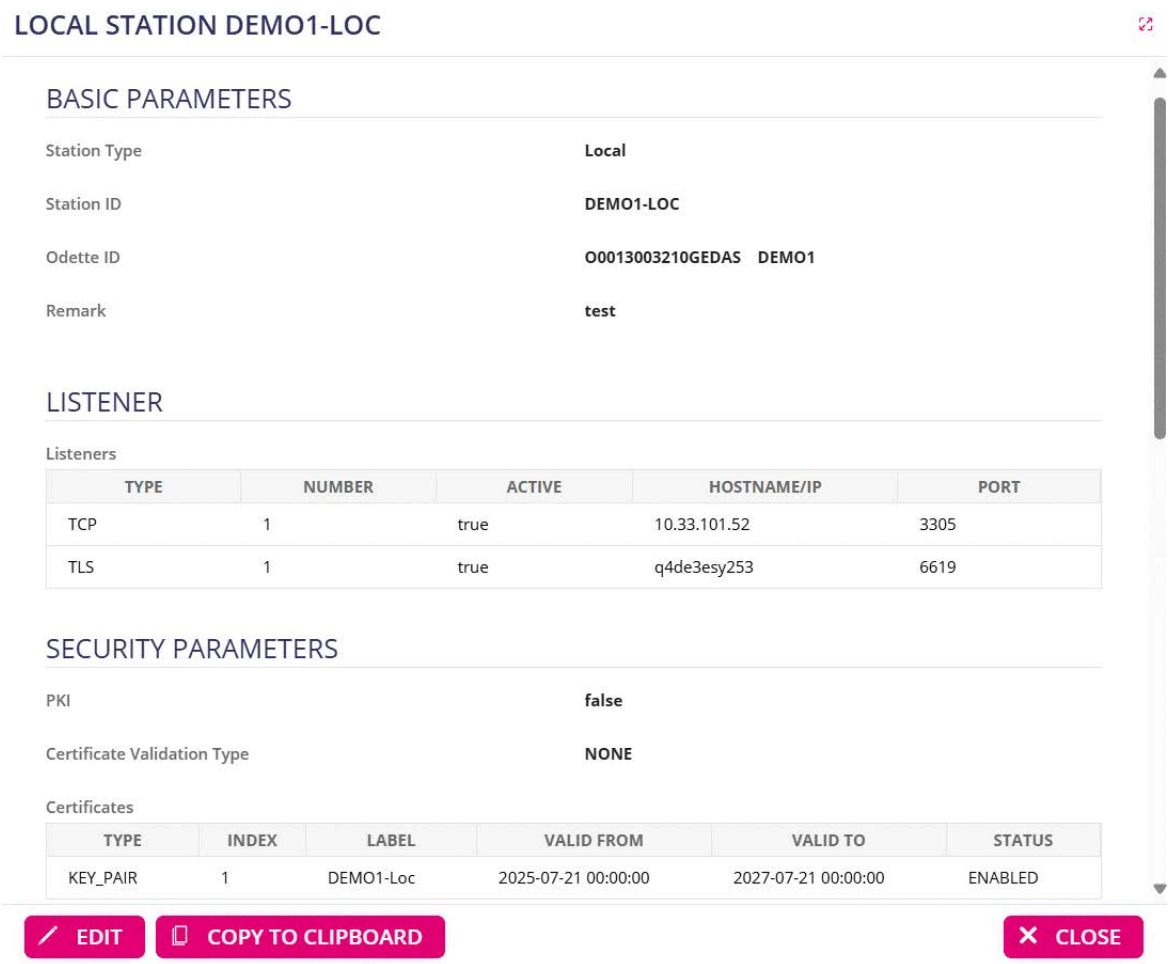
You get detailed information of a station via a double click on this station.

6.2 Local Station

Normally the local station parameters are already set during the installation. In this chapter you will find the explanation of all parameters concerning the local station. It is described how to edit the parameters.

6.2.1 Setting up of a Local Station

Open the detailed view of the local station to set up the parameters with a double-click on the station.



Exception: The Station ID and Station Type cannot be edited in the detaild view. Via right-click on the station in the list view and selecting `Change Station ID`, the SID can be set up.

If a keypair or a transmission job is assigned to the station, the SID can only be changed via cmd tool `changeStationId` (see chapter 25.7.1 "changeStationId").

6.2.2 Parameter Groups of Local Station

There are following parameter groups: **General**, **OFTP**, **Keys** and **Listeners**. With a single click on the small arrow on the left of the listeners, the receivers can be blended or blanked.

– **Basic parameters**: The mandatory parameters are **Station ID** and **Station Type**. The parameter Station ID is a ally unique station ID which can consist of up to sixteen characters. It is a strictly local definition; remote stations do not have access to these names. You can insert a text in the **Remark** field.

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– For the local station it is only necessary to set up the **ODETTE ID**. ODETTE ID is a worldwide unique identification of all nodes using the ODETTE file transfer protocol (OFTP). This 25 character name consists of

  – the letter O,

  – an 18 character organisation identifier provided by the ODETTE codification group, and

  – a 6 character computer sub address that is administrated by each organisation.

  Note: If you communicate within your own closed network only, the ODETTE ID may be freely chosen as long as it remains unique in your network.

  Please see Chapter 27 "PKI Connection" for description of **PKI** and **Certificate Validation Type** parameters.

– **LISTENER**: Your local station can receive the data of a neighbourstation via several data lines and networks. In this group the already configured listeners (receivers) are displayed. You can add several listeners of one network or listeners of several networks for your local station. See the Chapter **7.1 "Create a New Listener"**

Hints:

To add a Proxy TCP/IP or Proxy TLS listener you need to set up a Proxy Bastion before (see chapter 23.3 "Bastion Instance")

Hint: At the partner station, the handling of a virtual station is the same as the handling of a routed station.

– SECURITY PARAMETERS: PKI, Certificate Validation Type

– **CERTIFICATE USAGE DEFINITION**: In this parameter group the certificates are listed.

Key Usage Parameters of Local Station:

Parameter	Description
OFTP2 File Signature	Select the key using for file signature by default.
OFTP2 Response Signature	Select the key using for EERP/NERP-signature by default.
TLS-1	Certificate which is used for TLS encryption by default

– **CONTACT INFORMATION:** these parameters are optional and allow you to manage the contact information.

Hint: To share station data with your partners via PDX (Partner Data Exchange), the field „**Enterprise**“ must be entered.

Changes will only be saved when you click the **SAVE** button. To discard changes, click the **CANCEL** button.

6.2.3 Display Listerner of the Local Station

The detail view of the local station, shows a list of the listener configured for that station. Listener can only be edited via the sidebar under **NETWORKS**. See the Chapter **7 "Management of Networks"**

6.2.4 Display Certificates of the Local Station

The detail view of the local station, shows a list of the Certicates configured for that station. Certificates can only be edited via the sidebar under **CERTIFICATES**. See the chapter **CERTIFICATES**

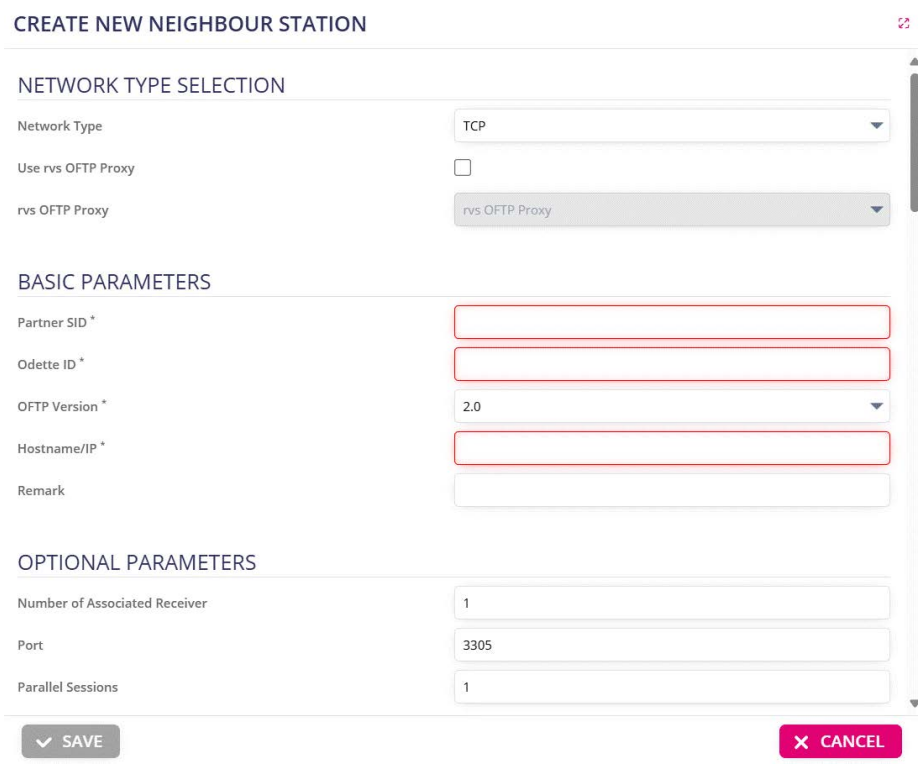
6.3 Neighbour Stations

This chapter shows how to add or delete a neighbour station and how to set up the parameters of a neighbour station.

6.3.1 Add a Neighbour Station

Follow the description below to add a neighbour station:

- Click the button **CREATE NEW** and select **Neighbour Station**



- Select a network for the communication. Possible are TCP/IP, TLS. A TLS connection is needed for line encryption. Clicking the option **Use rvs OFTP Proxy** activates the rvs OFTP Proxy input field. Here you can select the OFTP proxy.
- insert the Partner SID, Odette ID, OFTP Version, Hostname/IP of your neighbour station. In the Remark field you can insert a text.
- If needed, define the optional parameters and click **SAVE** to add the neighbour station.

  The description of the network parameters is under:

  – “TCP/IP, TSL, Proxy TCP/IP and Proxy TLS Parameters (Partner)”
- if needed set up the OFTP Optional Parameters. (You can find the description of the OFTP parameters in table “Optional OFTP Parameters (Partner)” on page 61)
- The next possibility is to set up the OFTP security parameters.
- The next possibility to set up the parameters for OFTP File Service Module.
- The last parameters are contact data. Click **SAVE** to add the neighbour station.

Hint:

You can add a neighour station with identical settings to an existing station. Right-click the neighbour station that is to be duplicated and select `Duplicate Station` entry. The settings of the duplicated stations are used as default values of the new station. SID and Odette ID parameters must be changed. The remaining parameters can also be edited if neccesary.

6.3.2 Setting up of a Neighbour Station

Open the detailed view for setting up of a neighbour station. Edit the parameters and then click **Save**.

Exception: The station ID cannot be edited in the detailed view.

Via right-click on the station to be changed and thereafter selecting `Change Station ID` an input window opens and the SID can be set up. If a certificate exists or transmission job is assigned to the station the SID can only be changed via cmd tool `changeStationId` (see chapter 25.7.1 "changeStationId").

6.3.3 Parameter Groups of the Partner Station

There are different parameter groups:

Basic parameters, **Network parameters, OFTP**, **Security parameters, Certificate Usage definition, Send file security settings** and **Listeners**.

Basic parameters:

Parameter	Description
Station Type	Type of the station
Partner SID	Mandatory: locally unique station ID which can consist of up to sixteen characters. 'A-Z', '0-9', '-', '_' and '.' are possible characters. '.' may not rank first. Partner SID is a strictly local definition
Odette ID	ODETTE ID is a worldwide unique identification of all nodes using the ODETTE file transfer protocol (OFTP). This 25 character name consists of - the letter `O`, - an 18 character organization identifier provided by the ODETTE codification group, and a 6 character computer sub address that is administrated by each organization.
OCS DocDate	Timestamp of the ocs.xml file imported via PDX
Receive Password	The password rvsEVO expects from the neighbour station.
Send Password	The password rvsEVO sends to the neighbour. ODETTE password exchange between two neighbour stations and verification always occurs while a sessio n is being established. **Hint:** With `activateStation` tool (chapter 16.3.1) this password can be overwritten temporarily for the unique connection.
Exchange Buffer Size	Maximal size of ODETTE Exchange Buffer in Bytes. Possible values: `1` - `99999` (default: `10000`). No entry means that the default value is used. **Hint:** For data exchange via ISDN network you should indicate max. 4000 Bytes because otherwise the transmission of big files could cause problems.
Exchange Buffer Credit	Maximal number of sent blocks (Exchange Buffer) without an acknowledgment. Possible values: `1` - `999` (default: `999`). No entry means that the default value is used.
Session Authentication	Activate the check box for authentification on ODETTE level at the beginning of an OFTP session. This setting must be made with both partners. Default: no authentication

Parameter	Description
OFTP Version	which OFTP version should be used. Possible values: 1.3, 1.4 and 2.0
Remark	insert a text

Network Parameters

Parameter	Description
Network Type	Type of the network
Number of Associated Receiver	Receiver number used for communication with this partner station. Default: `1`
Use rvs OFTP Proxy	Use rvs OFTP Proxy or not
rvs OFTP Proxy	OFTP Proxy
Hostname/IP	IP address or DNS name of the neighbour station.
Port	Port of neighbour station; default: `6619`.
Alternative Hostname/IP	IP address which is used for a second attempt to connect with the partner if the connection fails with the value defined in **IP Address** parameter.
Alternative Port	Used port if the connection fails with the port defined in **Port** parameter.
Active Connection	define if your local station establishs the connection automatically or not. When activating the check box the connection to the station is set up automatically if a send job is started. On the contrary, the partner station is to establish the connection and fetch the available files when required. The partner bears the costs for the connection.
Parallel Sessions	indicate the max. Number of Sessions to this station at the same time

OFTP: This group contains the parameters which refer to ODETTE protocol. In the following table you will find detailed explanation of the ODETTE group parameters.

Optional OFTP Parameters (Partner)

Parameter	Description
ERP in	Procedure for receiving receipts (EERPs/ NERPs). Possible values: `NORMAL`: The transmission is successfully terminated after receiving of EERP. `NEVER`: no EERP is needed. The transmission can be terminated without receiving a receipt. **Hint**: some operator use EERP destination and EERP originator in a different way. This causes problems with the identification of EERPs. Set the parameter **rvs_evo.job.additional_erp_reverse_search** =true in `$RVS_HOME\conf\rvs-system.properties` file to resolve this problem. Default: `NORMAL`
ERP Out	Procedure for sending receipts (EERPs/NERPs). `NORMAL`: Generation of a receipt after successful file reception and immediate active transmission. `NEVER`:no Generation of a receipt. The job is changed to ENDED status. `HOLD`: Generation of a receipt after successful file reception. The receipt, however, is only sent after having been released. `SYNC`: The OFTP session between sender and receiver station will be kept alive until the receipt is received on sender side. Default: `NORMAL` (see also parameter **EERP in**) **Hint:** this setting does not take effect to ACX transmissions. ERP is sent automatically.
Session File Direction	This parameter defines whether reception, distribution or reception and distribution of files to this partner is allowed. ERP jobs may always be transmitted. Possible values: - `B`: send and receive (default) - `S`: only direction send - `R`: only direction receive
VDSN charset	For receive jobs only: allowed character set for VDSN. Possible values: - `ODETTE`: ODDETTE character set is allowed only (default). - `ALL`: all ASCII characters are allowed

Optional OFTP Parameters (Partner)

Parameter	Description
SFIDDESC As Filename	Activate the check box for using **SFIDDESC** (Start File ID Description) as filename (instead of VDSN) for the file receiption from this partner. **Hint:** During creating a send job, **SFIDDESC** can be filled by using the **File Description** parameter. By default VDSN is used for delivery.
Restart	This parameter defines whether a restart is allowed after an error message. Possible values: true (default) or false:
Reiceive Job Priority	Priority for receive jobs. Possible values: - `1` to `5` `1` is highest priority, `5` is lowest priority, default: `3` The value is handed over to: - ERP transmission to this station - routing send job (handed over from routing receive job) - jobstart process if **Parameter Handling** = `ENV`

Security Parameters (Partner):

Parameter	Description
PKI	Please read chapter 27.2 "Configuration of PKI Binding" for detailed information.
Certificate Validation Type	Please read chapter 27.2 "Configuration of PKI Binding"

Certificate Usage Definition

Parameter	Description
Default Label	Label to use as default
OFTP2 Authentication	Select the certificate of your partner which you want to use for authentication.
OFTP2 File Signature	By default, the key used for file signature is defined at loacal station. Select the key you want to use for signing files for this partner station, if this is different from the local station configuration.

Certificate Usage Definition

Parameter	Description
OFTP2 Response Signature	By default, the key used for response signature is defined at loacal station. Select the key you want to use for EERP/NERP-signature with this partner, if this is different from the local station configuration.

Send File Security Settings

Parameter	Description
Cipher Suite	File security Algorithms: (CipherSuite: Symmetric Encryption, Asymmetric Encryption, Asymmetric Signature, Hash Function) - **02:** AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-1 - **04**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-256 - **06**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-512 - **07**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA3-512
Compression	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
Encryption	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
Signature	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
ERP Signature	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.

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Receive File Security Settings

Parameter	Description
Cipher Suite	File security Algorithmus: (CipherSuite: Symmetric Encryption, Asymmetric Encryption, Asymmetric Signature, Hash Function) - **02:** AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-1 - **04**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-256 - **06**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-512 - **07**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA3-512
Compression	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
Encryption	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
Signature	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
ERP Signature	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.

OFTP File Service Module

Parameter	Description
File Service Module	Only for File Service Module; possible values: - `NONE` (default) - `INTERNAL` - `EXTERNAL` Please see chapter 24 "File Service Module" for detailed explanation.
Compression	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.

Parameter	Description
Encryption	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
Encryption Algorithm	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
File Signature	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.
ERP Signature	Send parameter on station level. Only available if parameter **File Service Module =** `EXTERNAL`. Please see table “Security parameters:” on page 39 for detailed explanation.

6.3.4 Delete a Neighbour Station

Right-click the neighbour station to be deleted and select **Delete 1 selected station** to delete a station.

6.3.5 Activate Neighbour Station

Right-click the station and select `Activate Station` to activate the station and receive files from this station. You need this functionalty to get files from your neighbour station if the parameter **Active Connection** (group **Line Type**) is disabled on neighbour station side.
For testing the connection to a neighbour station without file transmission use the option **Activate Station without Transfer** in the dropdown menu.
Hint: You can also use the activateStation script to activate a station.

6.3.6 Automatic Certificate Exchange

certificate request

Right-click the neighbour station and select `Certificate Reqest to one Station`. In the opening window select the **station ID** of the sender and receiver and the certificate which should be ed to your partner station.
Please see also chapter 11.9 "Automatic Certificate Exchange via rvsEVO-GUI"

certificate

Right-click the neighbour station and select `Certificate to one Station` to send a certificate to your neighbour station. In the opening

window select the station ID of the sender and the receiver and the certificate which should be delivered to your partner station.
Please also see the chapter 11.9 "Automatic Certificate Exchange via rvsEVO-GUI"

6.3.7 Further Functionalities

Click in the detail view the button **ACTIONS** which offers you the following additional options:

– Activate Station - see the chapter 6.3.5 "Activate Neighbour Station"

– Activate Station without Transfer - see the chapter 6.3.5 "Activate Neighbour Station"

– Certificate Request to Station (only for User with the role Administrator) - see the chapter 11.9 "Automatic Certificate Exchange via rvsEVO-GUI"

– Certificate Deliver to Station (only for User with the role Administrator) see the chapter 11.9 "Automatic Certificate Exchange via rvsEVO-GUI"

– Change Station Type to Routed

– Add Routed Station hinzufügen (only for User with the role Administrator)

– Duplicate Station (only for User with the role Administrator)

6.4 Routed Stations

A routed station is reachable via a neighbour station. For this reason you must already have set up a direct neighbouring node about which the routed station is to be added.

Hint:

Via **Gateway Station** parameter (parameter group **General**) in the detailed view, you can replace the neighbour station. Select the station ID from the drop-down menu.

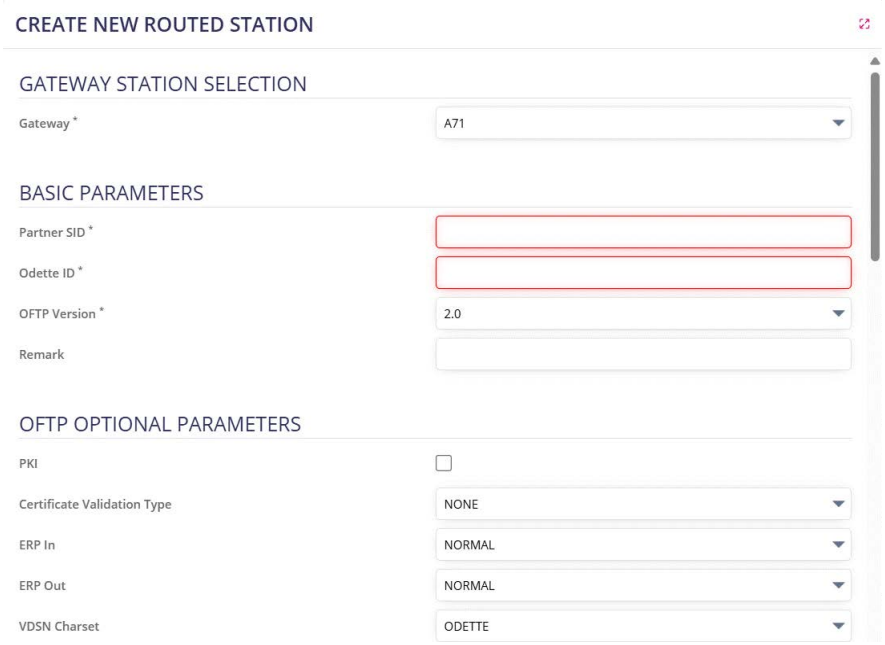
6.4.1 Add a Routed Station

Add a routed station by a right-click on the neighbour station and thereafter selecting the item `Add routed station` from the context menu.

The dialog below opens and you can set up the mandatory parameters:

– **SID** (Station ID of routed Station) and

– **Odette ID** (please see OFTP parameter group)



The other parameter groups of routed stations are: **OFTP Optional Parameters, OFTP File Service Module Parameters** and **Contact Details.**

Please see the Chapter 27 "PKI Connection" for getting information of **PKI** and **Certificate Validation Type** Parameters.

Hint:

You can add a routed station with identical settings to an existing station. Right-click the routing station that is to be duplicated and select the `Duplicate Station` entry from the combo box. The settings of the duplicated stations are used as default values of the new station. SID and Odette ID parameters must be changed. If neccesary, the remaining parameters can also be edited.

6.4.2 Setting up a Routed Station

Please see the chapter 6.3.2 "Setting up of a Neighbour Station".

The procedure is the same with all stations.

There is a special parameter in the parameter group **General**: **Gateway Station**. This parameter allows you to change the neighbour station about which the routed station is reachable.

6.4.3 Further Functionalities

In the detail view, click the button **ACTIONS** which offers the following additional options: `Certificate Request to Station` and `Certificate to Station and duplicate Station`.

The description of this functionalities you can find in chapter 6.3 "Neighbour Stations".

6.5 Virtual Stations

Virtual stations are independent OFTP instances. They are routed via local station. A unique ODETTE ID must be assigned to each virtual station. A virtual station can send and receive files.

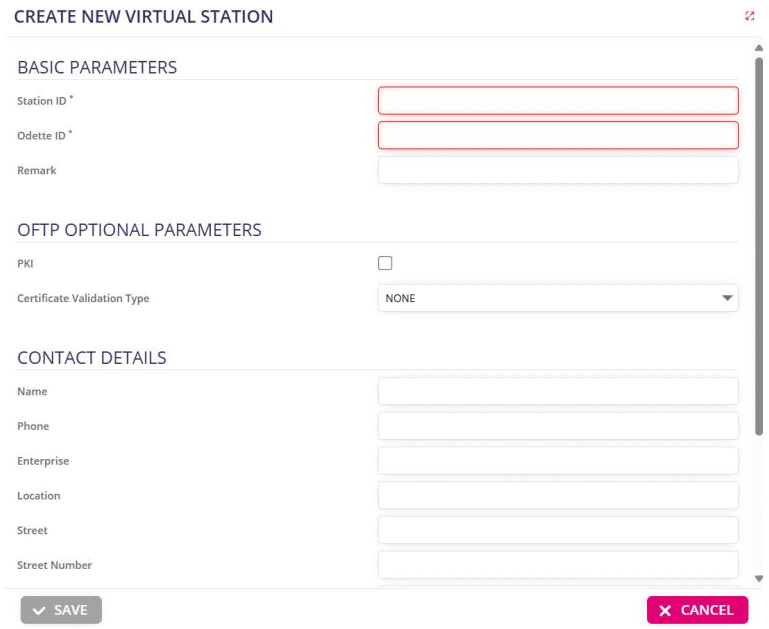
6.5.1 Add a Virtual Station

Add a virtual station by a right-click on the local station and thereafter selecting the item `Add Virtual Station` from the context menu.

The dialog box below opens and you can set up the **Mandatory Parameters**:

– **SID** (Station ID of virtual `Station`)

– **Odette ID** (please see OFTP parameter group in the chapter 6.2.2 "Parameter Groups of Local Station".)



Other parameter groups of virtual stations are **OFTP Optional Parameters** and **Contact Details.**

Please see the Chapter 27 "PKI Connection" for getting more information of **PKI** and **Certificate Validation Type** Parameters.

You will find the description of the other parameters in the chapter 6.2.2 "Parameter Groups of Local Station".

Hint:

You can add a virtual station with identical settings to an existing station. Right-click the virtual station that is to be duplicated and select the entry `Duplicate Station` from the combo box. The settings of the duplicated stations are used as default values of the new station. **SID** and

Odette ID parameters must be changed. If neccesary, the remaining parameters can also be edited.

6.5.2 Setting up a Virtual Station

Please see the chapter 6.3.2 "Setting up of a Neighbour Station".The procedure is the same with all stations.

6.5.3 Further Functionalities

In the detail view, click the button **ACTIONS** which offers you the following additional option: `Duplicate Station`.

Hint: At the partner station, the handling of a virtual station is the same as the handling of a routed station.

6.6 Station Configuration via rvsStationlist.xml

The station configuration can also be done via the XML station configuration file `$RVS_HOME/conf/rvsStationlist.xml`.

The `StationLoc` element in the configuration file is equivalent to the Local Station in the rvsWeb-GUI, `StationNeighbour` is Neighbour Station, `StationRouted` is the Routed Station and `StationVirtual` is the virtual station in the rvsWeb-GUI.

Abstract of `rvsStationlist.xml` file:

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<RVS_STATION_CONFIG>
-<StationLoc>
 . . .
-</StationLoc>
-<StationNeighbour>
 . . .
-<StationNeighbour>
-<StationRouted>
. . .
-</StationRouted>
-<StationVirtual>
. . .
-</StationVirtual>
</RVS_STATION_CONFIG>
```

Lines starting with (<!--) and ending with (-->) are interpreted as comments.

Edit the `rvsStationlist.xml` file if you have to change mandatory parameters (ODETTE ID, TCPIP_Basic) or if you wish to assign values to other optional parameters such as Contact.

Use a text editor (e.g. Edit, TextPad) to do so. Please make sure to save your XML files as valid XML documents after editing as otherwise rvsEVO will not be able to read them and may fail to start correctly.

The changes in the GUI will be visible immediately (after saving) in the XML station configuration file; the changes in the XML station configuration file will be visible in the GUI only after a new start of the rvsEVO (command startGUI).

Please see chapter 16.2 "updateStationList" for getting information about setting up the stations at run time.

Note: For correct TCP/IP communication you must ensure that the IP ports for RMI (1099) and for Odette (e.g. 3305) are free.

For the description of the parameters in the `rvsStationlist.xml` file please refer to the description of the GUI parameters.

The parameter label is not configurable via the GUI:

Parameter of rvsStationList.xml

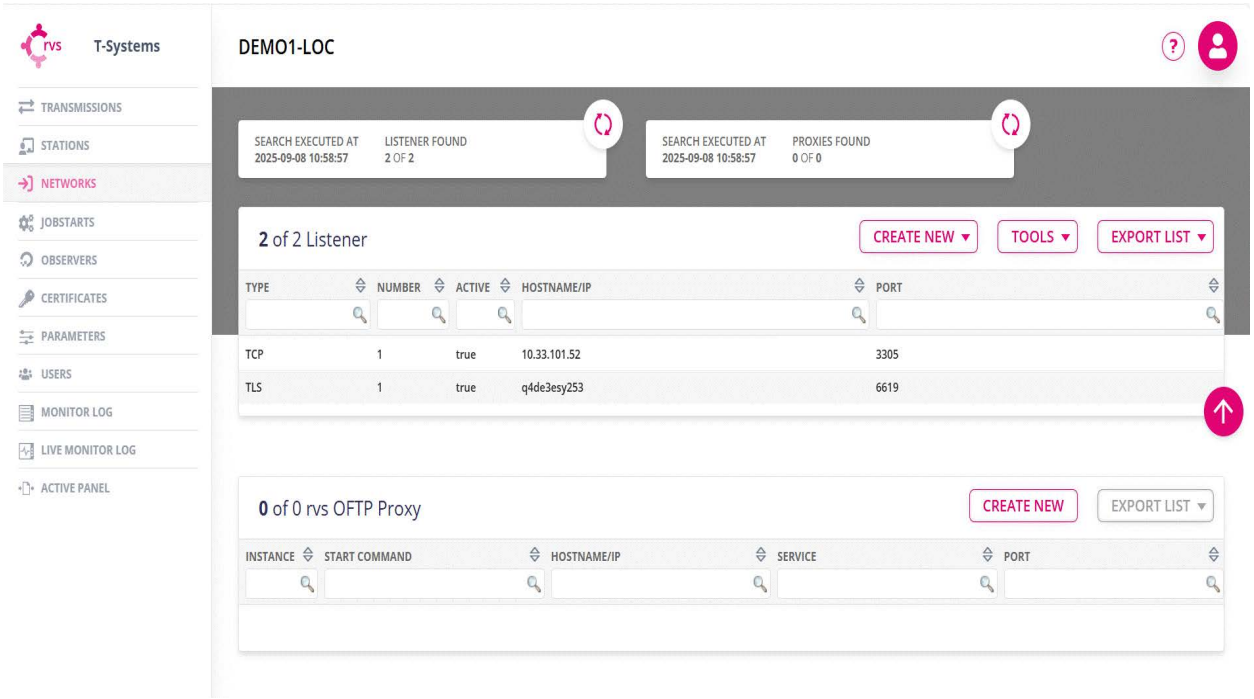
Parameter	Description
-Label	Free text; use this parameter for labeling the station

7 Management of Networks

The Listeners can be accessed via the sidebar under **NETWORKS**.

Your local station can receive the data of a neighbour station via several data lines and networks - **LISTENER**.

In the default view, the network management is structured as shown in the following picture and the already configured listeners (receivers) are displayed:



7.1 Create a New Listener

User the button **CREATE NEW** to create a new listener. The following types are available:

– TCP

– TLS

– Proxy-TCP

– Proxy-TLS

A TLS connection is required if all data transmitted is to be encrypted!

You can only add a **Proxy-TCP or Proxy-TLS** connection once a Proxy Bastion instance has already been configured.

Hint: The management of networks is available to users with the role of **Administrator** or **Operator**. The user with the role **User** kann open these data in read-only mode.

TTCP/IP and Proxy TCP/IP

In the next table you will find the description of the TCP/IP and Proxy TCP/IP parameters:

TCP/IP- and Proxy TCP/IP Parameters (Local)

Parameter	Description
ReceiverNumber	Number for differentiating the various receiving channels through which the local station can be reached. rvsEVO automatically assigns and manages this number.
Enabled	This parameter defines whether or not the TCP/IP listener is enabled or disabled: Default: disabled
Hostname/IP	IP address or DNS name of the own station. In case of PROXY TCP: IP address or DNS name of Proxy Bastion.
rvs OFTP Proxy	Only for Proxy TCP/IP: you can select the Bastion from a list of the Bastion instances created in rvsEVO in which the receiver / listener should be started
Port	Port on which a TCP/IP listener is to be started; `3305` by default.
Max. Incomming Sessions	Maximal number of simultaneously active receipt processes on the same channel. Maximum: `999`.
Time Out (ms)	Time Out in milliseconds after which the communication program the connection closes, if the partner station does not answers. Default: `60000` `0` means that the default value is used.

TLS and Proxy TLS

The TLS and Proxy TLS parameters are necessary for encrypted communication with a partner. TLS (Transport Layer Security) is an encryption protocol for data transmission via internet. It provides encryption during the communication way, on session level.

Please see the chapter 10 "Secure Transmission with OFTP2" for further information about encryption.

TLS and Proxy TLS Parameters (Local)

Parameter	Description
ReceiverNumber	Number for differentiating the various receiving channels through which the local station can be reached. rvsEVO automatically assigns and manages this number.

TLS and Proxy TLS Parameters (Local)

Parameter	Description
Enabled	This parameter defines whether or not the TLS listener is enabled or disabled: Default: disabled
Hostname/IP	IP address or DNS name of the own station. In case of PROXY TCP: IP address or DNS name of Proxy Bastion.
rvs OFTP Proxy	Only for Proxy TCP/IP: you can select the Bastion from a list of the Bastion instances created in rvsEVO in which the receiver / listener should be started
Port	Port on which a TCP/IP listener is to be started; `3305` by default.
Max. Incomming Sessions	Maximal number of simultaneously active receipt processes on the same channel. Maximum: `999`.
Time Out (ms)	Time Out in milliseconds after which the communication program the connection closes, if the partner station does not answer. Default: `60000` `0` means that the default value is used.
Client Authentication	Sometimes is for the TLS connection necessary, that a client (in this case a partner, from whom you receive data) should be authenticated. With this parameter you can choose, if this is: – `NEEDED` - `NONE` (no authentication) or - `WANTED` **Note**: The authentication should be done with the X.509 certificates.

You can add several listeners of one network or listeners of several networks for your local station.

7.2 Additional functionalities for Listener

By right-clicking on a Listener in the list view, a selection box will open, that offers different options, depending on your user role:

– Restart 1 selected Listeners

– delete 1 selected row

By clicking the button **TOOLS,** you can also start the functionality **Restart all Listener** (available to users with the role of **Administrator** or **Operator**)

7.2.1 Restart Listener

Select the listener you want to restart and right-click to open the selection box. Select the function `Restart 1 selected Listeners.` You can cancel the restart using the buton **CANCEL**.

Hint:

This functionality is available to users with the role of **Administrator** or **Operator**.

7.2.2 Delete Listener

Select the listener you want to delete and right-click to open the selection box. Select the function `Delete 1 selected row.` You can cancel the delete using the button **CANCEL**.

Hint:

This functionality is available to users with the role of **Administrator** or **Operator**.

7.3 Proxy Bastion

You need to add a Proxy Bastion instance as pre-condition for using the functionality rvs® OFTP Proxy.

7.3.1 Add Proxy Bastion

User the button **CREATE NEW** to create a new OFTP Proxy.

The parameters to be entered have already been saved during the installation of the Bastion Instance.

In the following table you find the description of Proxy Bastion parameters:

Basic Parameters

Parameter	Description
Service	Name of the Bastion Service. Default: service **Important:** Please note that the service name must be unique in case of multiple Proxy Bastion instances.
Port	Port of the Bastion Instance for the RMI-communication, on which a listener is started.
Hostname/IP	Name or IP-address of the server, on which the Bastion Instance has been installed. When using NAT (Network Address Translation) procedure the public IP address.

OFTP Parameters

Parameter	Description
Start Command	Directory and Start Command of rvs® OFTP Proxy
Backup Hostname/IP	Name or IP address of a machine with a back-up installation of Proxy Bastion.
Backup Port	Port for RMI communication of the Backup installation on which a listener is started.
RMI Sender Hostname/IP	Hostname or IP address of the sender used by rvsEVO server for connection with the Proxy Bastion.
TLS for Proxy Access	Activate the check box for using an encrypted connection with the Proxy Bastion. Only with **TLS Truststore** parameter.
TLS Truststore	Path and name of the truststore which includes the certificate for TLS encryption. Only with **TLS for Proxy Access** parameter.

7.3.2 Display Proxy Bastions

Via double-click on an Proxy Bastion in the list you will get the detailed view of the Proxy Bastion.

7.3.3 Setting up Proxy Bastion

Open the detailed view for setting up the parameters and click on the button **EDIT**. Confirm your changings with a click on the button **SAVE**.

Restart rvsEVO server to enable the changes.

Important: It is to be considered, that these parameters are customised also in the start script `$RVS_HOME/bin/bastion.cmd` (`$RVS_HOME/bin/bastion.sh`).

Hint:

This functionality is available to users with the role of **Administrator** or **Operator**.

7.3.4 Delete Proxy Bastion

Right-click the Proxy Bastion in the list view and select the item `Delete 1 selected row` from the context menu to remove a Proxy Bastion.

Hint:

This functionality is available to users with the role of **Administrator** or **Operator**.

8 Management of Jobstarts

The Jobstarts include rules that allow specific processes to be triggered when corresponding incoming or outgoing files are detected.

For example, these can be programs (Jobstart scripts) that are executed.

Jobstarts can be configured as Jobstart after receive, after send attempt or after failure.

Hint:

Examples of Jobstart scripts can be found in the directory `$RVS_HOME/docs/jobstart-examples`.

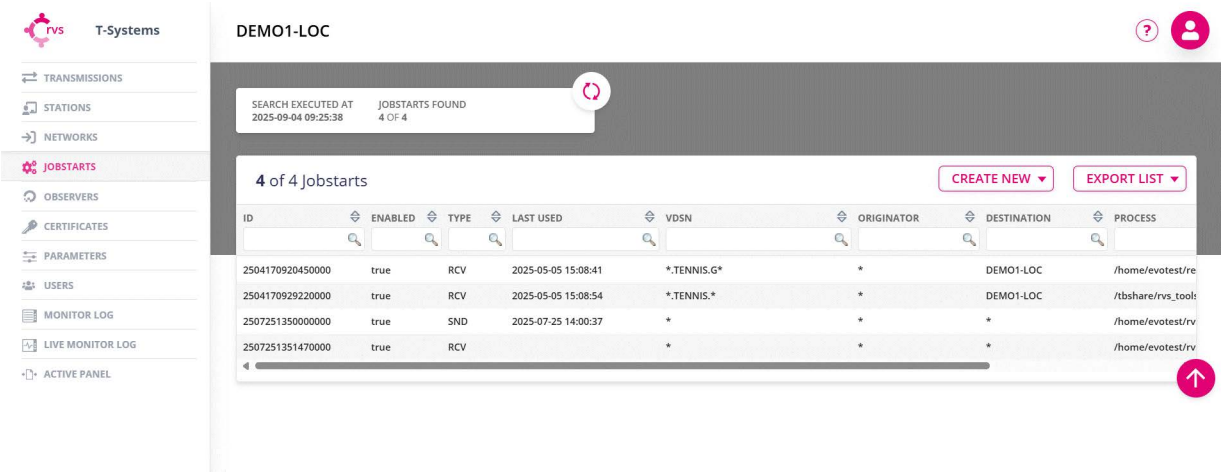
We recommend to save your own scripts in `$RVS_HOME/jobs` directory which has no other use and is taken into consideration in case of backup and recovery. Don't save changed scripts and your own one in `$RVS_HOME/bin` directory or subdirectories of `$RVS_HOME/bin` because these will be overwritten during update installations.

Job filter

If more than one jobFilter applies to the send job, receive job or failed job, the job is started whose jobFilter applies most exactly (eg while using wildcards: `testdoc*` is more exact than `testd*`).

8.1 The user interface

The Jobstart management is reachable via sidebar item `Jobstarts`.



If more than one job filter applies to a job, the program of the most specific job filter will be started. When using wildcards, for example, `Testdata` is more specific than `Test`.

The list displays the most important parameters of the jobstart:

– ID (automatically assigned Job ID by rvsEVO in format `YYMMDDh-hmmssCCCC`)

– Enabled (this parameter defines whether the jobstart is active)

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– Type (direction of communication, specified by rvsEVO: `SND, RCV, FAIL`)
– Last Used (date and time of the last use of the job, set by rvsEVO)
– VDSN (virtual data name)
– Originator (Station Id of source station (local or virtual Station))
– Destinator (Station Id of target station)
– Process (Program to be started when all filter conditions apply) Parameters, seperated by spaces, can be passed to the program. The following must be noted:

- Terms with blanc should be placed in quotation marks.
- By using cmd /: indicate cmd.exe only with the script name.
- If **Parameter Handling** is ARGS: indicate the user parameters first and thereafter the system parameters.
- With Linux shells the parameters are passed without quotation marks, with Windows the parameters are passed with quoations marks.

A defined set of the following job parameters is passed to the programs:
- 1 JobId
- 2 Station ID (of sender or recipient)
- 3 File name of the file sent or received
- 4 VDSN
- 5 Date of the job
- 6 Time of the job
- 7 Number of attempts to send.
- 8 Format of a file
- 9 Record length of a file
- 10 Transmitted Bytes

How these parameters should be used in a script, see the example at the end of this chapter
(file $RVS_HOME/bin/jobstart/jobstart_detailed.bat).

You can obtain a sorted list according to your needs by clicking on the parameter name in the column header after.Using the refresh icon in the upper right corner of the list view, you have the option to update the list.

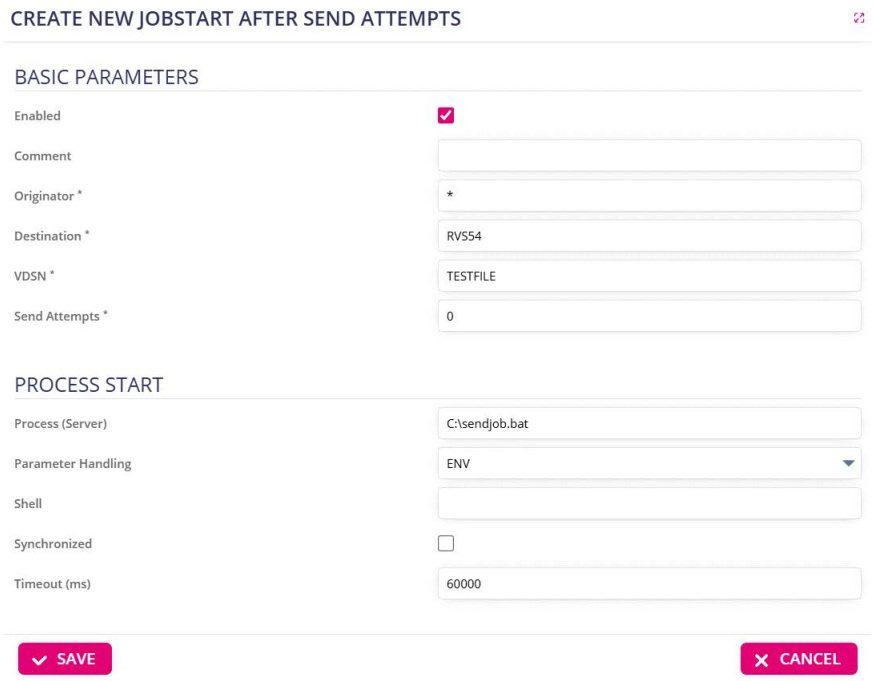
8.1.1 Create a new Jobstart

The following functionalities can be performed using the button **CREATE NEW**:

– Jobstart After SEND
– Jobstart After RECEIVE
– Jobstart After Fail

Jobstart After SEND — Click the **CREATE NEW** button

Select **Jobstart After SEND**, the following window will be loaded:



In the first part of the window, **BASIC PARAMETERS** the mandatory parameters (Enabled, Originator, Destination, VDSN and Send Attempts) are requested.

Under **PROCESS START** you are prompted to specify the parameters required to start a process.

You can complete your entries by clicking the **SAVE** button or discard them by clicking **CANCEL**.

In our example, the program `C:/sendjob.bat` will be started after a file with the virtual filename `TESTFILE` was successfully delivered (**Send Attempts=0**) from any station to the neighbour station `RVS54`.

It is possible to define a number of aborted send attempts after those a job is to be started. Therefor set the parameter **Send Attempts**. If several jobs are to be started after a different number of send attempts this is handled as follows.

Example: Program 1 (P1) shall be started after one send attempt and program 2 (P2) shall be started after three send attempts, then:

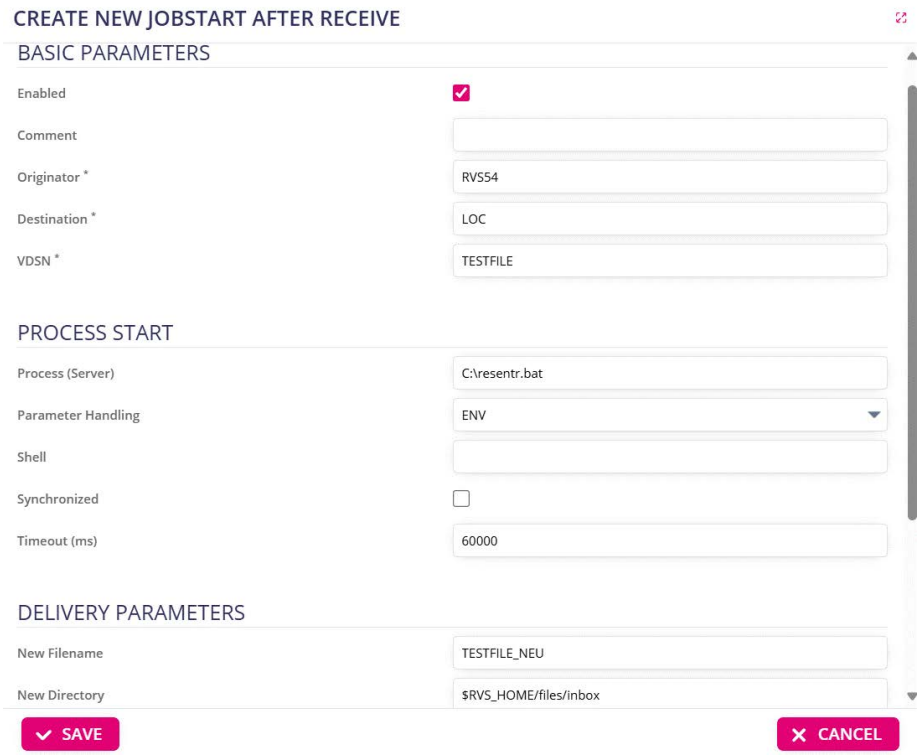
1. send attempt -> P1
2. send attempt -> P1
3. send attempt -> P2
4. send attempt -> P1
5. send attempt -> P1
6. send attempt -> P2

Hint: Please see the table “Jobstart parameters:” on page 82 for the description of the parameters.

Jobstart After Receive

Click the button **CREATE NEW**.

Select **Jobstart After Receive** the following window will be loaded:



A Jobstart After Receive is triggered after a file receive and before sending an EERP.

In the following example the program `C:/resentr.bat` will be started after the local station `LOC` receives a file with the virtual file name `TESTFILE` from the neighbour station `RVS54`. The file is renamed to `TESTFILE_NEW` and saved in `$RVS_HOME/files/inbox` directory.

In the first part of the window, **BASIC PARAMETERS** the mandatory parameters (Enabled, Originator, Destination, VDSN) are requested.

Under **PROCESS START** you are prompted to specify the parameters required to start a process.

You can complete your entries by clicking the **SAVE** button or discard them by clicking **CANCEL**.

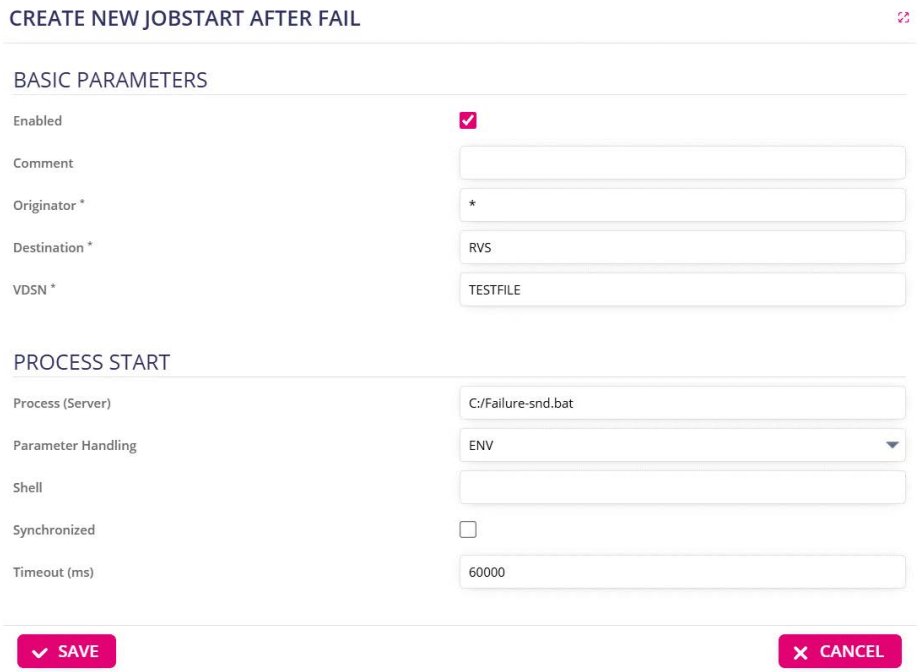
Under **DELIVERY PARAMETERS** you can enter the delivery parameters (New Filename, New Directory, Replace, TimeStamp, TimeStamp Format, Conversion Table and Record Handling).

Hint: Please see table “Jobstart parameters:” on page 82 for the description of the parameters.

Jobstart after Fail

Click the button **CREATE NEW**

Select **Jobstart After Fail**, the following window will be loaded



In the following example a file with the virtual filename `TESTFILE` should be send from local station `LOC` to neighbour station `RVS`. The program `C:\Failure-snd.bat` is to be started after receiving an error message after send attempt.

In the first part of the window the mandatory parameters must be entered (Enabled, Originator, Destination and VDSN).

Under **PROCESS START** you are prompted to specify the parameters required to start a process. (Process, Parameter Handling, Shell, Synchronized and Timeout (Sync.)).

You can complete your entries by clicking the **SAVE** button or discard them by clicking **CANCEL**.

Hint: Please see table “Jobstart parameters:” on page 82 for the description of the parameters.

Note: A JobStart after fail is launched if a job was changed to `FAILED` status or if active jobs were changed into an error status which requires an operator intervention (e.g. `Restart`). A JobStart After Fail is started also after a NERP was received.

The following situations activate a Jobstart After Fail:

Send transmission:

– preprocessing fails (status: `SP_FAILED`)

– partner station is missing

– Jobstart After Send Attempts fails (`ENDED_WITH_JS_ERROR`)
– SFNA (**S**tart **F**ile **N**egative **A**nswer) after connection establishment (e.g. unencrypted file transmission while the neighbour station demands encryption)
– `FATAL_ERROR` after SFNA (e.g. the file does not exist)
– Jobstart fails after successful send attempt (`ENDED_WITH_JS_ERROR`)
– EFNA error (**E**nd **F**ile **N**egative **A**nswer) after transmission
– receipt of NERP (Negative-End-to-End-Response)

Receive transmission:

– dispatch of SFNA
– dispatch of EFNA (e.g. the byte count of the received file is not conform to the byte count of the transmitted file)
– subsequent treatment fails; e.g. missing private key for decryption (`SP_FAILED`)
– JobStart after receive fails (`ENDED_WITH_JS_ERROR`)

8.1.2 Jobstart Parameters

Jobstart parameters:

Parameter	Description
ID	Jobstart ID set by rvsEVO
Enabled	With this parameter you decide whether the job is to be started or not.
Last Used	Date and time of last jobstart usage (set by rvsEVO)
Last Modified	Date and time of last modification. With RCP-GUI also name of user who changed the settings. (set by rvsEVO)
Last Modified By	Name of user who changed the settings (not with RCP-GUI; set by rvsEVO)
Comment	free Text
Originator	Station Id of source station (local or virtual Station)
Destination	Station Id of target station
VDSN	Virtual file name (regular expression as filter)

Jobstart parameters:

Parameter	Description
Send Attempts	Number of failed attempts to send. Successful file transmission is indicated by "0" here.
Process	Program to be started when all filter conditions apply. Parameters, seperated by spaces, can be passed to the program. The following must be noted: - Terms with blanc should be placed in quotation marks. - By using cmd /: indicate `cmd.exe` only with the script name. - If **Parameter Handling** is `ARGS`: indicate the user parameters first and thereafter the system parameters. - With Linux shells the parameters are passed without quotation marks, with Windows the parameters are passed with quoations marks. A defined set of the following job parameters is passed to the programs: - 1 JobId - 2 Station ID (of sender or recipient) - 3 File name of the file sent or received - 4 VDSN - 5 Date of the job - 6 Time of the job - 7 Number of attempts to send. - 8 Format of a file - 9 Record length of a file - 10 Transmitted Bytes How these parameters should be used in a script, see the example at the end of this chapter (file `$RVS_HOME/bin/jobstart/jobstart_detailed.bat`). (not for Remote GUI; please see parameter **process (Server)**).
Parameter Handling	With this parameter you can decide how to transmit the job data to the process. Possible values: - `ARGS`: job data are passed as arguments (default) - `ENV`: job data are - set as environment variables - `REPLACE`: for compatibility with rvs® portable. For further information see rvsEVO Operator Manual chapter „Customizing rvsEVO“ / „Pass Parameters to Jobstart Process“.

Jobstart parameters:

Parameter	Description
Shell	Command Shell for executions of the program e.g. `ksh`,`.csh`, ... on UNIX systems
Synchronized	With this parameter you decide whether the OFTP session between sender station and receiver station will be kept alive waiting until the process is finished or not. If the check box is activated the session will be kept alive. Default: not activated.
Timeout (ms)	If parameter **Synchronized** is activated: Time Out in milliseconds after which the communication program the connection closes, if the partner station does not answer. Default: 60000
New Directory	With this parameter you can save the received file in another directory than `Inbox` directory. (set by rvsEVO)
New Filename	With this parameter you can save the received file with another name. VDSN is used by deault. Timestamp is added like defined in **TimeStamp** parameter.
Replace	With parameter **Replace** you decide how to handle receiving files which own the same name like existing files. If the check box is activated the existing file is replaced. Otherwise a timestamp is added (default).
TimeStamp	With this parameter you define rules for timestamp creation. If the parameter is ... activated: timestamp is to be generated generally ... not activated: timestamp is to be added only if necessary (default)
TimeStamp Format	Format of the timestamp. Possible values: - `TIME`: creation time of the job and counter; format: `hhmmssccc` (default) - `DATETIME`: creation date and time of the job and counter; format: `YYMMDDhhmmssccc` - `SFID_DATETIME`: Odette timestamp of SFID; date, time and counter in format `YYMMDDhhmmsscccc` - `COUNTER`: 000000 - 999999. If the counter is not sequential, the gap is filled first before the counter is counted up.

Jobstart parameters:

Parameter	Description

Conversion table	For ASCII(ANSI) - EBCDIC conversion the following conversion tables are available:
	`ASCII-IBM037, ASCII-IBM273, ASCII-IBM500, ANSI-IBM037, ANSI-IBM273, ANSI-IBM500.`
	For EBCDIC - ASCII(ANSI) conversion the following conversion tables are available:
	`IBM037-ASCII, IBM273-ASCII, IBM500-ASCII, IBM037-ANSI, IBM273-ANSI, IBM500-ANSI.`
	Please read the „rvsEVO Operator Manual“ chapter „Customizing rvsEVO“ / „Conversion Tables“ for information about conversion tables in general as well as for using your own one.

Record Handling	With this parameter you decide, whether each line of a received file in **F** or **V** format is terminated by line feed. Possible values: `VFTXT`: Files in F or V format are saved in text mode. Line feed is inserted at the end of each record (default) `VTXT_FBIN`: Files in V format are saved in text mode; files in F format are saved in binary mode. `VBIN_FTXT`: Files in V format are saved in binary mode; files in F format are saved in text mode `VFBIN`: Files in F or V format are saved in binary mode.

8.1.3 Edit a Jobstart

Detailed view

With a double-click on an entry in the jobstart list you will get the detailed view of the jobstart. The parameters of a jobstart can be edited in the detailed view.

Clicking the **EDIT** button unlocks the fields that can be modified.

Save your changes via a mouse-click on the **SAVE** button.

Hint: This functionality is available to users with the role of Administrator or Operator.

8.1.4 Delete a Jobstart

You can delete a jobstart by right-clicking the jobstart in the list and thereafter selecting **Delete 1 selected row**, To delete several jobstarts select the jobstarts to be deleted while keeping the **ctrl** button pressed and select the jobstarts to be deleted with a left mouse-click.

Hint: This functionality is available to users with the role of Administrator or Operator.

9 Management of Observer

9.1 Observer

The feature Observer generates send entries for files, which are in a configurated directory. The Observer checks in regular time intervals a directory for the specified file names (mask). If such files are found, the send entry will be created. The options for send entry can be configurated, too.

If more than one Observers with the same Observer directory applies, that Observer with most exactly filter is applied.

Example:

File „Observertest“ is found in Observer directory. Value of **Mask** parameter of Observer 1 is „`Observ`*“ and of Observer 2 „`Observer`*“.

Observer 2 will be applied because the filter is more exactly.

Hint:

rvsEVO provides a thread pool for Observers. A thread is needed for each directory check. After termination of the control procedure the thread is enabled again. The pool contains 40 threads by default. This setting is configurable via Maintenance parameter **ObserverThreadPoolSize** (Please see chapter 12.4 "MAINTENANCE").

9.1.1 Display an Observer list

Observer lists

Observers can be listed via a single-click on the **OBSERVERS** entry in the sidebar.

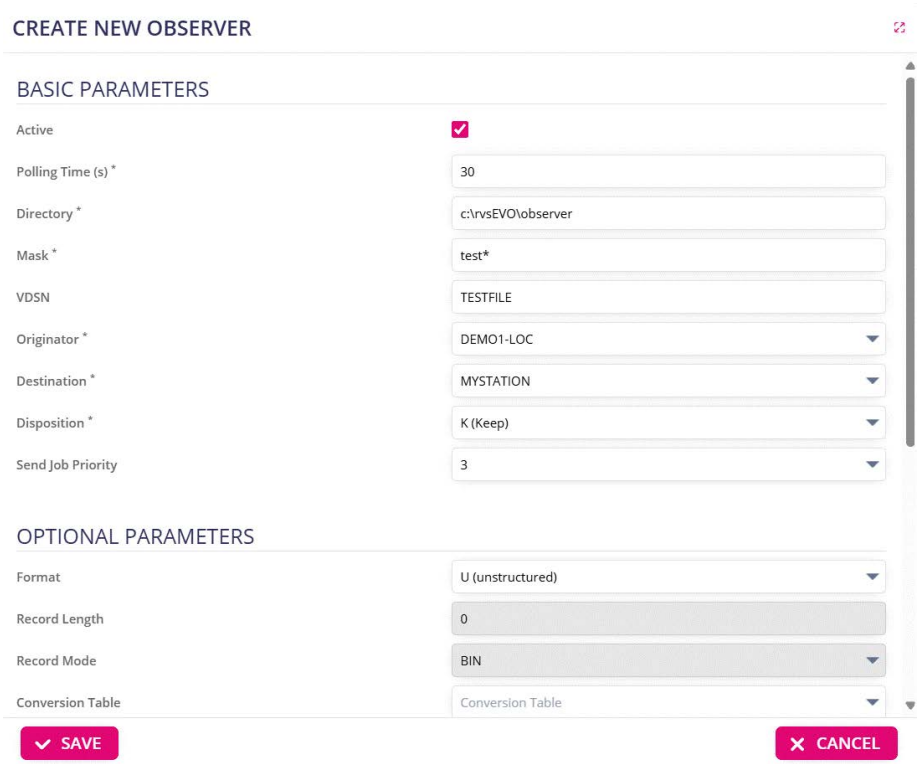
The following informations are given: ID (Number awards by rvsEVO in format `YYMMDDhhmmsscccc`, Active, Last Used, Directory, Mask, VDSN, Destination, Security and File Info).

You can get an output sorted according to your needs by clicking the arrow icon next to the column header.

You can refresh the list using the refresh icon in the upper-right corner of the Observer view 

9.1.2 Create a new Observer

Click the button **CREATE NEW**, the following window will be loaded:



In our example the directory `c:\rvsEVO\observer` will be scanned every 30 seconds for files which name begins with `test` and the detected files will be send with VDSN `TESTFILE` to the partner station MYSTATION. After transmission the files are not deleted from the Observer directory.(Keep)

You must now enter the Mandatory Parameters from the **BASIC PARAMETERS** (Directory, Mask, VDSN, Originator, Destination, Disposition).

You can next insert the **OPTIONAL PARAMETERS** (Format, Record Length, Record Mode, Conversion Table, Serialization, Label and File Description).

Insert next the **SECURITY PARAMETERS** (Cipher Suite, Compression, Encryption, File signature and Request signed EERP/NERP).

You can complete your entries by clicking the **SAVE** button or discard them by clicking **CANCEL**.

Hint: Another way to add an Observer is to duplicate an existing Observer and customize the parameter values. You will find the option `Duplicate Observer` via right-click on an Observer in the Observer list.

In the following table the Observer parameters are described.

Observer parameters

Parameter	Description
Active	This parameter defines, if the feature Observer is active or not.
Polling Time (s)	Time interval between two checks. Default: 30 Minimum: 1
Directory	Directory, which should be scanned.
Mask	The regular expression, which should be applied. The wildcards '?' (for one sign) and '*' (for one or more signs) and are permitted. If it finds them, it will create a send entry.
VDSN	Virtual file name (regular expression as filter)
Originator	Station Id of source station
Destination	Station ID of destination.
Disposition	Defines whether the file is removed or not. Possible values: - `D`: File is removed - `K`: File is not removed
Send Job Priority	Priority for send jobs. Possible values: `1` - `5`
Format	File format. Possible values: - `T`, `F`, `V`, `U`
Record Length	Maximum record length for files in format `F` or `V`
Record Mode	Mode of files in format 'F' or 'V'. Possible values: - `TXT` - `BIN`
Conversion Table	Name of conversion table.
Serialization	Serialization Possible values: - `true, false`

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Observer parameters

Parameter	Description
Label	Label for serialization. All the files which have been sent in the same group, should have the same ID (label). If this parameter in case of serialization=Y(true) is not set, VDSN is used as a label.
File Description	Comment
Cipher Suite	File security Algorithms: (CipherSuite: Symmetric Encryption, Asymmetric Encryption, Asymmetric Signature, Hash Function) - **02:** AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-1 - **04**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-256 - **06**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA-512 - **07**: AES_256_CBC, RSA_PKCS1_15, RSA_PKCS1_15, SHA3-512
Compression	Compression Possible values: - `true, false`
Encryption	File encryption Possible values: - `true, false`
File Signature	File is signed before sending Possible values: - `true, false`
Request signed EERP/ NERP	

Please see chapter 5.5.1 "Status Values for Send Jobs and Receive Jobs" for further information.

Notes:
You can use the wildcard '*' in the Observer parameter **VDSN** for taking over segments of the original filename, which are separated by points, into the new VDSN.

Examples:

Original filename:	AAA.BBB.CCC
VDSN (Observer):	PART1.*.PART2
New VDSN:	PART1.BBB.PART2

Original filename:	DEH.KMUKQY.AS.111.ASN.D123.T456
VDSN (Observer):	DEH.KMUKQY.AS.*.ASN
New VDSN:	DEH.KMUKQY.AS.111.ASN

Original filename:	DEH.KMUKQY.AS.111.ASN.D123.T456
VDSN (Observer):	*.*.*.*.*
New VDSN:	DEH.KMUKQY.AS.111.ASN

You can also use the original filename as file description. This function is controlled by the properties file `$RVS_HOME/conf/rvs-system.properties` with the definition `observer.sfid-desc.usefilename=true`.

Detailed view

With a double-click on an entry in the Observer list you will get the detailed view of an Observer.

9.1.3 Edit an Observer

The parameters of an Observer can be edited in the detailed view. Save your changes via a mouse-click on the **Save** button.

Hint: This functionality is available to users with the role of Administrator or Operator.

9.1.4 Enable or Disable Observer

Select the Observers to be enabled or disabled in the Observer list and open the drop-down menu via right-click on selected Observers. The options `Enable Observer` and `Disable Observer` are offered via drop-down list.

You can also enable or disable a single Observer via `Active` checkbox in the detailed view of the Observer.

9.1.5 Delete an Observer

You can delete an Observer by right-clicking the Observer in the list and thereafter selecting **Delete 1 selected row**. To delete several Observers select the Observers to be deleted while keeping the **ctrl** button pressed and select the Observers to be deleted with a left mouse-click.

Hint: This functionality is available to users with the role of Administrator or Operator.

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10 Secure Transmission with OFTP2

Basics

For a secure transmission risks such as the attack on the confidentiality, intergrity and authentication of your data, must be excluded. Also the traceability of the transmission must be ensured. In rvsEVO this is achieved with the following features:

- line encryption via TSL protocol
- OFTP2 authentication via AES and RSA
- file encryption via AES and RSA
- file signature
- signature of EERP/NERP

10.1 Line Encryption

Line Encryption is the encryption of the network connection between two direct neighbour stations during the transmission. In rvsEVO the TLS protocol (**T**ransport **L**ayer **S**ecurity) is used therefore.

10.2 File Encryption

The file to be send is encrypted by the sender and decrypted by the receiver.

CMS (Cryptographic Message Syntax) is an Internet Standard (rfc 2630) for the syntax of an encrypted message. This standard is implemented in the OFTP version 2.

CMS encryption format in rvsEVO combines the benefits of symmetrical and asymmetrical techniques: the high speed of the symmetrical and the security level of the asymmetrical technique. rvsEVO uses the following techniques:

- **AES** as symmetrical technique for CMS
- **RSA** as asymmetrical technique for CMS

10.2.1 Principle and Sequence of rvsEVO Encryption

The principles of the encryption, which will be explained in the following part, are the same for all encryption formats, mentioned in this chapter. Each participant in encrypted communication creates a key pair, comprising the **public key** and the **private key**.

Distributing the public key / safely storing the privat key

Each participant provides the public key to each partner he expects files from. This allows the data to be exactly encrypted for the partner who sent this public key. You can safely distribute the public key since this key alone is not sufficient for decryption.

Each participant keeps his private key and stores it safely.

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Three keys are required for decryption (the own key pair and the partner's public key). It will no longer be possible to decrypt files sent by the partner if one of the three necessary keys is lost. The private key is also needed for creating a signature.

10.3 Digital Signature

rvsEVO uses digital signature to increase security. The signature ensures that data do not undergo any unnoticed changes during the transmission. The digital signature of the End-To-End-Response (EERP / NERP) ensures the traceability of the data transmission.

10.4 Key Usage

If more than one key or certificate exists for a station the following key/ certificate with default label is used by default:

– File signature: private key with eldest „valid-from date“

– ERP signature: private key with eldest „valid-from date“

– Authentication: Certificate of the partner with newest „valid-from date“

– File encryption: Certificate of the partner with newest „valid-from date“

Hints:

This settings can be changed in Stations perspective via parameter group **Keys** of the relevant station.

For receiver of files with signature rvsEVO provides the possibility to ignore a failed validation. In this case the job will be ended without fault. Via defintion `rvs_evo.crypto.filesignature.verification.weak_sids in $RVS_HOME/conf/rvs-system.properties` file you can customize this function. (Please see rvsEVO Operator Manual, chapter „Customizing rvsEVO“ for further information)

10.4.1 Necessary Steps for Encrypted Transmission via rvsEVO:

- Create your own key pair (see chapter 11.3 "Generate a Key Pair")
- Export your own public key in form of X.509 certificate
- Send the own public key to your partner (via email or rvsEVO)
- The partner has to import your X.509 certificate (public key) and send his X.509 certificate to you.
- You also have to import the partners' X.509 certificate into the key management of rvsEVO
- Associate your own key via Stations Management (detailed view of local station -> parameter group **Keys**).
- Associate the certificate of your partner via the parameter group **Keys** of the partner station)

Hint for TLS encryption: The private key can only be associated if a TLS listener was created

Please read the chapter 11.9 "Automatic Certificate Exchange via rvsEVO-GUI" for information about automatic certificate exchange.

In chapter 8.1.1 "Create a new Jobstart"you get information about transmitting encrypted files.

10.4.2 Using of CA Certificates

This chapter describes the procedure for working with CA (**C**ertificate **A**uthority) certificates. CA Certificates are issued by a TrustCenter like the Odette certification authority.

Generate CSR

- Generate a key pair as described in chapter 11.3 "Generate a Key Pair"
- Generate a certification request (CSR - **C**ertificate **S**igning **R**equest) via a right-click on the key pair in the list view and select `Create CA Signing Request` or using the button **TOOLS**. Define name and directory in which the request (.csr file) should be saved. By default the file is generated in PKCS#10 format.
- Send the request to the TrustCenter (CA). With Odette CA you can make your application online: `https://www.odetteca.com`. Please see `https://www.odette.org/repository/odettesecure_digital_certs_help.pdf` for more information. You will receive your server certificate (your public key signed by CA), a CA root- and an intermediate certificate if all entries of your certificate order are correct.

Import of own certificates

- Import the Root certificate via button **TOOLS,** then select `Import CA Certificate`.
- Import the intermediate CA certificate a via **TOOLS** on `Import Intermediate Certificate` entry.
- Import the server certificate with the `Import CA Reply` entry which is reachable via a right-click on the private key in the list view.

Send the certificates to your partner and import the certificates of your partner as described below:

Import of the partner's certificate

- Import the Root certificate of your partner via a right-click on entry `CA Certificates` in the navigation tree and selecting `Import CA Certificate` from the drop-down menu.
- Import the intermediate CA certificate of your partner via right-click on `Intermediate Certificates` entry and selecting `Import Intermediate Certificate` entry from the drop-down menu.
- Import the server certificate of your partner via right-click on `Partner Certificates` entry in the navigation tree and selecting `Import Partner Certificate` from the drop-down menu.

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Hints:

With TLS encryption either Root or Intermediate certificate must be imported.

With TSL encryption CertPath validation takes place by default. Please see chapter27 "PKI Connection" for more information.

Renew a certificate

Before reaching the expiration date, your server certificate must be renewed. Please follow the description below to renew your certificate:

- Generate a CSR file from your current signed server certificate and send it to your TrustCenter.
- Import the certificates returned by the TrustCenter into your database as described above and send them to your partner stations.
  The old certificates are not overwritten and can continue to be used.
- Do not delete the old certificates until you are sure that all partners are using the new certificates, preferably only after the expiration date has been reached.
  Note that the CA root and intermediate certificates can also be used to verify other certificates.

11 Certificate and Key Management in rvsEVO

This chapter describes how to create, export and import keys, as well as you get information about using CA certificates.

These functionalities are available:

– simplified procedure for creating own key pairs, independend of the functionality.

– functional assignment of the keys / certificates via management of stations

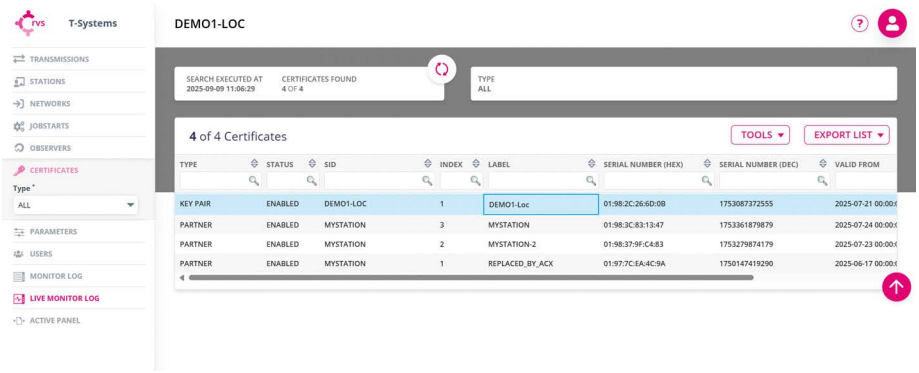
– saving of the keys / certificates in the database

– ACX Request, ACX Deliver and ACX Replace

– Only the administrator can manage the private keys

11.1 User Interface of Key Management

The key management is reachable via sidebar and the entry `CERTIFICATES`. The illustration below shows the default view of the key management:



In the search field **Type** in the sidebar, the following filters are available for the list output:

– ALL

– Keypair

– Partner

– CA Intermediate

– CA ROOT

The default setting is Type = ALL

The following information for certificates is displayed: Type, Status, SID, Index, Label, Serial Number (HEX), Serial Number (DEC), valid from and valid to.

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The **Index** is a unique identifier of the certificate and is assigned by rvsEVO when a certificate is created or imported.

Using the button **TOOLS**, you can start also the following functionalities:

- ACX Request Certificate
- ACX Deliver Certificate
- ACX Replace Certificate
- Import Trusted Service List
- Import Key Pair
- Import Partner Certificate
- Import CA Intermediate Certificate
- Import CA ROOT Certificate
- Generate Key Pair

Hint: The button **TOOLS** is only available to users with the role „Administrator“ or „Operator“.

Detailed view

With a double-click on an entry in the certificate list you will get the detailed view of the certificate.

In the detailed view, the parameters **Label** and **Station ID** can be changed.

Hint: Own key pairs can only be assigned to your local station, virtual stations or internal partner stations (see **File Service Module** parameter of table “Certificate Usage Definition” on page 62)

11.2 Import Trusted Service List

The **Import Trusted Service List** function allows you to import a Trusted Service List (TSL) into the rvsEVO database.

For OFTP file transfers, ODETTE publishes a Trusted Service List at the following address:

`https://www.odette.org/TSL/TSL_OFTP2.XML`

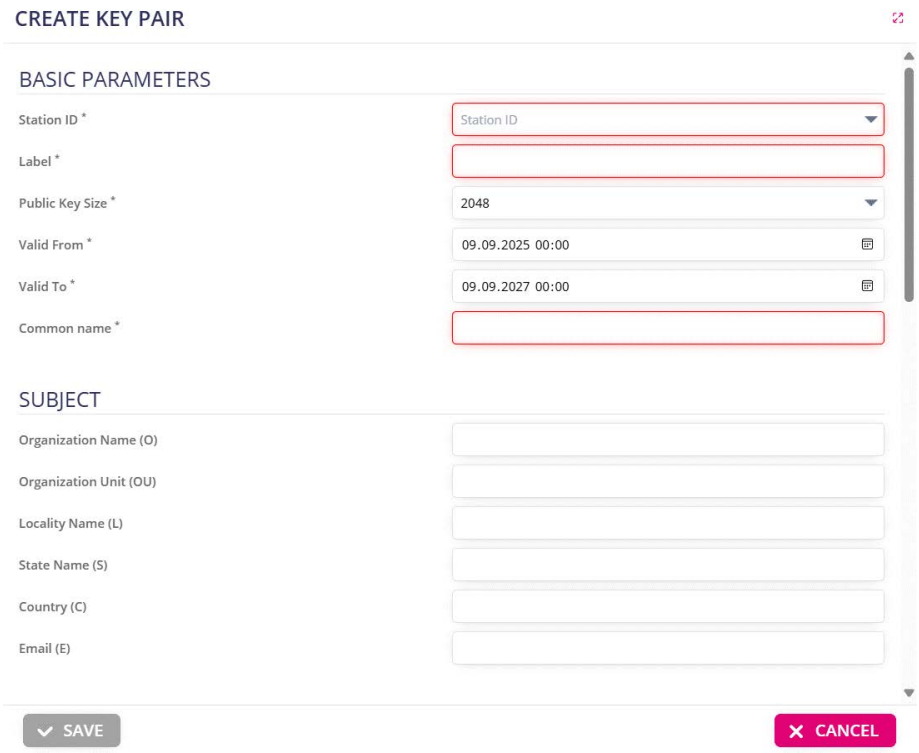
This file must first be downloaded and saved to your file system.

11.3 Generate a Key Pair

Hint: This functionality is only available to users with the role „Administrator“.

Use the button **TOOLS** and the entry **Generate Key Pair.**

The following window opens and you can insert the mandatory parameters.



In the table below you will find a description of the parameters:

Basic parameters

Parameter	Description
Stations ID	ID of the station the key pair is created for.
Label	Identifier for the functional assignment of the certificate.
Public Key Size	Specify the default value (1024 or a multiple of 1024).
Valid from	Start of the keys validity date. Default: current date
Valid to	End of the keys valitity date. Dafault: current date plus tree years
Common Name	The Common Name parameter applies in case of a connection to an existing PKI (**P**ublic **K**ey **I**nfrastructure)

The parameters **SUBJECT** are optional and describe the address and contactdata for your organisation.

The **EXTENDED** parameters are optional and concerns the X.509 certificate:

– User Odette ID
– Hostname
– IP Address

Via **Key Usage Type** in **CRITICAL EXTENSION** parameter the usage of the certificate can be restricted to the activated purposes. With activating **Key Usage Type** check box the **Key Usage** settings are enabled.

The table below gives a short description of **Key Usage Type** parameter.

You can found further information under the following URL: `http://tools.ietf.org/html/rfc5280`.

KEY USAGE TYPE and EXTENDED KEY USAGE TYPE

Parameter	Description
Digital Signature	Certificate is used for verifying digital signatures (other than signatures on certificates or CRLs)
Non Repudiation	Certificate is used for verifying manual signatures (other than signatures on certificates or CRLs)
Key Encipherment	Certificate is used for key management (encipherment of private and secret keys at key exchanging)
Data Encipherment	Certificate is used for enciphering raw user data.
Client Auth	With TLS communication: certificate is used for client authentication through the server.
Key Agreement	Certificate is used for certificate exchange (see also parameters **Decipher Only** and **Encipher Only**).
Key Cert Sign	Certificate is used for verifying certificates.
CRL Sign	Certificate is used for verifying CRLs.
Encipher Only	If **Key Agreement** is activated: Certificate is used for enciphering certificates during certificate exchange.
Decipher Only	If **Key Agreement** is activated: Certificate is used for deciphering certificates during certificate exchange.
TLS Server Autehtication	user of the certificate for server authentication by the client
TLS Client Authentication	user the certificate for client authentication by the server
Code Signing	Certificate is used for signing program codes.
Email Protection	without function
Time Stamping	Certificate is used for extending the hash of an object with timestamp.
OCSP Signing	Certificate is used for signing OCSP responses.

Save the parameters using the button **SAVE**.

11.4 Detailview and Actions for Key Pair

Detailview

With a double-click on an Key pair you will get the detailed view.

In this view the parameters **Label** and **Station ID** can be changed, after clicking the button **EDIT**, the fields will be activated.

Hint: This functionality is only available to users with the role „Administrator“ and „Operator“.

From the detailed view, using the button **ACTIONS** you can start more functionalities:

- Reset validation settings from certificate
- validate certificate
- create CA signing request
- Import CA reply
- Show PEM
- Show X.509 certificate
- Export Certificate
- Export private key

Export of the own Private Key

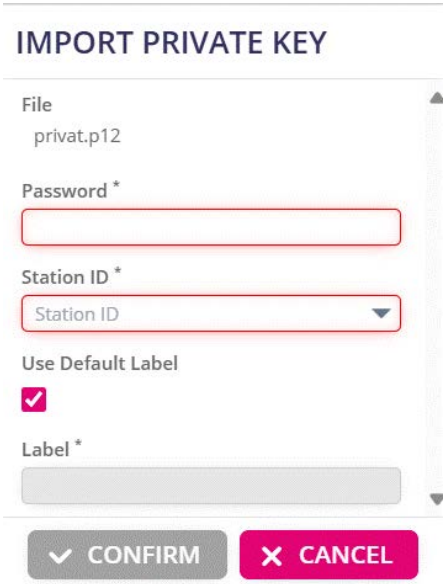
Right-click the key pair in the list view and select `Export private key` from the drop-down menu to export your public key into a file (X.509 certificate). A window opens and you can enter a password. Finish the export via a single-click on the **CONFIRM** button.

11.5 Import and Export of a Private Key Pair

Import

Own RSA key pairs can be imported into rvsEVO database from `.p12` files. The minimum length of the keys must be 1024 bit.

- click the button **TOOLS** and select `Import Key pair` from the drop-down menu for importing a private key. Select the key you want to import and confirm this with the **Open** button.
- the following window will open:



- Select the station ID for which the key should be imported. The private key can be imported for a local station, a virtual station or an internal partner station (if using OFTP File Service Module). Insert in input field **Password** the password of the key management file. By default the check-box **Use Default Label** is activate. For setting another label than the default label deactivate **Use Default Lable** and indicate the label. Press the **CONFIRM** button to import the key.

Hint: The first key pair gets the station ID as label by default and is saved as key pair with default label. The default label can be changed in parameter group `Keys` of the local station. (See chapter 6.2.2 "Parameter Groups of Local Station" for more information).
Further key pairs gets the defaultlabel as label by default.

Export

Right-click the key to be exported in the list view and select `Export Private Key` from the drop-down menu. A dialog box opens and a password is requested. Press the **CONFIRM** button to export the key into the specified directory and file.

11.6 Import and Export of a Partner Certificate

rvsEVO enables to import trusted certificates from partners into the key management and to export already imported certificates from the key management into a file.

Import

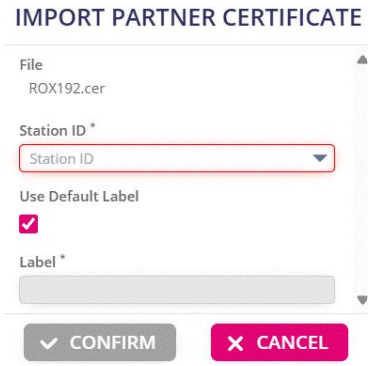
Public keys (in X.509 certificate format) of RSA partner key pairs can be imported from .cer or .pem files into rvsEVO database. The minimum length of the certificate must be 1024 bit.

Click the button **TOOLS** and select `Import Partner Certificate` from the drop-down menu for importing a partner certificate.

Select a certificate and after clicking the button **Open** the following window will be shown, where you can insert the **station ID** for which the

certificate should be imported, Press the **CONFIRM** button for importing the certificate.

By default the check-box **Use Default Label** is activate. For setting another label than the default label deactivate **Use Default Lable** and indicate the label.



Hints:

The first certificate of a partner station is saved as certificate with default label. This setting can be configured in parameter group `Keys` of the partner station. (See chapter 6.3.3 "Parameter Groups of the Partner Station" for more information).

The same certificate can be imported several times for different partners.

Export

Right-click the certificate in the list view and select `Export Certificate` from the drop-down menu for exporting a partner certificate into a file.

11.7 Detail View and Actions for Partner Certificate

Detailview

With a double-click on an partner certificate you will get the detailed view.

In this view the parameters **Label** and **Station ID** can be changed, after clicking the button **EDIT**, the fields will be activated.

Hint: This functionality is only available to users with the role „Administrator“ and „Operator“.

From the detailed view, usinf the button **ACTIONS** you can start more functionalities:

- Reset validation settings from certificate
- validate certificate
- Show PEM
- Show X.509 certificate
- Export Certificate

11.8 Deletion of a Certificate / Private Key Pair

Right-click the certificate or key pair which should be deleted in the list view and select `Delete 1 selected row` from the drop-down menu. Confirm the subsequent security query to remove the key.

11.9 Automatic Certificate Exchange via rvsEVO-GUI

The functionalities `ACX Deliver Certificate`, `ACX Request Certificate` and `ACX Replace Certificate` can be executed via the button **TOOLS** in the function bar.

Instruction for entering the parameter values:

Station IDs can be written into the input feld or selected from the combo box. If you enter the value dircetly into the input field you can use the wild-cards '*' and '?'.

A check box is to be enabled or disabled by single-click in the check box.

Deliver Certificates

Use this feature to deliver certificates to partner stations and import them at the partner stations.

- Open the wizard for ACX Deliver certificate via a single-click on the fbutton **TOOLS** and `ACX Deliver Certificate`.
- Select the station ID of the certificate's originator via `SID Originator`.
- Select the certificates to be delivered via `Select certificate`. Use the button `Select all` and all certificates of the indicated originator will be delivered.
  Hint: Selection of the certificate is done via certificate index. You find the index in the list view and detailed view of a certificate.
- Select the station IDs of the neighbours to whom the certificate is to be sent via `SID destinations`. Select the button `Select all` and all the specified certificates are delivered to all neighbours.
- click the **CONFIRM** button to execute the process.

Request Certificate

Use this feature to request cerificates from the neighbour station and to deliver an own one to this station.

- Open the wizard for `ACX Request certificate`
- Select the station ID of the certificate's originator via `SID Originator`.
- Select the certificate to be sent via `Select certificate`
  Hint: Selection of the certificate is done via certificate index. You find the index in the list view and detailed view of a certificate.
- Select the station ID of the neighbour to whom the request is to be sent via `SID destination`.
- click the **CONFIRM** button to execute the process.

Hint: The Request answer contains all certificates with default label of your partner. The imported certificates gets the default label by default.

Replace Certificate

Use this feature to send a certificate to partner stations and replace the existing certificates with this stations.

Open the wizard for `ACX Replace certificate` via a single-click on the button **TOOLS**.

- Select the station ID of the certificate's originator via `SID Originator`.
- Select the certificate which shall replace the current certificates via `Select certificate`.
  Hint: Selection of the certificate is done via certificate index. You find the index in the list view and detailed view of a certificate.
- Select the station IDs of the neighbour stations whose certificates are to be replaced via `SID destination`. Use the button `Select all` iand all the certificates are replaced at all neighbour stations.
- click the **CONFIRM** button to execute the process.

11.10 Using of CA Certificates

This chapter describes the procedure for working with CA (**C**ertificate **A**uthority) certificates. CA Certificates are issued by a TrustCenter like the Odette certification authority.

Generate CSR

- Generate a key pair as described in chapter 11.3 "Generate a Key Pair"
- Generate a certification request (CSR - **C**ertificate **S**igning **R**equest) via a right-click on the key pair in the list view and select `Create CA Signing Request`. Define name and directory in which the request (`.csr` file) should be saved. By default the file is generated in PKCS#10 format.
- Send the request to the TrustCenter (CA). With Odette CA you can make your application online: `https://www.odetteca.com`. Please see `https://www.odette.org/repository/odettesecure_digital_certs_help.pdf` for more information.

You receive your server certificate (your public key signed by CA), a CA root- and an intermediate certificate if all entries of your certificate order are correct.

Import of own certificates

- Import the Root certificate via **TOOLS** and then select `Import CA Certificate` from the menu.
- Import the intermediate CA certificate a via `Import Intermediate Certificate` entry from the drop-down menu.
- Import the server certificate with the `Import CA Reply` entry which is reachable via a right-click on the private key in the list view. You can

also execute this function via the `Import CA Reply` button in the key's detailed view.

Send the certificates to your partner and import the certificates of your partner as described below:

Import of the partner's certificate

- Import the Root certificate of your partner via **TOOLS** and `Import CA Root Certificate` from the menu.
- Import the intermediate CA certificate of your partner via right-click on `Intermediate Certificates` entry and selecting `Import Intermediate Certificate` entry from the drop-down menu.
- Import the server certificate of your partner via right-click on `Partner Certificates` entry in the navigation tree and selecting `Import Partner Certificate` from the drop-down menu.

Hints:

With TLS encryption either Root or Intermediate certificate must be imported.

With TSL encryption CertPath validation takes place by default. Please see chapter 27 "PKI Connection" for more information.

Renew a certificate

Before reaching the expiration date, your server certificate must be renewed. Please follow the description below to renew your certificate:

- Generate a CSR file from your current signed server certificate and send it to your TrustCenter.
- Import the certificates returned by the TrustCenter into your database as described above and send them to your partner stations.
  The old certificates are not overwritten and can continue to be used.
- Do not delete the old certificates until you are sure that all partners are using the new certificates, preferably only after the expiration date has been reached.
  Note that the CA root and intermediate certificates can also be used to verify other certificates.

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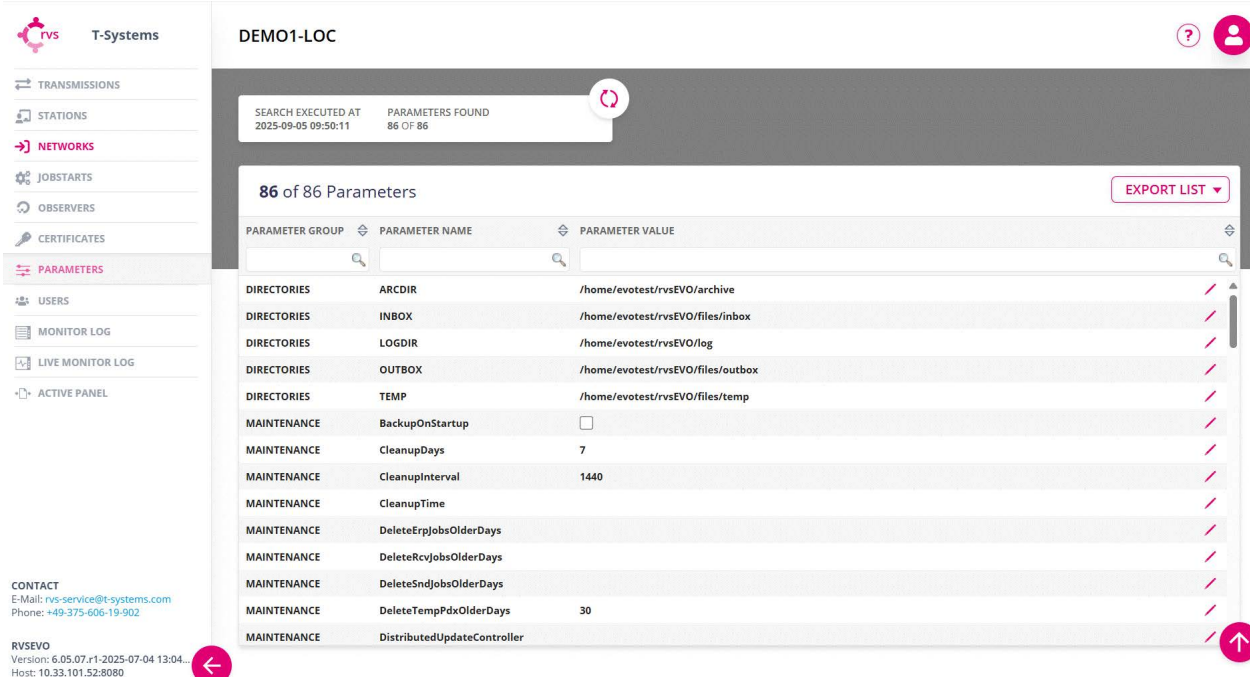
12 Customizing the rvsEVO Parameters

This chapter describes how to customize rvsEVO.

12.1 The User Interface of Parameter

The parameter management is reachable via sidebar `Parameter`.

The illustration below shows the default view of the parameterlist:



The navigation tree on the left side of `Parameter` view contains the following entries:

- DIRECTORIES: paths of directories which includes the transfer and jobdata
- MAINTENANCE: parameters for archiving of completed or failed jobs.
- NOTIFICATION: parameters refering to the feature `SNMP-Agent`
- ODETTE: OFTP Parameters
- RESOURCES_CHECK. Parameters for error handling relating to resource check and expiring certificates.
- SECURITY: security settings

Via a single-click on a parameter group the datailed view of this group is shown on the right side of the window.

The parameters can be set up in this view.

12.2 BASIC Parameters

From version 6.05.07 on you can set the BASIC parameters **RMIService-Host**, **RMIServiceName**, **RMIServicePort**, **RMITLSEnabled**, **RMITLSKey-store** and **RMITLSKeystorePassword** via `rvs-api-rmi.properties` file in `$RVS_HOME\conf` directory.

12.3 DIRECTORIES

In this group you can indicate the directory paths for your data.

Parameter	Description
ARCDIR	Archive directory where archive files and backup files are saved. Default: `$RVS_HOME\archive`. `RevisionLog.xml` files contain the entries of processed send or receive jobs. `RevisionLog.txt` files contain the entries of processed send or receive jobs in `txt` format `.jar files` contain the backup data.
INBOX	Directory where completely received files are stored. Default: `$RVS_HOME\files\inbox`
LOGDIR	Main directory for trace files in `trace` subdirectory.
OUTBOX	Temporary directory for files to be sent. Default: `$RVS_HOME\files\outbox`
TEMP	Directory for temporary use.

12.4 MAINTENANCE

In this group the parameters for security of important data and deletion of data from the database can be defined. See chapter 28 "Backing Up and Recovering rvsEVO Data" for information about Backup and Recovery.

Parameter	Description
BackupOnStartup	Use this parameter to specify an automatic backup, performed on each time you start rvsEVO. Possible values: - `Y`: each time when rvsEVO will be started the directories `$RVS_HOME/conf`, `$RVS_HOME/system` and `$RVS_HOME/jobs` are saved in a .jar file in `$RVS_HOME/archive` directory. Also database data are saved. The name is a composition of „EVOB-“,date and time in format `EVOB-YYYYMMDD-hhmmss.jar` - `N (default)`: no automatic backup
CleanupDays	Specify days for cleanup of completed or failed jobs. The `archiveJobs` program will delete any jobs older than the time specified in this parameter. Default: `7` days Note: You can set only one of cleanup parameters: **cleanupdays** or **cleanuptime**
CleanupInterval	Time interval in minutes between two cleanup cycles. This parameter has no relation to the other two cleanup parameters: **CleanupDays** or **CleanupTime** Default: `1440`
CleanupTime	Specify time in the format HH:mm:ss (24 hour format) for cleanup of completed and failed jobs. Any jobs older than the time specified will be deleted. Note: You can set only one of cleanup parameters: **cleanupdays** or **cleanuptime**.
DeleteErpJobsOlderDays	Interval in days for deletion of active ERP jobs which are older than indicated in this parameter. The jobs will be set to category ENDED and state DELETED. The jobs will be checked once an hour. Default: No automatic deletion of jobs.
DeleteRcvJobsOlderDays	Interval in days for deletion of active receive jobs which are older than indicated in this parameter. The jobs will be set to category ENDED and state DELETED. The jobs will be checked once an hour. Default: No automatic deletion of jobs.

Parameter	Description
DeleteSndJobsOlder-Days	Interval in days for deletion of active send jobs which are older than indicated in this parameter. The jobs will be set to category ENDED and state DELETED. The jobs will be checked once an hour. Default: No automatic deletion of jobs.
DeleteTempPdxOlder-Days	Interval in days Files in the directory **$RVS_HOME/files/pdx/temp** older than indicated in this parameter, will be automatically delete
DistributedUpdateController	SID of controller station which is allowed to execute Distributed Update
MailLocalAddress	Mail address of the sender of warnings. See also **Mail** parameter in chapter 12.7
MailSMTPHost	Hostname or IP address of the mail server if warnings are sent. See also **Mail** parameter in chapter 12.7.
MaxRevisionLogCount	Number of RevisionLog (.xml and .txt) files that can be generated. Default: `1000`
MaxRevisionLogSize	Maximum file size of the RevisionLog (.xml and .txt) file in lines (1 line has maximal 10 kbytes). Default: `1000000`
MonlogInDatabase	In this parameter you define whether the log messages should be written into the database. **Important hint:** The database table is not cleaned up by rvsEVO. (I.e. the user has to clean up the table because otherwise it would fill up.) Possible values: - `Y`: the log messages are displayed - `N`: the log messages are not displayed (default)
MonlogInDatabase-Cleanup	Only relevant if **MonlogInDatabase**=`Y`. In this parameter you define whether the log messages should be deleted automatically from database or not. Possible values: `NONE` (Default): log messages are not deleted automatically. `AUTO`: All log messages older than 7 days will be deleted every 24 hours, by default. This configuration can be set up via the parameters **CleanupDays**, **CleanupInterval** and **CleanupTime**.

Parameter	Description
ObserverThreadPool-Size	Size of Observer thread pool (default: 40). Please see also chapter 9.1.2 "Create a new Observer".
PermissionTempScript	File permission of the temporary script file which is copied into temp directory after a jobstart process was started. Only if Jobstart parameter **Parameterübergabe**=`REPLACE`. `770` = read/write/execute for owner and group `777` = read/write/execute for all users
PersistenceArchive	Name of the xml file with the transfer data (statistic file). Default: `$RVS_HOME/archive/RevisionLog.xml`.
PersistenceArchiveText	Name of the text file with the transfer data (statistic file). Default: `$RVS_HOME/archive/RevisionLog.txt`.
RedoLog	no function

12.5 NOTIFICATION

The parameters in this group refer to the feature rvsSNMP-Agent.

rvsSNMP-Agent is an application that can respond to Network management system (NMS) queries and send rvsEVO status information and monitor messages to NMS.

Since version 6.04 rvsSNMP-Agent is provided by rvsEVO in `$RVS_HOME/snmp-agent` directory.

For more information about this feature, please read the rvsSNMP-Agent User Manual which can be find in `RVS_HOME/snmp-agent/doc` directory.

The following parameters have to be edited in rvsEVO to be able to work with rvsSNMP-Agent:

Parameter	Description
AgentActive	This parameter defines whether or not rvsSNMP-Agent is enabled. Possible values: - `Y` (Yes) - `N` (No/default)

Parameter	Description
AgentHeartbeat-Intervall	This parameter defines the interval (in milliseconds) at which rvsEVO sends a Heartbeat message to the Agent UDP address (see parameter **AgentHostname** and **AgentPort**). Default: `30000`
AgentHostName	Agent computer name (or IP address). Default: `localhost`.
AgentLogLevel	This parameter defines which kind of log messages rvsEVO sends to the Agent. Possible values: - `0`: no log messages are sent - `1`: error messages are sent - `2`: warnings and error messages are sent - `3`: information, warnings and error messsages are sent.
AgentPort	Agent IP port. Default: `3744`.

12.6 ODETTE Parameters

In this group you can set the parameters for the Odette transfer.

Parameter	Description
CMSTracingStationIds	Allows to write CMS-TRACE files for indicated station ID to `$RVS_HOME\files\temp` directory. Several SIDs have to be seperated by comma.
ConnSetupFailWaitTime	Time in milliseconds rvsEVO waits after a connection failed to be established before rvsEVO tries to establish the connection again. Default: `10000`
ConnSetupProxyFail Count	Number of failed connection attempts with rvs® OFTP Proxy after which the script, defined in **ConnSetupProxyFailScript** parameter should be started.
ConnSetupProxyFail Script	Job to be started after the number of failed connections, which is defined in **ConnSetupProxy FailCount** parameter is reached.

Parameter	Description
LooptestNeighbourSID	ID of the station via which the loop test (transmission of a file to the own local or virtual station) is performed. The file is to be transmitted to the own station, and rvsEVO sends this file via a neighbour station back to the own station. If no StationID is specified the file is delivered directly to `inbox` of local station and not routed via a neighbour station.
MaxSessions	Maximum number of simultaneously running OFTP sessions. Default: `20`; Maximum number is restricted by system resources and the licensekey.
OFTPTimeout	Time out in milliseconds at ODETTE level; default: `30000`, no maximum
SessionAliveTimeout	Time in milliseconds to consider a connection active; Default: `600000`, no maximum.
SessionWaitTime	**SessionWaitTime** is the time in milliseconds the OFTP session between the sender and the receiver station will be kept alive after an ended transmission for waiting for another send job. Default: `0`
Timestamp	Defines whether or not a time stamp is added to a file name name when it is received. Possible values: - `N` (default): Time stamp is added by default, only if the file name already exists; - `Y`: File name always receives the time stamp.
TraceItem	Parameter that enables tracing. The following values are possible: - `O`: for Odette level, - `N`: not active (default) See chapter 17.1 "Trace Files" for further information.
TraceNetworkData	only if **TraceItem** = `O` Parameter that enables tracing of network data. Possible values: - `N`: no tracing of network data (default) - `Y`: tracing of network data
TraceStationIds	only if **TraceItem** = `O` Limit Odette tracing for given station IDs. Seperate the station IDs by comma.

Parameter	Description
TransmissionFailWait-Time	Time in milliseconds for a transmission restart after a failure. Minimum: `100` Default: `30000`

12.7 RESOURCES_CHECK

This parameter group refers to resource check of rvsEVO directories and checking the expiration dates of certificates. This functionality allows the operator to react early enough to lack of free disk space and expiring certificates.

Resource check

All active directories of rvsEVO will be checked: `DB, TEMP, INPUT, OUTPUT, ARCDIR and LOGDIR`.

In case of the low disk space you will see in the Monitor-Log a message, that informs you in which rvsEVO directories has not enough disk space. At the same time an E-mail to the responsible administrator will be sent.

There are three levels of resource lack:

– first level: a warning will be dispayed.

– second level: the file reception will be blocked (and activated if the needed disk space is available)

– third level: all processes and rvsEVO itself will be ended.

Certificate check

Checking of expiration dates of your own keypairs and the certificates of your partners. If a certificate expires in preset time a message will be issued in the Monitor-Log.

Parameter	Description
CertificateExpirationError	If certificate expiration is reached in less days than defined in this parameter an error message will be issued in Monitor-Log and, if indicated, will be sent (see also parameter **CertificateExpirationMail**) Default: `7`
CertificateExpirationMail	With this parameter you define whether an email is sent in case of certificate expiration. Possible values: `Yes` / `Y`: email is to be sent `No` / `N`: email is not to be sent

Parameter	Description
CertificateExpiration-Warning	If certificate expiration is reached in less days than defined in this parameter a warning will be issued in Monitor-Log and, if indicated an email will be sent (see also parameter Mail). Default: `30`
CertificateExpira-tionCleanupDays	This parameter deletes certificates that have expired for a longer period than specified here. The cleanup occurs at the same time as the check for expired certificates, i.e., at the interval specified by the CertificateExpirationCheckTime parameter, and once per server startup. Only partner, CA intermediate, and CA root certificates are deleted. Key pairs are ignored. Value: 0-999 Field empty = no certificates delete
CertificateExpira-tionCheckTime	Time in hours between two certificate checks (see parameters CertificateExpirationWarning, CertificateExpirationError and CertificateExpira-tionCleanup). The first check of the day will always be performed at 0:00. Default: 1
CriticalDiskSpace	Number of kilobytes, that has to be free in every checking rvsEVO directory. With this parameter the critical limit for the resource deficit should be configured. If the number of free kilobytes is less than a value of this parameter all rvsEVO processes and the rvsEVO server will be stopped. You have a possibility to start a script (see parameter `System` in this table) Default: `100000`
DiskSpace	Number of kilobytes, that has to be free in every checking rvsEVO directory, before a warning in Monitor-Log will be issued. Default: `150000` kB.
DiskSpaceCheckEn-abled	With this parameter you define whether the disk space is to be checked or not. Possible values: `Y`: disk space check should be done (default). `N`: none disk space check

Parameter	Description
Mail	Email address, to which the warning is to be sent at the server startup if 1. the number of free disk space is less than defined in **DiskSpace** parameter. The warning will be sent cyclically as defined in **SuspendTime** parameter. 2. the values defined in parameters **CertificateExpirationWarning** and **CertificateExoirationError** are underrun. Subsequently the email is sent once a day (after first check of the day). See also parameters **MailLocalAddress** and **MailSMTPHost** “” on page 111 (only if the parameter **CertificateExpirationMail = Y**)
RcvStopDiskSpace	Number of kilobytes, that has to be free in every checking rvsEVO directory, before a warning in Monitor-Log will be issued. If the number of free kilobytes is less than the given value, an incoming SFID (**S**tart **F**ile **ID**) is replied with SFNA (**S**tart **F**ile **N**egative **A**nswer) and the file reception is blocked. Default: `120000` kB. **Hint:** The receiption is automatocally re-activated after the disk space problem is eliminated.
SuspendTime	Time between the resourse checks after the value given in **RcvStopDiskSpace** parameter was underrun. Default: `600`
System	a script, which should be executed, if the critical limit (level) is reached. See **CriticalDiskSpace** parameter in this table.
Time	Time in seconds between two resource checks. Default: `600`.

Note:

You do not need to restart the server to save the changes to this parameter group.

12.8 Security Parameters

In this group you can configure proxy settings for the monitoring of certifications (validation via CRL and OCSP). Please see chapter 27 "PKI Connection" for more information about certificate validation.

Parameter	Description
CRLAutoDownload	Only available if parameter **Certificate Validation Type**=`CRL` (see table “Parameters for PKI Binding” on page 247). Possible values: - `Y` = Yes - `N` = No (default) If **CRLAutoDownload**=`Y` an automatic download of CRL and import into the database takes place (first import into the database).
CRLAutoUpdate	Only available if parameter **Certificate Validation Type**=`CRL` (see table “Parameters for PKI Binding” on page 247). Possible values: - `Y` = Yes - `N` = No (default) If **CRLAutoUpdate**=`Y` an automatic update of CRL takes place in regular time intervals. (See also parameter **CRLAutoUpdateInterval**)
CRLAutoUpdateInterval	Time interval for automatic CRL update. A description of the format you can find under `http://quartz-scheduler.org`. The whole URL you can find below the table. Exmples: „0 0 0 * * *“: The update is executed daily at 12:00 am (default). „0 15 10 ? * *“: The update is executed daily at 10:15 am. „0 0 0,6,12,18 * * ?“: The update is executed daily at 12:00 am, 6:00 am, 12:00 pm and 6:00 pm. (See also parameter **CRLAutoUpdate**)
CRLHttpProxyHost	http-Proxy server to be used for the connection to the external CRL URIs
CRLHttpProxyPassword	Password for http-Proxy in case of user/ password access (for CRL download)
CRLHttpProxyPort	Port for the connection to the Http-Proxy-Server (for CRL download)
CRLHttpProxyUser	User for http-Proxy in case of user/password access (for CRL download)
CRLLocalAddress	Sender address of the CRL download request.

Parameter	Description
CRLOftpProxyInstance	Selection of a Bastion instance, configured in rvsEVO (for CRL-Download). **Hint:** This parameter will only be regarded if **CRLUseOftpProxy**=Y
CRLUseHttpProxy	If **CRLUseHttpProxy**=**Y** a Http-Proxy server is to be used for the CRL Download. In this case please, also consider parameters **CRLHttpProxyHost**, **CRLHttpProxyPassword**, **CRLHttpProxyPort**, **CRLHttpProxyUser** and **CRLLocalAddress**.
CRLUseOftpProxy	If **CRLUseOftpProxy=Y** an OFTP-Proxy server is to be used for the CRL Download. **Hint:** If **CRLUseOftpProxy** is activated also **CRLOftpProxyInstance** is to be set.
OCSPHttpProxyHost	http-Proxy server to be used for the OCSP check
OCSPHtt-pProxyPassword	Password for http-Proxy in case of user/ password access (for OCSP check)
OCSPHttpProxyPort	Port for the connection to the Http-Proxy-Server (for OCSP check)
OCSPHttpProxyUser	User for http-Proxy in case of user/password access (for OCSP check)
OSCPLocalAddress	Sender address of the OCSP Requests
OSCPOftpProxyInstance	Selection of a Bastion instance, configured in rvsEVO (for OCSP check).
OSCPUseHttpProxy	If **OCSPUseHttpProxy**=Y a http-Proxy server is to be used for the OCSP check.
OSCPUseOftpProxy	If **OCSPUseOftpProxy**=Y an Oftp-Proxy server is to be used for the OCSP check.

Parameter	Description
TSLAutoUpdate	If **TSLAutoUpdate** = `Y` **TSLAutoUpdateNext** parameter is checked like indikated in **TSLAutoUpdateInterval** parameter: - If **TSLAutoUpdateCheckURI** parameter is indicated: Value of **TSLAutoUpdateLast** is checked against the time stamp of the latest TSL (see parameter **TSLAutoCheckURI**) . If a new TSL is uploaded to the providers server rvsEVO downloads the TSL from the address specified in **TSLAutoUpdateURI** parameter. - If **TSLAutoUpdateCheckURI** parameter is not indicated: rvsEVO downloads the TSL from the URI indicated in **TSLAutoUpdateURI**, if the time of parameter **TSLAutoUpdateNext** is reached. After download, the certificates of TSL are imported into the rvsEVO database and the parameters **TSLAutoUpdateLast** and **TSLAutoUpdateNext** are updated. Possible values: `Y` = Yes `N` = No (default)
TSLAutoUpdate-CheckURI	URI of the providers web side with the time stamp of the latest TSL. Default: `http://www.odette.org/TSL/TSL_OFTP2.UPD.`
TSLAutoUpdateInterval	Time interval for checking parameter **TSLAutoUpdateNext**. A description of the format you can find under `http://quartz-scheduler.org`. The whole URL you can find below the table. Exmples: „0 0 0 * * *“: The update is executed daily at 12:00 am (default). „0 15 10 ? * *“: The update is executed daily at 10:15 am. „0 0 0,6,12,18 * * ?“: The update is executed daily at 12:00 am, 6:00 am, 12:00 pm and 6:00 pm. (See also parameter **TSLAutoUpdate**)
TSLAutoUpdateLast	Time of last automatic TSL update. Set by rvsEVO.
TSLAutoUpdateNext	Time of next automatic TSL update. Set by rvsEVO.

Parameter	Description
TSLAutoUpdateURI	URI for downloading the TSL; default: `http://www.odette.org/TSL/TSL_OFTP2.xml`. **Hint:** With HTTP connection and using http-Proxy server the parameters **CRLHttpProxyHost**, **CRLHttpProxyPort**, **CRLHttpProxyUser** and **CRLHttpProxyPassword** should be setted.
TlsCertificateValidationBehavior	Possible values: - `NORMAL` (default): Only the Root or CA certificate has to be imported for TLS encryption (identical to rvsEVO version 5.xx). In this case, with CRL validation the validation URI must be entered in intermediate or root certificate. **Hint:** Value `NORMAL` replaces `LEGACY` - `PARTNER_CERTIFICATE_NEEDED`: In addition to the root and server certificate the user certificate has to be imported. (See chapter 10.4.2 "Using of CA Certificates" for more information)
TlsCertificateValidationType	Only effective if parameter **TlsCertificateValidationBehavior**=`NORMAL` With this parameter the validation type can be indicated. Possible values: - `CERT_PATH` – `OCSP` – `CRL` (See chapter 27.3 "Validation Methods" for further information)
TlsPerfectForwardSecrecy	Algorithms, used. Possible values: - `FORCED`: Cipher Suites List contains DHE algorithms - `CLIENT_PREFERED`: Cipher Suites List contains all algorithms, but DHE algorithmus is preferably used. - `JVM_DEFAULT`: Cipher Suites Liste remains unchanged. After changing a restart is necessary.
TlsPerfectForwardSecrecyFilter	Regular expression; Default: `((.*)_DHE_(.*))|((.*)_ECDHE_(.*))` After changing a restart is necessary.

Parameter	Description
TlsVersions	Possible values of TLS protocol: - `TLSv1` - `TLSv1.1` - `TLSv1.2` You can specify more than one version seperated by comma and spaces. Default: `TLSv1.2, TLSv1.1, TLSv1` After changing a restart is necessary.

URL for format description of **CRLAutoUpdateInterval** parameter:

`http://quartz-scheduler.org/documentation/quartz-2.1.x/tutorials/crontrigger`

12.9 Customizing via rvs-system.properties File

Please see the rvsEVO Operator Manual chapter „Customizing rvsEVO“ for information about configuration options via `rvs-system.properties` file.

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13 User Management

Through the sidebar, you can navigate to the user management.By clicking on the USERS entry, the user list will be displayed.

Hint:

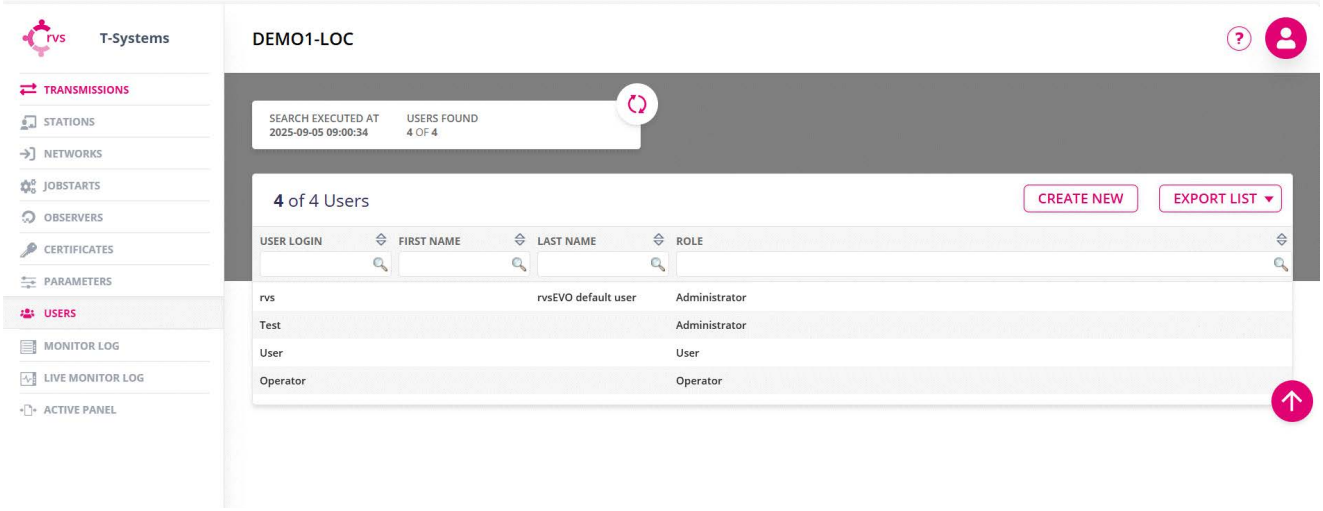
The User Management view is only available to users with the role „Administrator“.

For the public user, the user management is not available.

13.1 User Interface for Administrators

Select the item **USERS** in the sidebar to open the User management.

The picture below shows the default view:



On this page, user data can be created, edited or deleted.

The data of the default user cannot be edited or deleted.

13.2 Display Users

A double-click on an user in the list opens the detail view.

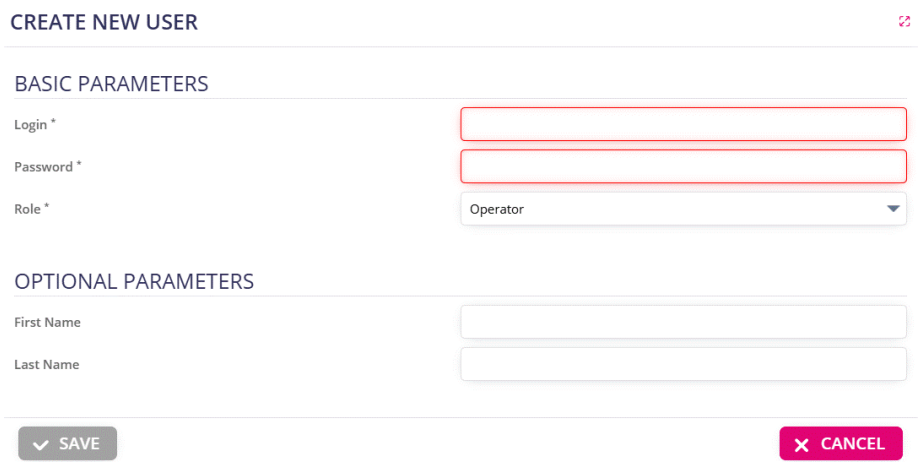
A right-click on a user opens a drop-down menu which offers the following additional options `Delete User`, `Change password`, `Duplicate User`.

Hint:

These functionalities are available for the defaultuser and the users with the role „Administrator“. A regular user or operator cannot use these functionalities.

13.3 Create New User

To create a new user, click the button CREATE NEW:



The following parameters are mandatory:

– `Login` (user account consisting of 7-bit ASCII characters except colon)

– `Password` (for login; consisting of 7-bit ASCII characters)

– `Role` (Administrator, Operator or User)

Optional parameters are the `First Name` and `Last Name` of the user.

After setting up the parameters, click the **SAVE** button to add the user.

You can add an user with identical settings to an existing user. Right-click the user that is to be duplicated in the list view and select `Duplicate User` entry from the list.

The settings of the duplicated user are used as default values of the new user. The value of **Login** parameter must be changed. The remaining parameters can also be edited if neccesary.

Hint:

A basic **User** may use the rvsEVO program to send and receive files.

An **Operator** may execute operator commands.

An **Administrator** has all the rights of the user and the operator and may also manage private keys and users and also configure the rvsEVO system.

13.4 Edit User Data

Open the detailed view via double-click on the user in list view to edit the user data.

All parameters can be edited. After changes, press the **SAVE** button to confirm the changes.

Change Password

The user password can be changed via **Change Password**.

Additional the default user cannot change his own password via rvsWeb-GUI.

Furthermore a password can be changed via `rvscli` script with `updateUser` command. Please see chapter 16.5 "rvscli" for further information about `rvscli` script.

Hint: For default user the function „change password“ is only available.

13.5 Delete a User

Open the list view of the user and select the user you want to delete. After a right-click select the entry `Delete User`.

You can delete several users at once by highlighting the users to be deleted (holding down the control key).

Hints:

The function `Delete User` is not available for the default user.

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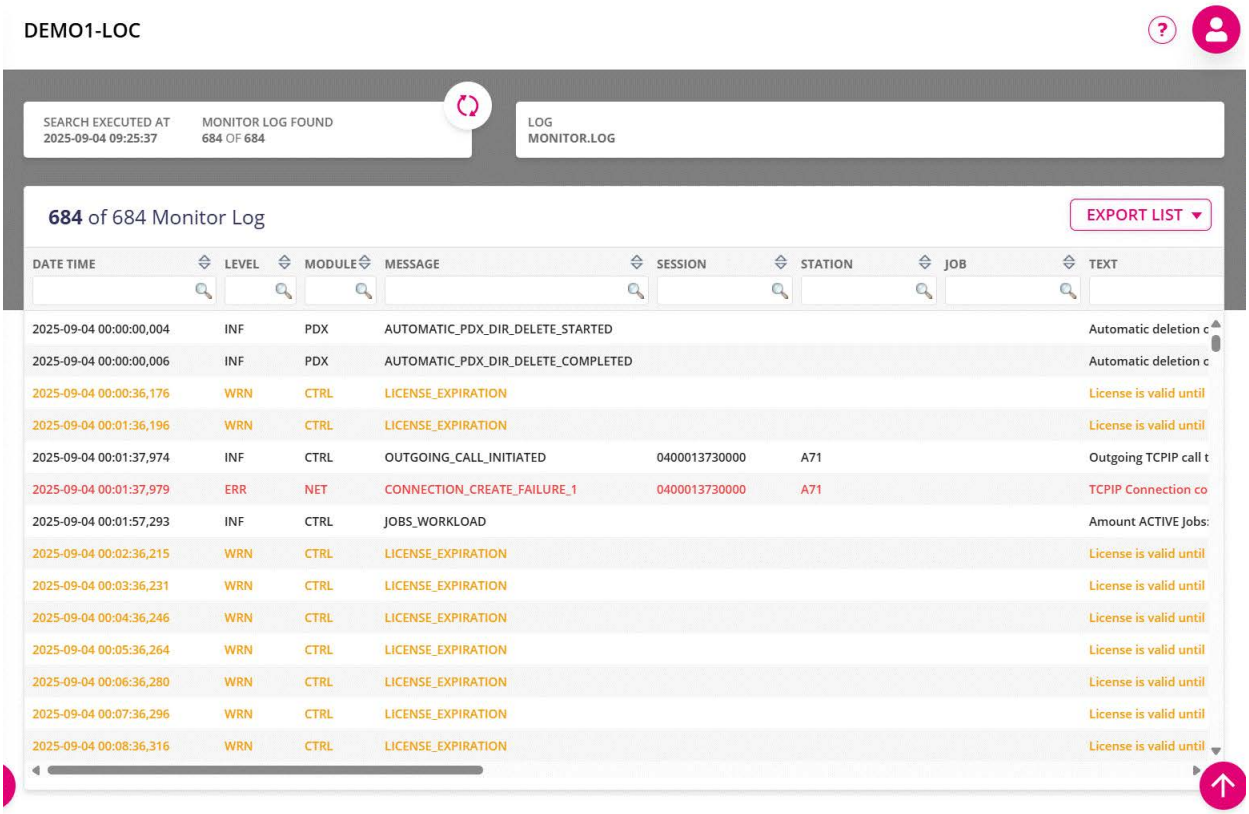
14 Monitoring

This chapter provides an overview of the available output options for messages and warnings in rvsEVO.

14.1 User Interface of Monitor-Log

Current and previous messages can be viewed via the sidebar under **MONITOR-LOG**.

The following picture illustrates the default view of `Monitoring` perspective:



The log files containing the current and previous messages can be selected via the **Log*** search field.

In the list view, you have the option to filter the output.

14.2 Live Monitor-Log

Via a single-click on the **LIVE MONITOR LOG** entry in the sidebar the current messages will be displayed.

Language Setting

The language setting of the Live Monitor-Log can be aligned via `user.language` parameter in `$RVS_HOME/conf/rvs.properties` file. Changes only take effect after a restart of rvsEVO server.

Configuration

The latest 1000 messages are displayed by default. This setting can be configured via the definition `<entry key="client.monitorlog.livelog.pagesize" value="1000"/>` in `$RVS_HOME/conf/rvsEvoClient.prefs` file.

Anymore it is possible to highlight the display of information (`INF`), warnings (`WRN`) and error messages (`ERR`). This is configurable via the definitions `<entry key="client.monitorlog.[INF|WRN|ERR].colour.[r|g|b]" value="[255|0]"/>` in `$RVS_HOME/conf/rvsEvoClient.prefs` file.

Examples:

```
<entry key="client.monitorlog.ERR.colour.r"
value="255"/>
```

In the example above the error messages are displayed red (`colour.r`).

```
<entry key="client.monitorlog.WRN.colour.g"
value="255"/>
```

In the example above warnings are displayed in green (`colour.g`).

Output Parameters of Log Messages:

Parameter	Description
Date-Time	Date and time of the entry (format `yyyyMMdd-HHmmss`)
Level	Kind of message. Possible values: - `INF`: Information - `WRN`: Warning / importent hint - `ERR`: ERROR
Module	Module of rvsEVO, which transmitted the message. Possible values: - `CTRL`: controller - `NET`: network - `OFTP`: Odette protocol - `ACX`: automatic exchange of certificates - `SP`: Service Provider - `OBSR`: Observer - `PROXY`: rvs® OFTP Proxy
Message	Message key
Session	ID of OFTP session
Station	SID of neighbour station
Job	rvsEVO jobID
Text	Text message from message key (GUI) or parameter for text message (log file)

14.3 Deletion of Monitor Log Entries from Database

If adjustment of MAINTENANCE parameter **MonlogInDatabase** = `Y` the database table must be cleaned up regularly because otherwise it would fill up.

You can automate clean-up via **MonlogInDatabaseCleanup** parameter (please see chapter 12.4 "MAINTENANCE") or execute clean-up via command line tool `cleanup` (please see chapter 16.7 "cleanup: Deletion of Jobs and Monlog Data from the Database" for more information).

14.4 Active Panel

This tool gives you the additional details of the status of an active transfer.

A mouse-click on the `Active Panel` entry opens a list of current transfers.

The following information is available:

Active Panel parameters

Parameter	Description
Neighbour	Station ID of the neighbour station
State	`Initiator` (own station is an active side of the communication process); `Responder` (own station is the passive side of the communication process)
Command	`Create Session`: creation of session `Start File`: beginning of the transfer `Transmit File`: transfer of the file `End File`: end of the transfer
Originator	SID of originator
Destination	SID of destination
Direction	`Sending` `Receiving`
File Name	virtual file name
Line Speed	transfer rate in kB/sec
Progress	percentage closed content of the transfer
Start File	Time of the beginning of the transfer.
SessionID	Is automatically signed for each transfer
Start Session	Time of the session start
Line Type	Network type

Hints:

You can get a sorted output by clicking on the parameter (column headings) you want to sort.

Also a filter function is provided. Use the symbol for refreshing at the upper right-hand border for refreshing the list.

15 REST-API

From version 6.04.00 rvsEVO provides the possibility to create a send job via command line with http command `curl`.

15.1 Setting up of REST-API

As well as rvsWEB REST-API can be activated and set up via `$RVS_HOME/conf/rvs-rest.properties` file. Please read in the rvsEVO Operator Manual, chapter „Customizing rvsEVO“ / „Customizing rvsWeb and REST-API“ for further information.

15.2 Create a Send Job via REST-API

A send job can be created with existing authorization token which you get with login to rvsEVO-Server or by login to rvsEVO-Server with each send job.

15.2.1 Create a Send Job with Existing Token

At first you need to login to rvsEVO-Server. Thereby you send a request to the rvsEVO server and receive an authorization token with the response. This token is required for creating the send job. Login is also performed via `curl` command.

Format:

```
curl -k -v -u <rvs-user>:<rvs-password> -X POST
'https://<host>:<port>/rvs/sessions'
```

Example (Login including response):

```
[evotest@q4de3esy190 ~]$ curl -k -v -u rvs:rvs -X POST
'https://10.33.101.135:8081/rvs/sessions'
* About to connect() to 10.33.101.135 port 8081 (#0)
*   Trying 10.33.101.135...
* Connected to 10.33.101.135 (10.33.101.135) port 8081
(#0)
* Initializing NSS with certpath: sql:/etc/pki/nssdb
* skipping SSL peer certificate verification
* SSL connection using TLS_ECDHE_RSA_WITH_AES_256_GC-
M_SHA384
* Server certificate:
*       subject: CN=rvsEVO HTTPS
*       start date: Nov 08 15:16:35 2021 GMT
*       expire date: Nov 08 15:16:35 2023 GMT
*       common name: rvsEVO HTTPS
*       issuer: CN=rvsEVO HTTPS
```

```
* Server auth using Basic with user 'rvs'
> POST /rvs/sessions HTTP/1.1
> Authorization: Basic cnZzOnJ2cw==
> User-Agent: curl/7.29.0
> Host: 10.33.101.135:8081
> Accept: */*
>
< HTTP/1.1 200 OK
< Authorization: Bearer
```

Authorization token

```
eyJhbGciOiJIUzUx-
MiJ9.eyJzdWIiOiJydnMiLCJleHAiOjE2NjM2Mjg0NzksIlJvbGUi
OiJBZG1pbmlzdHJhdG9yIn0.TcxLA_UIAKCuz3yit_nXFSel-
no9AobF-F2VA0LbtY2k-mL79zA_-wbhe_Ndx0pN6bZS3M8-8rrR-
wZ8A9Q4_Kog
```

```
< Content-Length: 0
<
* Connection #0 to host 10.33.101.135 left intact
```

```
[evotest@q4de3esy190 ~]$ curl -k -u rvs:rvs -X POST
'https://10.33.101.135:8081/rvs/sessions'
```

Hint:

You find the authorization token which is required for creating a send job in the login response behind „Authorization: Bearer“.

Each job parameter and each parameter value is indicated with quotation marks (“), the value is seperated with colon from parameter name. More than one parameters are seperated by comma.

The response of rvsEVO Server contains all job data.

The createSendjob commad is composed as follows:

```
curl -v -i -k -X POST -H 'Authorization: <Token>' -H
"Content-Type: application/json" -d '{<Jobparameter>}'
'<URL>'
```

Example (Create send job including response):

```
[evotest@q4de3esy190 ~]$ curl -i -k -X POST
-H 'Authorization: eyJhbGciOiJIUzUx-
MiJ9.eyJzdWIiOiJydnMiLCJleHAiOjE2NjM2Mjg0NzksIlJvbGUi
OiJBZG1pbmlzdHJhdG9yIn0.TcxLA_UIAKCuz3yit_nXFSel-
no9AobF-F2VA0LbtY2k-mL79zA_-wbhe_Ndx0pN6bZS3M8-8rrR-
wZ8A9Q4_Kog' -H "Content-Type: application/json" -d
'{"vdsn":"KDETESTFILE",
"destination":"173","rvsType":"EVO","filename":
```

```
"test21.txt","filePathType":"OUTBOX"}'
'https://10.33.101.135:8081/rvs/api/job'

HTTP/1.1 200 OK

Content-Type: application/json

Content-Length: 939
```

Job data:

```
{"rvsType":"EVO","id":"2209191509550000",
"direction":"SND","priority":3,
"transmissionStartTime":"",
"transmissionEndTime":"","transmittedBytes":0,
"retryCount":0,"serialization":false,"label":"",
"category":"ACTIVE","status":"QUEUED","lastChange":
"2022-09-19 15:09:55","filename":"/home/evotest/kde/
rvsEVO/files/outbox/test21.txt.2209191509550000",
"fileSize":80,"fileDescription":"","recordFormat":
"U","recordLength":0,"recordMode":"",
"conversionTable":"","encryption":false,
"encryptionAlgorithm":"DES_EDE3_CBC",
"fileSignature":false,"erpSignature":false,
"compression":false,"errorNumber":"",
"errorDescription":"","nerpReasonCode":"",
"nerpReasonText":"","nerpCreatorOdetteId":"",
"rvsType":null,"objectType":"job.send","vdsn":
"KDETESTFILE","originator":"LOC","destination":
"173","sfidDateTimeCounter":
"20220919  150955 0001","sfidDescription":null,
"sfidFileSize":0,"sfidOriginalFileSize":null,
"sfidFileFormat":null,"sfidRecordLength":null}
[evotest@q4de3esy190
```

15.2.2 Create a Send Job with Login to rvsEVO Server

If you prefer to login to rvsEVO Server with each createSendjob command indication of rvsEVO user and whose password is required instead of authorization token. Apart from that the command is composed as described above.

```
curl -u <rvs-user>:<rvs-password> -i -k -X POST -H
"Content-Type: application/json" -d '{<Jobparameter>}'
'<URL>'
```

Send job parameters

The tables below describe the send job parameters. You can find a more detailed description in chapter 16.3.3 "createSendJob".

Required Send Parameters:

Parameters	Desctiption
destination	Station ID of destination.

Required Send Parameters:

Parameters	Desctiption
filename	Name of the file to be send.
rvsType	Value for rvsEVO: `EVO`
vdsn	Virtual file name, max. 26 characters.

Optional Send Parameters:

Parameters	Description
originator	Station ID of sender
cipherSuite	Encoding algorithm (Please see „Cipher Suite table“ in chapter 14.4.3)
compression	Compression Possible values: - `true, false`
conversionTable	Name of conversion table.
disposition	Defines whether the file is removed or not. Possible values: - `D`: File is removed - `K`: File is not removed
encryption	File encryption Possible values: - `true, false`
encryptionAlgorithm	Encryption algorithm (deprecated; please see parameter **cipherSuite**)
erpSignature	Requests singed ERP. Possible values: - `true, false`
externalJobID	external job ID
fileDescription	Comment
filePathType	Defines whether the file to be send is saved in `$RVS_HOME/files/outbox` folder or not. Possible values: - `ABSOLUTE` (absolute path is to be indicated) - `OUTBOX` (file is saved in outbox folder)
fileSignature	File is signed before sending Possible values: - `true, false`

Optional Send Parameters:

Parameters	Description
hold	Send entry is created in `HOLD` mode. Possible values: - `true, false`
label	Group name of serialized send jobs
priority	Priority for send jobs. Possible values: `1` - `5`
recordFormat	File format. Possible values: - `T, F, V, U`
recordLength	Maximum record length for files in format `F` or `V`
recordMode	Mode of files in format 'F' or 'V'. Possible values: - `TXT` - `BIN`
scheduleDateTime	Defines the timepoint when the send entry is to be executed (format `<yyyy-mm-dd hh:mm:ss>`)
serialization	Serialization Possible values: - `true, false`
noCopy	IF **noCopy**=`true` the file is not copied into outbox directory. Possible values: - `true, false`

Verwendete curl-Parameter und Methoden

Parameters	Description
-d	Data (send parameters)
-i	Output contains header
-k	Insecure SSL connection is permitted
-v	Verbose output
-H	Header parameters
-X	Request method (POST)

16 Command Line Tools

In this chapter you get an overview and the description of the command tools in **$RVS_HOME/bin** directory.

Hints:

Consider that directories and filenames which include blancs are always specified in double quotes.

For starting the command line tools from remote machine the remote mode parameters are required (excluded `showMonitorLog.bat` und `showMonitorLogFile.bat`).

16.1 getStationList

Use this tool to export the stationlist from the database into an XML file.

Syntax:

```
getStationlist -f <File> [-h][--verbose] [-su <user> -
sp <password> -sh <host:port>]
```

Required Parameter:

Parameter	Description
-f <File> **--xml-filename <File>**	Path and name of the file in which the stationlist is to be imported.

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.2 updateStationList

With the updateStationList program you have the possibility to set up the stations with a command tool at run time. For this an XML file, which contains all information about the stations you would like to modify, must be generated. The content of this file is described in a XML pattern (`$RVS_HOME/system/data/xsd/stationList.xsd`). For each station, an additional XML element (function) is necessary. Via `function` element the procedure to be executed is indicated. Position the function element between station type (e.g. `<StationNeighbour>`) and `<Contact/>`.

Have a look at the end of this chapter for an example.

Possible directives:

– `insert`: add a new station

– `update`: modify existing stations

– `delete`: delete a station (the own station cannot be deleted)

Syntax:

```
updateStationList -f <file name> [-r  <file name>]
[-h] [--verbose] [-su <user> -sp <password>
-sh <host:port>]
```

Requested Parameters:

Parameter	Description
-f, --xml-filename <file name>	Name of the XML element which contains information about the stations.

Optional Parameter:

Parameter	Description
-r, --failed-stations-xml-filename <file name>	Name of the XML file which contains the stations with failed update.
h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client

Remote Mode Parameters:

Parameter	Description
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

Example of XML file:

```
<?xml version="1.0" encoding="UTF-8" ?>
  <rvsStationConfig>
    <StationNeighbour>
    <Function>insert</Function>
      <Contact/>
      <DefaultCertificateLabel>114</DefaultCertificateLabel>
      <FileSecParams>
        <ReceiveCipherSuite>01</ReceiveCipherSuite>
        <ReceiveFileCompression>OPTIONAL</ReceiveFileCompression>
        <ReceiveFileEncryption>OPTIONAL</ReceiveFileEncryption>
        <ReceiveFileSignature>OPTIONAL</ReceiveFileSignature>
        <ReceiveResponseSignature>OPTIONAL</ReceiveResponseSignature>
        <SendCipherSuite>02</SendCipherSuite>
        <SendFileCompression>OPTIONAL</SendFileCompression>
        <SendFileEncryption>OPTIONAL</SendFileEncryption>
        <SendFileSignature>OPTIONAL</SendFileSignature>
        <SendResponseSignature>OPTIONAL</SendResponseSignature>
      </FileSecParams>
      <Fileservice>
        <Proxy>NONE</Proxy>
        <SecuritySet>OPT</SecuritySet>
      </Fileservice>
      <Line>
        <Instance>2</Instance>
        <ReceiverNumber>1</ReceiverNumber>
        <Sessions>20</Sessions>
        <TcpBasic>
          <IPAddress>10.33.101.75</IPAddress>
          <AlternativePort>0</AlternativePort>
          <Port>16711</Port>
        </TcpBasic>
        <Timeout>60000</Timeout>
        <Type>REMOTE_TLS</Type>
      </Line>
      <Pki>
        <CertificateValidationType>NONE</CertificateValidationType>
        <PkiEnabled>false</PkiEnabled>
      </Pki>
      <Protocol>
        <Oftp>
          <ActiveConnSetup>Yes</ActiveConnSetup>
          <Authentication>false</Authentication>
```

```
        <BufferSize>10000</BufferSize>
        <Compression>ODETTE</Compression>
        <Credit>999</Credit>
        <EerpIn>NORMAL</EerpIn>
        <EerpOut>NORMAL</EerpOut>
        <Level>2.0</Level>
        <OdetteID>O001300ZTU210   114</OdetteID>
        <Restart>true</Restart>
        <SessionFileDirection>B</SessionFileDirection>
        <UseDescAsFilename>false</UseDescAsFilename>
        <VdsnCharset>ODETTE</VdsnCharset>
      </Oftp>
      <Type>OFTP</Type>
    </Protocol>
    <ReceiveJobPriorityAsOriginator>3</ReceiveJobPriorityAsOriginator>
    <Sid>114</Sid>
  </StationNeighbour>
</rvsStationConfig>
```

16.3 File Transmission

This section contains the command tools for file transmission and display of job data.

16.3.1 activateStation

Use the `activateStation` program to activate a direct neighbour station or to activate the station and induce file transmission from this station.

Syntax:

```
activateStation -s <SID> [-p <password>] [-nt] [-h]
[--verbose] [-su <user> -sp <password>
-sh <host:port>]
```

Required Parameter:

Parameter	Description
-s <SID>	Station ID to be activated

Optional parameters:

Parameter	Description
-p <password>	Set this parameter for temporary overwriting of the Odette **Send Password** in stationlist. Only this connection is set up with this password.
-nt	Set this parameter to activate the neighbour station without introducing file transmission.

Optional parameters:

Parameter	Description
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

Hint: If the activating is not successful, please notice the messages in command prompt window beginning with **Message** or **Error**.

16.3.2 testConnection: Testing of TLS Connection

Use this tool for testing and tracing a TLS or Proxy TLS connection to your neighbour station. The information is saved in `$RVS_HOME/log/trace/tls` directory.

Syntax:

```
testConnection -s <SID> [-trace <type> [-p <arg>]]
[-h] [--verbose] [-su <user> -sp <password>
-sh <host:port>]
```

Required parameter:

Parameter	Description
-s <SID>	Station ID whose connection is to be traced.

Optional parameters:

Parameter	Description
--trace <type>	Trace-Typ. Possible Values: - `NONE`: no trace files is created (default) - `TLS`: file for TLS tracing is created - `NETWORK`: without function - `ODETTE`: without function
-p <arg>	Parameter for TLS tracing if **-trace** = `TLS`. Possible values are the values of Djavax.net.debug.

Optional parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.3.3 createSendJob

With the `createSendJob` tool you can create a send job. This functionality is identical with the `Create New Transmission` function via GUI.

Syntax:

```
createSendJob -d <filename> -s <receiver sid> |
-i <receiver oid> -v <vdsn> [-so <originator sid>]
[-f <originator oid>] [-Sd <init date>] [-St
<init time>] [-sh] [-t <table name>] [-S <serialize>]
[-p <priority>] [-l <label>] [-D <disposition>] [-F
<format>] [-M <length>] [-rm <record mode>] [-xid
<ext. job ID>] [-idf <filename>] [-nocp] [-sfs <set
id> [-c] [-Y] [-Yalg <algorithm>] [-Ycsn <serial>]
[-Ycin <issuer>] [-sif] [-sifcsn <serial>] [-sifcin
<issuer>] [-rsr] [-desc <description>]] [-h]
[--verbose] [-rd <remote filename>] [-uft <file type>]
[-su <user> -sp <password> -sh <host:port>]
```

Required createSendJob parameters:

Parameter	Description
-d <filename>	Name of the file to be send.
-s <receiver sid>	Station ID of destination
-i <receiver oid>	Odette ID of destination (alternative to Station ID)

Required createSendJob parameters:

Parameter	Description
-v <vdsn>	Virtual file name; the length of the file name used for ODETTE transfer must not exceed 26 characters. **Hint:** Via defintion `rvs_evo.controller.underscore_in_vdsn_allowed=true` in $`RVS_HOME/conf/rvs-system.properties` file you can configure that underscores are allowed in VDSN.

Optional Send Parameters:

Parameter	Description
-so <originator sid>	Station ID of originator: local (default) or virtual station
-f <originator sid>	Odette ID of originator: local (default) or virtual station
-Sd <init date>	With this parameter you can define what date the send job should be launched. Format: `<yyyy-mm-dd>`
-St <init time>	With this parameter you can define what time the send job should be launched. Format: `<hh:mm:ss>`
-Sh	With this parameter the send job is created in `HOLD` status (Release: please see chapter 16.3.9 "releaseJob").
-t <table name>	Name of your own conversion table: For ASCII(ANSI) - EBCDIC conversion: `ASCII-IBM037, ASCII-IBM273, ASCII-IBM500, ANSI-IBM037, ANSI-IBM273, ANSI-IBM500.` For EBCDIC - ASCII(ANSI) conversion: `IBM037-ASCII, IBM273-ASCII, IBM500-ASCII, IBM037-ANSI, IBM273-ANSI, IBM500-ANSI.` **Hint**: Not available for files in format `'T'`
-S <serialize>	The files will be sent in the same order, as the send jobs were created. The next job will only be sent, if the previous is completely finished. All send jobs for the serialization must have the same label (see parameter **-l**).

Optional Send Parameters:

Parameter	Description
-p <priority>	Priotity for sendjobs. Possible values: - 1 to 5 1 is highest priotrity, 5 is lowest priority, default: 3
-l <label>	Name of group of serialized send jobs. User specified (descriptive) label for this job. (See parameter **Serialization**)
-D <disposition>	Disposition of local file after successful send attempt. Possible values: - K (Keep): file will not be deleted after sending - D (Delete): file will be deleted after sending.
-F <format>	Format of the file to be sent: - T=text file; a stream of ASCII characters - U=unstructured (binary); byte stream - V=variable; variable record length - F=fixed; fixed record length **Hint**: set also the **-M** parameter for files in 'V' or 'F' format. Please see also the parameter **-rm**.
-M <length>	Maximal record length is to be set for files in format 'F'or 'V'. **Hints:** With format 'V' and record mode 'TXT' the record length will be determined automatically if **-M** is not set. With format 'T' and 'U' **-M** should not be set. Please see also the parameter **-rm**.
-rm <record mode>	In this parameter you indicate the mode of files in format 'F' or 'V'. Possible values: - TXT: text file - BIN: binary file Text File in format '**F**': If the record length deviate from the specified one (Parameter **MaxRecl**) the line will be filled with blanks or will be cut up to defined value. Text File in format 'V': longer records will be cut up to defined value. Binary files: The length of all records is the max. record size except the length of the last one. Binary files in format '**F**': If the size of the last record is shorter than the specified one the line will be filled to the defined value.
-xid <external job id>	Parameter for an external JobID, which can refer to several rvsEVO JobIDs. The external JobID is a string of alphanumeric characters.

Optional Send Parameters:

Parameter	Description
-idf <filename>	The ID of the job is written in this file after successful transmission (return code/error level = 0)
-nocp	By default, creating a send entry the files to be send are copied to Outbox directory of rvsEVO. With option -nocp the original file will be deleted after transmission was successfull. (Not if the format of the file to be send is `'U'`) **Attention**: If you have installed a server farm, this parameter must not be activated. The transmitted files need to be copied to outbox directory on the NFS drive, so that each node is able to access the file. (see also chapter 19 "Server Farm of rvsEVO").
-sfs <set-id>	Please see parameter **Security Feature Set** in table "Security parameters:" . Possible values: - `1` / None - `4` / OFTP 2.0 (CMS)
-cs	Number of OFTP2 Cipher Suite, definines the Encryption Algorithm. Possible values: 02, 04, 06, 07 (see Cipher Suite table)
-C	ODETTE compression (offline compression).
-Y	File will be sent encrypted
-Yalg <encryption algorithm>	See the description of the parameter **Encryption Algorithm** for GUI. Possible values: - `3DES` (not available; switch to AES) – `AES`
-Ycsn <serial no>	Certificate serial number for encryption (only for **sfs** 4)
-Ycin <issuer name>	Certificate issuer name for encryption (only for **sfs** 4)
-sif	File signature; only for **sfs=4** (OFTP 2.0).
-sifcsn <serial no>	Certificate serial number for file signature (only for **sfs** 4
-sifcin <issuer name>	Certificate issuer name for file signature (only for **sfs** 4)

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Optional Send Parameters:

Parameter	Description
-rsr	Active only in connection with OFTP 2.0 (CMS) (see parameter **-sfs** in this table). In OFTP 2.0 there is a possibility to request the signature of the EERP or of the NERP. Your partner must support OFTP 2.0 in this case.
-desc <description>	Comment, free text, available only in connection with OFTP version 2. If your partner does not support OFTP version 2, this field will be ignored.
-j <start job>	without function
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

:

Remote Mode Parameters:

Parameter	Description
-rd <file name>	Remote file name: **-rd** has different functions: 1. Create transmission remote: name of the file to be send. The file must be in outbox directory (**-d** and **-uft** must not be set). 2. Target file name in case of creating transmission local. The file will be uploaded with this name into outbox directory (**-d** and **-uft** must be set).
-uft <file type>	File Type for upload. Possible values: - `TXT`: text file - `BIN`: binary file
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

Cipher Suite table

Cipher Suite	Encryption Algorithm	Hash Algorithm
02	AES256_CBC	SHA-1
04	AES256_CBC	SHA-256
06	AES256_CBC	SHA-512
07	AES_256_CBC	SHA3-512

Examples:

```
createSendJob -d C:/text.txt -s RVS -v test
```

In this example the `C:/text.txt` file is sent to the station RVS with a virtual name `test33`. The virtual file name parameter (option `-v`) is mandatory.

```
createSendJob -d C:/text.txt -s RVS -v OFTP_TEST -F F
-M 80
```

In this example the `C:/text.txt` file is sent to the station RVS with a virtual name `OFTP_TEST`, this file is a text file of which each line has a length of 80 characters without CR/LF (`-F F -M 80`).

```
createSendJob -d C:/part.txt -s RVS -v PART  -S Y -l
AUTO
```

In this example the `C:/part.txt` file is sent to station `RVS` with a virtual name `PART`, this file belongs to the serialized group of files with the label (for the whole group) AUTO.

16.3.4 sendNext: Transmit Job Next

Use this tool for setting up a job in category ACTIVE to highest priority (`0`). Thereby the job will be transmitted with next free session.

A big number of active jobs may lead to delay. If necessary you can accelerate the process by terminating an active session via `terminateSession` tool or via rvsEVO GUI.

Syntax:

```
sendNext -n <JobID> [-h] [--verbose] [-su <user> -sp
<password> -sh <host:port>]
```

Erforderlicher Parameter:

Parameter	Beschreibung
-n <JobID>	ID of the job which should be sent next

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.3.5 handleEERP: Deleting or releasing EERPs

Use the program `handleEERP` to delete or release EERPs/NERPs.

Syntax:

```
handleEERP -r | -n | -d <JobID> [-c <code>] [-t
<text>] [-h] [--verbose]
[-su <user> -sp <password> -sh <host:port>]
```

Required Parameters:

Parameter	Description
-d <JobID>	ID of the job for which an EERP is to be deleted.
-n <JobID>	ID of the job for which an NERP is to be deleted.
-r <JobID>	ID of the job for which the EERP is to be released.

Optional Parameters:

Parameter	Description
-c <code>	NERP reason code (default='99')
-t <text>	NERP reason text
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.

Remote Mode Parameters:

Parameter	Description
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.3.6 holdJob

Use `holdJob` to pause a job. The job changes to status HOLD until it is released again.

Syntax:

```
holdJob -n [-h] [--verbose] [-su <user> -sp <password>
-sh <host:port>]
```

Required Parameters:

Parameter	Description
-n	The jobID of the job to be held

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.3.7 deleteJob

Use the program `deleteJob` for deletion of an active job. Without using `-f` parameter the job must be in status `HELD`.

Syntax:

```
deleteJob -n <jobID> [-f] [-r] [-h] [--verbose] [-su
```

```
<user> -sp <password>   -sh <host:port>]
```

Required Parameter:

Parameter	Description
-n	ID of the job to be deleted

Optional Parameters:

Parameter	Description
-f, --forced	This parameter allows to delete an active job independent of status
-r, --nerp	For routing send jobs, only: Addional NERP is sent to the originator and the corresonding routing receive job is deleted
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.3.8 deleteJobsOlder: Delete Old Jobs

Use this tool to delete jobs in category ACTIVE depending on age, direction and state. By default, send jobs in WF_EERP state are deleted. The deleted jobs are set to category ENDED.

Syntax:

```
deleteJobsOlder -m <months> | -d <days> | -h <hours> |
-i <minutes> [-s <state>] [-dir <direction>]
[-t <timestamp>] [-h] [--verbose]
```

Requiered Parameters

Parameters	Description
-m <month>	Jobs older than given number of months are deleted.
-d <days>	Jobs older than given number of days are deleted (alternative to -m).
-h <hours>	Jobs older than given number of hours are deleted (alternative to -m).
-i <minutes>	Jobs older than given number of minutes are deleted (alternative to -m).

Optional Parameters:

Parameter	Description
-s <status>	State of jobs to be deleted. Possible values: - each possible state - `ALL`: all possible states. Default: `WF_EERP`
-dir <direction>	Direction of the job. Possible values: - `SND`: send job (default) - `RCV`: receive job - `ERP`: end response - `ALL`: all directions
-t <timetype>	Time type for check. Possible values: `UPDATE`: last change (default) `CREATE`: creation date
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

16.3.9 releaseJob

Use this program to release a send job in status `HELD`.

Syntax:

```
releaseJob -n [-h] [--verbose] [-su <user>
-sp <password> -sh <host:port>]
```

Required Parameter:

Parameter	Description
-n	ID of the job to be released

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.3.10 restartJob

Use `restartJob` to restart a job in status `FAILED_WITH_SFNA` or `SP_FAILED`.

Syntax:

```
restartJob -n [-h] [--verbose] [-su <user> -sp
<password> -sh <host:port>]
```

Required Parameter:

Parameter	Description
-n	ID of the job to be restarted

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.

Remote Mode Parameters:

Parameter	Description
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

Hint:

The following information is for encrypted transmission only:

Jobs which can not be decrypted or encrypted by the Service Provider were changed to the `SP_FAILED` status. Use the button **Error** to display the error number and the description. After bug fixing you can activate the job with the tool `restartJob`.

16.3.11 terminateSession

With this tool an active session can be terminated.

Syntax:

```
terminateSession -s <session id> [-h] [--verbose]
[-su <user> -sp <password> -sh <host:port>]
```

Required Parameter:

Parameter	Description
-s, --sessionId <session id>	ID of the job to be terminated.

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

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16.4 Display Job Data

16.4.1 getJoblist: Listing of Send and Receive Jobs

The `getJoblist` program lists the active jobs, by default. Alternatively the ended or failed jobs can be listed.

Syntax:

```
getJobList [-a][-e][-f][-ac <option>][-h][--verbose]
[-su <user> -sp <password> -sh <host:port>]
```

Optional Parameters:

Parameter	Description
-a	Detailed information on jobs currently being processed (in addition display of creation date, SID of neighbour station, and VDSN).
-e	Information on determinated jobs
-f	Information on failed jobs
-ac <option>	Additional information. Possible values: - **S**: information on original filename - **D**: information on file description
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

If you call getJobList without parameter the display contains a list of all jobs currently being processed.

Examples:

```
getJobList -e
```

Result:

```
C:\rvsEVO_60001\bin>getJobList -a
************************************************************
************************
Job/Transmission      Status      CreatedAt           SIDORG
     SID:VDSN
************************************************************
************************
1402131138150000 (SND) SCHEDULED       13.02.14 11:38:15   LO
         EV054:TESTFILE.TXT
1402131139370000 (SND) SCHEDULED       13.02.14 11:39:37   LO
         EV054:TEST54
1312161250560000 (SND) PREPROCESSING_FAILED       16.12.13 12
    EV054          EV054:CLUSTER.PROPERTIES
************************
```

`getJobList -a`

Result:

```
C:\rvsEVO_60001\bin>getJobList -e
************************
job 1402131136590000 (SND): state=ENDED
job 1402131137170000 (SND): state=ENDED
job 1402131147200000 (RCV): state=ENDED
job 1402131148490000 (RCV): state=ENDED
job 1402131149280000 (SND): state=ENDED
************************
List ended. size=5
************************
```

16.4.2 getJob: Information on a Job Entry

Use `getJob` program to receive information on a particular job.

Syntax:

```
getJob -n <jobid> [-a] [-h] [--verbose] [-su <user> -
sp <password> -sh <host:port>]
```

Required Parameter:

Parameter	Description
-n <jobid>	Information on a send or receive job with ID `<jobid>`.

Optional Parameters:

Parameter	Description
-a	All available job information is given.
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

Example:

```
getJob -n 040329175603000
```

Result: `job 040329175603000 (SND): state: WF_EERP`

16.4.3 getJobInfoList: Create an XML File for an External JobID

Use the program getJobInfoList to create an XML-file for an external JobID. The detailed information of all jobs which refer to the external JobID are stored in this file.

Syntax:

```
getJobInfoList -xid <external Job ID> [-f <filename>]
[-h] [--verbose] [-su <user> -sp <password> -sh
<host:port>]
```

Required Parameter:

Parameter	Description
-xid, --external-jobid <external Job ID>	external job ID. The ID can be stated with creating a transmission. (See chapter 5 "Transmission with rvsEVO")

Optional Parameter:

Parameter	Description
-f , --xml-filename <filename>	Filename
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

The XML schema `JobInfoList.xsd` is stored in the directory `$RVS_HOME/system/data/xsd`.

JobInfoList

The root element is `JobInfoList`. The XML element `JobExtended-Detail` contains all job information. The XML element `JobExtended-Details` contains `JobDetails` and `JobError`.

The XML element `JobError` is optional. It contains `ErrorIdentifier`, `ErrorText` and `ErrorType`.

JobInfoList: Explanation of the job attributes:

Explanation of the job attributes

Parameter	Description
externalJobId	JobID for external application
jobNumber	rvsEVO-JobID
category	Categoriy of the job. Possible values: `ACTIVE`, `FAILED` oder `ENDED`
compression	Compressing; possible values: true(yes) or false(no). Default: false
conversionTable	path to conversion table
creationDate	Date and time (from SFID) of creating a dispatch order in the form `JJJJ-MM-DD HH:MM:SS`
counter	If several send jobs were created at the same time: sequential counter from SFID beginning with 0001
deleteAllowed	If **deleteAllowed**=`true` deleting a job is allowed dependent on its status. Possible values: `true` (Yes) or `false` (No). Default: `false`
direction	Direction; possible values SND (sending) or RCV(receiving).

Explanation of the job attributes

Parameter	Description
dispositiion	Whereabouts: This parameter decides whether the file to be sent remains as it is, or is deleted locally after sending. Possible values: `K` (Keep): File remains as it is, after sending.
encryption	Encoding; possible values: true(yes) or false(no). Default: false.
encryptionCertificateIs-suerName	Publisher name of the certificate. A value is displayed, if a certificate is indicated for use explicit. It is not displayed which certificate is used in fact.
encryptionCertificateSe-rialNumber	Serial number of the certificate. A value is displayed, if a certificate is indicated for use explicit. It is not displayed which certificate is used in fact.
fileDescription	File-description. Text comment, possible after the OFTP version 2. If your destination device is not OFTP 2.0, this field is ignored.
filename	The entire path of the file, which is sent from rvsEVO outbox directory. The file is copied initially from original directory to outbox directory before transmission. **Hint:** This filename can be different to original filename.
filenameSrc	The entire path of the original file; this file is copied in the rvsEVO-outbox-directory.
filePos	Number of the read bytes of the dispatch file.
holdAllowed	This parameter gives information as to whether the stopping a job, depending on the status, is allowed. Possible values: - `true` (yes) - `false` (no).
inputCharset	Without function
label	Label for serialization. All the files which have been sent in the same group, should have the same ID (label). If this parameter in case of serialization=Y(true) is not set, VDSN is used as a label.
lastByteRead	Value of the last read byte of a file to be sent in decimal representation as a character string. Default: `-1`.

Explanation of the job attributes

Parameter	Description
lastByteSend	Value of the last sent byte of a file to be sent, in decimal representation, as a character string. Default: `-1`
lastStateChange	Time of the last change in the job status in the format `DD.MM.YY HH:MM:SS`.
lengthOriginFile	Length of the original file while sending in bytes, before processing through the service provider. This parameter is set to 0 in case of receiving.
nerpCreatorOdetteID	OdetteID of the station sending a NERP.
nerpReasonCode	Code of NERP
nerpReasonText	Description of NERP
oidDest	OdetteID of the target station.
oidNeighbor	OdetteID of the neighboring station.
oidOrig	OdetteID of the sending station (Originator)
outputCharset	without function
priority	Priotity of sendjobs. Possible values: - `1` to `5`, default: `3`
recCount	Number of the data records sent (records).
recordFormat	Format of the file to be transferred: - `T` (Text): a consequence of ASCII-characters - `F` (Fixed): fixed record length - `V` (variable): variable record length - `U` (unstructured): binary file.
recordLength	Maximum record length.
recordMode	Satzmodus: Gibt bei Dateien im Format 'F' oder 'V'an, ob es sich um eine Textdatei oder um eine Binärdatei handelt. Mögliche Werte: - `TXT`: Textdatei - `BIN`: Binäre Datei
releaseAllowed	This parameter gives information as to whether releasing a jobs depending on the status is allowed. Possible values: - `true` (yes) - `false` (no). Default: f`alse`.

Explanation of the job attributes

Parameter	Description
restartPos	restart position, in case of which a new transfer of the file should be started..
retryCount	Number of failed send attements
scheduleDateTime	Planned time for the job start in the form `DD.MM.YY HH:MM:SS`
sendAttempts	Number of faulty dispatch attempts. If there is a "0", it means that this is a successful file dispatch.
serialization	This option means that your files are there in a serial order. All the files which should be sent in the same group, should have the same ID (Label). Possible values: - `true` (yes) - `false` (no). Standard value: f`alse`
SID	SID depending on the direction: While sending: SID of the recipient; While receiving: SID of the Originator station.
sidDestination	SID of the target station
sidOriginator	SID of source station
sidResponseCreator	SID which is sending a NERP.
signCertificateIssuer-Name	Publisher name of the signature A value is displayed, if a certificate is indicated for use explicit. It is not displayed which certificate is used in fact.
signCertificateSerial-Number	Serial number of the signature A value is displayed, if a certificate is indicated for use explicit. It is not displayed which certificate is used in fact.
status	Job status. For possible status values, please see the rvsEVO-user manual (chapter 4.5 dispatching a file).
timeFirstSendAttempt	Time of first connection setup to the neighbour station.
timeStartFile	Time of the transfer receipt (system time in milli-seconds)
transferFileLength	Sending: Size of the file to be sent. Sending: Size of the file to be sent.

Explanation of the job attributes

Parameter	Description
transmittedBytes	Number of the actually transferred bytes
user	Number of the actually transferred bytes
VDSN	Virtual file name for Odette-transfer.
virtualSID	For future applications.
waitTime	without function
jobSecurityAndSign	without function

Explanation of jobSecurity Parameters

Parameter	Description
securityFeatureSet	Security features; Possible values: - `1 (none)` - `4 (OFTP 2.0)`
encryptionAlgorithm	Encoding algorithm; possible values: blank (no algorithm), DES_EDE3_CBC (3DES) or AES256_CBC..
sign	Signature; possible values: - `true` (yes) - `false` (no). Default: `false`.
signEERP	Requesting signed EERP; possible values: - `true` (yes) - `false` (no). Default: `false`.

Explanation of jobError Parameters

Parameter	Description
errorID	If the error type is transmission, these error codes are identical with the ones from ESID, EFNA, SFNA of the Odette-log.
errorText	If the error type is transmission, these error texts are identical with the ones from ESID, EFNA, SFNA of the Odette-log.
errorType	Type of error; possible values: Internal, Transmission, MissingReceipt, Undefined.

Example:

```
<JobInfoList>
    <JobExtendedDetails>
        <JobDetails>
            <externalJobId>111</externalJobId>
            <JobID>
                <direction>0</direction>
                <jobNumber>1801161247360000</jobNumber>
            </JobID>
            <category>ACTIVE</category>
            <compression>true</compression>
            <conversionTable>IBM037-ANSI</conversionTable>
            <creationDate>2018-01-16 12:47:36</creationDate>
            <counter>0001</counter>
            <deleteAllowed>true</deleteAllowed>
            <disposition>K</disposition>
            <direction>SND</direction>
            <encryption>true</encryption>
            <encryptionCertificateSerialNumber></encryptionCertificateSe-
rialNumber>
            <encryptionCertificateIssuerName></encryptionCertificateIssu-
erName>
            <fileDescription></fileDescription>
            <filename>C:\rvsRT1\tenants\4011/files/
outbox\1801161247360000.spp</filename>
            <filenameSrc>C:\Daten\desti.txt</filenameSrc>
            <filePos>0</filePos>
            <holdAllowed>false</holdAllowed>
            <inputCharset></inputCharset>
            <label></label>
            <lastStateChange>2018-01-16 12:47:39</lastStateChange>
            <lastByteRead>-1</lastByteRead>
            <lastByteSend>-1</lastByteSend>
            <lengthOriginFile>24</lengthOriginFile>
            <nerpReasonCode></nerpReasonCode>
            <nerpReasonText></nerpReasonText>
            <nerpCreatorOdetteID></nerpCreatorOdetteID>
```

```
                    <oidDest>O4012</oidDest>
                    <oidNeighbor>O4012</oidNeighbor>
                    <oidOrig>O4011</oidOrig>
                    <outputCharset></outputCharset>
                    <recCount>0</recCount>
                    <recordFormat>V</recordFormat>
                    <recordLength>22</recordLength>
                    <recordMode>TXT</recordMode>
                    <releaseAllowed>false</releaseAllowed>
                    <restartPos>0</restartPos>
                    <scheduleDateTime></scheduleDateTime>
                    <sendAttemps>0</sendAttemps>
                    <retryCount>0</retryCount>
                    <serialisation>false</serialisation>
                    <SID>4012</SID>
                    <sidOriginator>LOC</sidOriginator>
                    <sidDestination>4012</sidDestination>
                    <signCertificateSerialNumber></signCertificateSerialNumber>
                    <signCertificateIssuerName></signCertificateIssuerName>
                    <status>WF_EERP</status>
                    <timeFirstSendAttempt>2018-01-16 12:47:38</timeFirstSendAt-
tempt>
                    <timeStartFile>1516103259121</timeStartFile>
                    <transmittedBytes>653</transmittedBytes>
                    <transferFileLength>653</transferFileLength>
                    <user></user>
                    <VDSN>DESTI.TXT</VDSN>
                    <virtualSID></virtualSID>
                    <waitTime>0</waitTime>
                    <jobSecurityAndSign>
                        <securityFeatureSet>4</securityFeatureSet>
                        <encryptionAlgorithm>DES_EDE3_CBC</encryptionAlgorithm>
                        <sign>true</sign>
                        <signEERP>true</signEERP>
                    </jobSecurityAndSign>
                </JobDetails>
                <JobError>
```

```
            <errorId></errorId>
            <errorType>Undefined</errorType>
            <errorText></errorText>
        </JobError>
    </JobExtendedDetails>
</JobInfoList>#
```

16.5 rvscli

This tool provides the possibility to execute the GUI functions via command line.

You can start `rvscli` in cmd box on two ways:

– Start `rvscli` with input file which includes the `rvscli` commands

  Example of input file:

```
displayUser  --userLoginID  rvs
createUser --userLoginID test --password te --role
User
updateUser --userLoginID test --firstName test1
listUsers
deleteUser --userLoginID test
```

– Start `rvscli` first. After a successful start you can write the command and the parameters into the command line.

Syntax:

```
rvscli [-i <filename> [-exitOnError]
[-generateReport]] [-trace] [-h] [-su <user>
-sp <password> -sh <host:port> -sl -sr]
```

Optional Parameters:

Parameter	Beschreibung
-i <filename>	Filename of input script to execute
-exitOnError	Exit further script execution on error
-generateReport	To save information about rvscli processes in a csv file in `$RVS_HOME/log` directory. The csv file name constists of „`rvscli-report_`“ and timestamp and „`.csv`“ (Format: `rvscli-report_YYYYMMDD-HHmmss.csv`).
-trace	This parameter activates tracing
-h, -help, -?	Requests help information (usage).

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	host is the IP address or hostname of rvsEVO Server; port is the RMI port for client server connection
-sl	enforce local mode even -su,-sp,-sh are given
-sr	enforce remote mode with login parameter from `$RVS_HOME\conf\rvsEvoClient.prefs` file, even -su,-sp,-sh are not given.

`rvscli`-Commands:

Use the `help` command for listing all `rvscli` commands.

```
C:\rvsEVO5416-2\bin>rvscli
*=============================================================================*
*                                                                             *
*                  RVS Command Line Interface                                 *
*                                                                             *
*=============================================================================*

                         Version:6.01.00

rvscli>help
!                       Command to Execute Operating System (OS) Commands
activateStation         Command to Activate Station.
acxDeliverCertificate   Command to Deliver ACX Certificate.
acxReplaceCertificate   Command to Replace ACX Certificate.
acxRequestCertificate   Command to Request ACX Certificate.
cleanupJobs             Command to Cleanup Jobs.
cleanupMonitorLogInDB   Command to Cleanup Monitor Log from Database.
cls                     Command to Clear Console.
createCSR               Command to Create Certificate Request.
createJobQuery          Command to Create Job Query.
createJobstart          Command to Create New Jobstart.
createObserver          Command to Create Observer.
createProxy             Command to Create Proxy Bastion.
createReceiver          Command to Create Receiver.
createSendJob           Command to Create New Job in Local Mode.
createSendJobRemote     Command to Create New Job in Remote Mode.
.
.
```

Insert ' `--help`' after a command for displayng the requested and optional parameters of the command.

Syntax:

```
activateStation --help
```

Insert the `help` command and a keyword for receiving a list of all commands which refer to this keyword. Insert `help --` and TAB key for listing the possible keywords.

`help --jobstart` displays a list of all commands which refer to a jobstart:

```
rvscli>help --jobstart
createJobstart  Command to Create New Jobstart.
deleteJobstart  Command to Delete Jobstart.
displayJobstart Command to Display Jobstart Details for
                given Jobstart ID.
listJobstarts   Command to Display Jobstart List.
updateJobstart  Command to Update existing Jobstart.

* Parameter value containing space should be enclosed with " " (for e.g. "r
i").

rvscli>
```

Insert the first characters of a command and the TAB key for listing all commands beginnig with the entered characters.

If you enter `cre` [TAB] the following output is displayed:

```
rvscli>create
createCSR          createJobQuery     createJobstart     createObserver
createProxy        createReceiver     createSendJob      createStation
createUser
rvscli>create
```

16.6 restartListener:

Use the `restartListener` tool to restart a listener.

Syntax:

```
restartListener [--restart] [--start] [--stop]
[-l <line> -n <number>] [-h] [--verbose]
```

Optional Parameters:

Parameter	Description
--restart	With this parameter all listeners are restarted (default). **Hint**: Do not use parameters **-restart**, **-start**, and **-stop** together.
--start	With this parameter all listeners are started. **Hint**: Do not use parameters **-restart**, **-start**, and **-stop** together.
--stop	With this parameter all listeners are stopped. **Hint**: Do not use parameters **-restart**, **-start**, and **-stop** together.
-n <number>	If **-l** is set: Listener number for starting or stopping the indicated listener (see network parameters of local station)

Optional Parameters:

Parameter	Description
-l <line>	Network type for starting or stopping a listener of the indicated type. Possible values: TCPIP, TLS, REMOTE_TCP, REMOTE_TLS (In conjunction with **-n**)
-h, -help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.7 cleanup: Deletion of Jobs and Monlog Data from the Database

Use this program to remove ended and failed transmission jobs or/and monlog data which are older than the defined date.

Syntax:

```
cleanup -d <date> [-r] [-t <type>] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-d, --date <date>	This option needs a date in the following format: `yyMMddHHmmss` or `yyMMdd`. The jobs or monlog data, which are older than the entered date will be removed from the database.

Optionale Parameters:

Parameters	Descriptiong
-r **--remote**	Remote, availables deletion during rvsEVO is running.

Optionale Parameters:

Parameters	Descriptiong
-t **--type**	Type of entry to be deleted. Possible values: - `JOBS` (default): data of ended or failed jobs - `MONLOG`: monlog data - `ALL`: jobs and monlog data
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

16.8 archiveJobs: Deletion of Job Data from Database

Hint: The name of the program is historical. Up to rvsEVO version 6.00 `archiveJobs` was used for archiving jobs.

Use this program to delete ended and failed transmission jobs from the database which are older than the defined date.

Syntax:

```
archiveJobs -d <date> [-r] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-d, --date <date>	This option needs a date in the following format: `yyMMddHHmmss` or `yyMMdd`. The jobs, which are older than the date will be deleted from the database.

Optional Parameters:

Parameter	Description
-r **--remote**	remote, availables deletion during rvsEVO is running.
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

16.9 archiveJob: Deletion of an Individual Job from the Database

Warning: This command is deprecated, please use the `cleanup` tool instead.

The name of the program is historical. Up to rvsEVO version 6.00 `archiveJob` was used for archiving jobs.

This program removes the data of a specific job in category ENDED or FAILED from the data base.

Syntax:

```
archiveJob -j <args> [-h] [--verbose] [-su <user>
-sp <password> -sh <host:port>]
```

Required Parameter:

Parameter	Description
-j, --jobID <args>	ID of the job/jobs to be deleted. Several job IDs have to be seperated by comma.

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.10 Key Management with rvsEVO

16.10.1 deliverCertificate

With this tool certificates of local or virtual station can be delivered to partner stations and imported into there database. Virtual file name for ODETTE transfer is `ODETTE_CERTIFICATE_DELIVER`.

Syntax:

```
deliverCertificate -s <stationID> -r <statio-
nID1,stationID2> [-i <keyIndex1,keyIndex2>]
[-c <referenceIndex>] [-h] [--verbose] [-su <user> -sp
<password> -sh <host:port>]
```

Required Parameters:

Parameter	Description
-r <stationID>	ID of the partner station, to which the certificate should be sent (several SIDs have to be seperated by comma).
-s <stationID>	StationID of the station, whose certificate should be sent.

Optional Parameters:

Parameter	Description
-i <keyIndex>	Index of the certificate to be sent. If not indicated the certifacte with the newest „valid-from date“ with default label is sent. You find the index in the list view and detailed view of a certificate. Several indexes have to be seperated by comma.
-c, --referenceIndex <index>	Index of the reference certificate. At your partner station the delivered certificate is added to the label group of this certificate. If the reference certificate is unknown at destination the delivered certificate gets the default label. The certificate reference is transmitted via SFID-DESC to the destination. **Hint:** For this use case only one index is allowed for parameter `-i`.
-h, --help, -?	Requests help information.
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.10.2 requestCertificate

Use this tool for delivering a certificate to a partner station and request certificates from your partner station. Virtual file name for ODETTE transfer is `ODETTE_CERTIFICATE_REQUEST`.

Syntax:

```
requestCertificate -s <stationID> [-i <index>
-o <SID>] [-h] [--verbose] [-su <user> -sp <password>
-sh <host:port>]
```

Required Parameter:

Parameter	Description
-s <stationID>	ID of the station, from which the certificate should be requested.

Optional Parameters:

Parameter	Description
-i <keyIndex> **--index <keyIndex>**	Index of the certificate that will be send in request. If not indicated the certificate with default label with the newest „valid-from date“ is sent.
-o <stationID> **--ownerSid <stationID>**	ID of the station whose certificate is to be sent. If not indicated the ID of local station is used.
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.10.3 replaceCertificate

With this tool a certificate of local or virtual station can be sent to partner stations and replace the already existing certificates with this stations. Virtual file name for ODETTE transfer is `ODETTE_CERTIFICATE_REPLACE`..

Hint: For replacing certificates of a special label group parameter **-c** is to be set.

Syntax:

```
replaceCertificate -s <station ID> -i <keyIndex>
-r <receiver SIDs> [-c <referenceIndex>] [-h] [--
verbose]
[-su <user> -sp <password> -sh <host:port>]
```

Required Parameters:

Parameter	Description
-r <stationID>	ID of the partner sation, to which the own certificate is to be sent. The old certificates are replaced and cancelled. Several SIDs have to be separated by comma.
-s <stationID>	StationID of the own station, whose certificate is to be sent.
-i <keyIndex>	Index of the certificate, which should be sent. You find the index in the list view and detailed view of a certificate.

Optional Parameters

Parameter	Description
-c, --referenceIndex <index>	Index of the reference certificate. At your partner station all certificates are replaced which belong to the label group of this certificate. If the reference certificate is unknown at destination the delivered certificate is imported with default label. The certificate reference is transmitted via SFIDDESC to the destination.
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.10.4 getCertificateList

The tool `getCertificateList` shows the certificate list of a special stationID.

Syntax:

```
getCertificateList -sid <stationID> [-h] [--verbose]
[-su <user> -sp <password> -sh <host:port>]
```

Required Parameter:

Parameter	Description
-sid, --stationid <stationID>	ID of the station whose certificates should be listed.

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.10.5 importTSL

Use this tool to import a TSL (**T**rust **S**ervice **L**ist) into the rvsEVO database.

Syntax:

```
importTSL -f <TSL filename|URI> | -o [-h] [--verbose]
[-su <user> -sh <host:port> -sp <password>]
```

Required Parameter:

Parameter	Description
-f, --file name <TSL filename|URI>	Path of the file which includes the TSL or URI* for CRL download.

Required Parameter:

Parameter	Description
-o **--odette**	(Alternative to parameter **-f**) Download and Import of TSL from Odette organisation* (`http://www.odette.org/TSL/TSL_OFTP2.XML`).

* Set the security parameters **CRLUseHttpProxy**, **CRLHttpProxyHost**, **CRLHttpProxyPort** before importing a TSL from the internet.

Optional Parameters:

Parameter	Description
-h, --help, -?	Fordert Hilfe (Usage) an.
--verbose	Ausgabe von ausführlichen Meldungen.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.10.6 showCRLs

Use this tool to display a list of the imoprted **CRLs** (**C**ertificate **R**ovocation **L**ist).

Syntax:

```
showCrls [-h] [--verbose] [-su <user>
-sp <password>  -sh <host:port>]
```

Optional Parameters:

Parameter	Description
-h, --help, -?	Fordert Hilfe (Usage) an.
--verbose	Ausgabe von ausführlichen Meldungen.

Optional Remote Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.11 Display Monitor Messages

16.11.1 showMonitorLog: Trace Current Monitor Messages

Use the `showMonitorLog` program to trace the current monitor messages and to analyze error messages. When starting the program, the last 25 messages are displayed.

Syntax:

```
showMonitorLog [-h] [--verbose] [-su <user> -sp
<password>  -sh <host:port>]
```

Optional Parameters:

Parameter	Description
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.11.2 showMonitorLogFile: Show Earlier Log Messsages

Use `showMonitorLogFile` to view earlier Monitor-Log messages. The log files are saved in `$RVS_HOME/log` directory.

Syntax:

```
showMonitorLogFile -i <filename> [-h] [-verbose]
```


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Required Parameter:

Parameter	Description
-i	Path and name of the log file you would like to display.

Optional Parameters:

Parameter	Description
-h, -help, -?	Requests help information (usage).
-verbose	Verbose message output.

16.12 showDnsCache: Show DNS Cache and TTL Settings

Use this tool for displaying the DNS cache and TTL settings and for DNS name resolution.

Syntax:

```
showDnsCache -all | -pc | -nc | -ttl | -l <name> | -c
<name>] [-P <instanceId>] [-h] [--verbose] [-su <user>
-sp <password> -sh <host:port>]
```

Required Parameters:

Parameter	Description
-all	Display of all entries (at least one of the following parameters is requiered: **-all, -pc, -nc, -ttl, -l, -c**)
-pc	Display of positve entries (at least one of the following parameters is requiered: **-all, -pc, -nc, -ttl, -l, -c**)
-nc	Display of negative entries (at least one of the following parameters is requiered: **-all, -pc, -nc, -ttl, -l, -c**)
-ttl	Display of TTL settings (at least one of the following parameters is requiered: **-all, -pc, -nc, -ttl, -l, -c**)
-l <name>	DNS name resolution of the given name and display of cache (at least one of the following parameters is requiered: **-all, -pc, -nc, -ttl, -l, -c**)

Required Parameters:

Parameter	Description
-c <name>	Deletion of entries from positive DNS cache. Possible values: - Hostname whose entry is to be deleted - „`all`“: deletion of all entries from cache (at least one of the following parameters is requiered: **-all, -pc, -nc, -ttl, -l, -c**)

Optionale Parameters:

Parameter	Description
-P <instanceID>	Display refers to rvs® OFTP Proxy. You can find the instanceID in Proxy Bastion List of rvsEVO.
-h, --help, -?	Requests help information (usage).
--verbose	Verbose message output.

Remote Mode Parameters:

Parameter	Description
-su <user>	Client user for remote login.
-sp <password>	Password of the client
-sh <host:port>	`host` is the IP address or hostname of rvsEVO Server; `port` is the RMI port for client server connection

16.13 Command Line Tools for Starting and Stopping rvsEVO

The description of the following programs, you can find in chapter 3 "Start and Stop of rvsEVO":

- `startGUI`
- `startServer`
- `rvsEVOService`
- `stopServer`

16.14 Command Line Tools for Odette Partner Data eXchange (PDX)

Via the following tools rvsEVO provides the possibility to add new partner stations and configure existing stations semi-automatic. Please read rvsEVO Operator Manual, chapter „Odette Partner Data eXchange (PDX)“ for further information about PDX.

- `processPdxInbound`
- `processPdxOutbound`

16.15 Command Line Tools for Internal Use

The following programs are only for internal use:

- `setclientcp`
- `setcp`

17 Log and Trace Files

rvsEVO messages and warnings are saved in the log files in directory `$RVS_HOME/log` by default. The list below shows the log files and it's type of information contained.

- `monitor.log` file: Monitor messages
- `rvs.log` file: Error messages and warnings from rvsEVO Server
- `rvsClient.log` file: Error messages from rvsEVO Swing GUI and command line tools.
- `sysinfo.log` file: System information
- `rvscli.log` file: Messages of command line tool `rvscli`

Hint: See rvsEVO Operator Manual chapter „Customizing rvsEVO" / „Customizing the Log Files" for information about defining the maximum file size and maximum number of log files and information about defining the path of the log files.

The trace files contain the transmitted Odette data packages (Odette tracing), information of TLS connection (TLS connection tracing) or information of used API methods.

Please see rvsEVO Operator Manual, chapter 'Monitoring of rvsEVO' for further information about log and trace files.

17.1 Trace Files

Odette tracing

Set parameter **TraceItem** = `O` (parameter group **ODETTE**) for enabling Odette tracing functionality.

The trace files with information about all transmitted Odette data packages are saved in `RVS_HOME/log/trace/odette` directory by default. The name is composed as follows: StationID and '_' and SessionID and '.oftp.trc' extension (e.g. `SID1_6372170.oftp.trc`).

With parameter **TraceNetworkData** (parameter group **ODETTE**) you define wether the network data should be traced and with parameter **TraceStationIds** you enable tracing per station (see chapter 12.6 "ODETTE Parameters").

Hints:

Please see the rvsEVO Operator Manual, chapter „Monitoring of rvsEVO" / „Tracing" / „Odette Tracing" for Odette parameter description.

If Odette tracing is configured permanently it is important to clear the directory regulary to prevent filling up (see also rvsEVO Operator Manual, chapter „Maintenance of Directories").

TLS connection tracing

The trace files which contain the information about TLS or Proxy TLS connections to a neihgbour station are saved in `$RVS_HOME/log/trace/tls` directory by default. The name is a composition of SID, „_",

session ID and the extension „.tls.trc“. (e. g. `RVSE-VO_2211055530000.tls.trc`).

Enable the TLS tracing functionality in `Stations` perspective via a right mouse click on the neighbour station and thereafter select `Trace TLS Connection Setup` from the drop down menu (see chapter 6.1 "User Interface of Stations") or via cmd tool `testConnection` (see chapter 16.3.2 "testConnection: Testing of TLS Connection").

API tracing

The API trace files which contain information about used API methods are saved in `$RVS_HOME/log/trace/api` directory by default. If the directory contains more than one trace.log file a counter is added.

You can activate API tracing via `rvs-system.properties` file.

18 rvsbat Batch Interface

`rvsbat` is for compability with rvs® portable. `rvsbat` gives you an additional possibility for **creating send jobs** and **managing jobstarts**.

18.1 Working with rvsbat

In order to create a send job or to manage jobstarts via rvsbat, follow the procedure given below:

- start rvsbat via command line
- enter the command for send jobs (`SEND`) or for management of jobstarts (`RESENTR`, `SENDJOB` or `FAILURE`):
  - via input file (see chapter 18.1.2) or
  - via the command prompt window (see chapter 18.1.3).

Hint: Commands and parameters are not case-sensitive.

18.1.1 Starting rvsbat

`rvsbat` tool is located in `$RVS_HOME/bin` directory and can be started via command line:

Syntax:

```
rvsbat [/c] [/l<language>] [/i<input file>] [/q]
```

Optional rvsbat Parameters:

Parameter	Description
/i <input>	Read commands from input file. The input file may contain the following elements: - Comment lines (starting with *) - Commands (may extend over several lines by specifying **+** as the last character in the line to be continued)
/c	Continue with `rvsbat` after an error occured during execution of a utility command. By default, `rvsbat` will terminate after an error.
/q	Execute user commands in quiet mode, i.e. do not echo them to standard output; feedback about success or failure of the operation will still be provided.
/l	Use message language given by „language“. Values: - e: English (default) - d: German

18.1.2 Import a Command via Input File:

Using the example of SEND/CREATE command we describe what steps are necessary to import the rvsbat commands and the parameters via input file:

- Provide the file to be transfered (example: test.txt)
- Provide the input file which contains the rvsbat command and send parameters.
- **Example** for an input file (input.txt):

  ```
  SEND /C SIDORIG=LOC DSN=C:\docs\test.txt (SID=EVO54E
  DSNNEW=TESTVDSN)
  ```

  In the above example a send job is created for sending the file C:\docs\test.txt from local station LOC to station EVO54E with VDSN TESTVDSN.
- Start rvsbat with the input file input.txt.

  Expample:

  ```
  rvsbat /Ic:\docs\input.txt
  ```

18.1.3 Call the Command via Command Prompt

To enter the rvsbat commands and the parameters into the command prompt follow the procedure given below (using the example of SEND/CREATE command):

- Call rvsbat via command prompt

  Example:

  ```
  rvsbat
  ```
- Input the commands and the send parameters into the prompt window after rvsbat was started successfully (message „Command 'START /USER' successfully executed“)

  Example:

  ```
  SEND /CREATE SIDORIG=LOC DSN=C:\docs\test.txt
  (SID=EVO54E DSNNEW=TESTVDSN)
  ```

Hint: If the command is called via command prompt **rvsbat** can be terminated by **<STRG> C**.

In the following chapters you find a detailed description of the commands.

18.2 Create a Send Job with SEND/CREATE Command

Use the SEND /CREATE (or SEND /C) command of rvsbat script for sending files.

You can create a send job by starting rvsbat and entering the command in the command promt or reading the command from an input file (see chapter 18.1.2 and 18.1.3).

The successful transfer is confirmed in both cases via the message „Command 'SEND /CREATE' successfully executed".

Syntax:

```
SEND /CREATE DSN=<filename> [SIDORIG=<originator sid>
OIDORIG=<originator OID> CODEIN=<file code>
DISP=<disposition> FORMAT=<format> MAXRECL=<length>
LABEL=<label> SERIAL=<set id> VFTYP=<set id>
INITTIME=<set id> NOCOPY=<set id>] (SID=<receiver sid>
OIDDEST=<receiver oid> DSNNEW=<vdsn> [CODEOUT=<set id>
CODETABLE=<table name> COMPRESSION=<set id> ENCRYP-
TION=<set id> SFS=<set id> ALG=<set id> CRYPT-
CERTSN=<serial number> CRYPTCERTIN <issuer name>
SIGN=<set id> SIGNCERTSN=<serial number> SIGN-
CERTIN=<issuer name> SIGNRESP=<set id>
FILEDESC=<file description> XID=<external job id>])
```

18.2.1 Send /Create Parameters Supported by rvsEVO

Required Send /Create Parameters:

Parameter	Description
ALG	Active only in connection with OFTP 2.0 (`sfs = 4` / OFTP 2.0/CMS). The following algorithms are possible: `3DES` (deprecated) `AES` (GUI name: **Encryption Algorithm**)
DSN	Path and name of local data set to be sent.
DSNNEW	Virtual file name; the length of the file name used for ODETTE transfer must not exceed 26 characters. **Hints:** The parameter is to be enclosed in parentheses. Via defintion `rvs_evo.controller.underscore_in_vdsn_allowed=true` in `$RVS_HOME/conf/rvs-system.properties` file you can configure that underscores are allowed. (GUI name: **VDSN**)
OIDDEST	Odette ID of receiver (enclosed in parantheses). Only one of the parameters must be specified: **OIDDEST** or **SID**.
SID	Station ID of receiver (enclosed in parantheses). Only one of the parameters must be specified: **OIDDEST** or **SID**.

Optional Send /Create parameters:

Parameter	Description
CODEIN	Code of local file. Possible values: - A=ASCII - E=EBCDIC Default: code of local system
DISP	Disposition of local file after successful send attempt. Possible values: - **K**=keep; file will not be deleted after sending (default) - **D**=delete. (GUI name: **Disposition**)
FORMAT	Format of the file to be sent. Possible values: - T=text file; a stream of ASCII characters - U= unstructured (binary); byte stream - F=fixed; fixed record length - V=variable; variable record length Default: U
INITTIME	With this parameter you can decide, what time the send job is to be created. Possible values: - date and time in format "YYYY-MM-DD HH:mm:ss" - H or HOLD: job is created with HOLD status. - N or NOW: job is started immediately (default) (2 GUI parameters: **Hold, Schedule**)
LABEL	User label (up to 20 characters) used to serialize on a proceeding send job (if **SERIAL**=Y) or which can be used for serialization by a subsequent send request
MAXRECL	Max. record length for the files in format F or V (see also FORMAT parameter)
NOCOPY	With this paramter you can decide whether the file which should be transmitted is to copy to Outbox directory or not. Possible values: - Y: no copy - N: copy is saved in Outbox dorectory (default) Note: not with format **U** and code conversion
OIDORIG	Odette ID of originator (OID of local station or OID of virtual Station)

Optional Send /Create parameters:

Parameter	Description
SERIAL	If you set **SERIAL**=Y, the files will be sent in the same sequence as the send jobs were created. The next job will only be sent, if the previous is completely finished. All send jobs for the serialization must have the same **LABEL**.Possible values: - `Y` (Yes) - `N` (No) (GUI name: **Serialization**)
SIDORIG	Station ID of originator; local station (default) or virtual station. (GUI name: **SID Originator**)
VFTYP	In this parameter you indicate the mode of files in format 'F' or 'V' (see also **FORMAT** parameter). Possible values: - `X`: text file (record mode ’TXT’ in the GUI) - `U`: binary file (record mode ’BIN’ in the GUI) Please see parameter **Record Mode** in chapter "Jobstart parameters:" for further information. (GUI name: **Record Mode**)

Optional Send /Create Parameters (enclosed in parentheses)

Parameter	Description
ALG	Active only in connection with OFTP 2.0 (**sfs = 4** / OFTP 2.0/CMS). The following algorithms are possible: - `3DES` (deprecated) – `AES` (GUI name: **Encryption Algorithm**)
CODEOUT	Desired code of data set at receiver; possible values: - `A`=ASCII - `E`=EBCDIC Example: `send /c dsn=C:\test22.dat CODEIN=A FORMAT=V(SID=RTT CODEOUT=E DSNNEW= FIX0GBE.TEXT)`

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Optional Send /Create Parameters (enclosed in parentheses)

Parameter	Description
CODETABLE	Defines the code table, which is to be used for the code conversion (see Parameter **Conversion table** in the table in “Basic parameters:” on page 38) . Indicate the alias name of conversion table. Example: `send /c dsn=c:\programs\rvsEVO\files\outbox\test22.dat FORMAT=V(SID=RTT CODETABLE=ASCII-IBM037 MAXRECL=80 DSNNEW=FIX0GBE.TEXT).` (GUI name: **Conversion Table**) **Hint**: Not available for files in format `'T'`
COMPRESSION	Compression during transmission; active only in connection with **sfs**=`4`. Possible values: - `Y` (Yes): the file is to be compressed before sending. - `N` (No/default) This parameter has to be placed after SID parameter. Example: `SEND /C DSN=\home\test\test11.txt (SID=RTZ COMPRESSION=Y).` (GUI name: **Offline Compression**)
CRYPTCERTSN	Serial number of the certificate for file encryption; active only in connction with **sfs**=`4` and **ENCRYPTION**=`Y` (GUI name: **Encryption Cert. Serial Number**)
CRYPTCERTIN	Issuer name of the certificate for file encryption; active only in connction with **sfs**=`4` and **ENCRYPTION**=`Y` (GUI name: **Encryption Cert. Issuer Name**)
ENCRYPTION	File encryption; active only in connection with **sfs**=`4`; Possible values: - `Y` (Yes): the file is to be encrypted before sending. - `N` (No): None encryption. This parameter has to be placed after SID parameter. Example: `SEND /C DSN=\home\test\test11.txt (SID=RTZ COMPRESSION=Y ENCRYPTION=Y).`
FILEDESC	File description, active only with **sfs**=`4` (OFTP 2.0) (GUI name: **File description)**

Optional Send /Create Parameters (enclosed in parentheses)

Parameter	Description
NOCOPY	With this parameter you can decide whether the file which should be transmitted is to copy to `Outbox` directory or not. **Attention:** If you have installed a server farm, this parameter must not be activated. The transmitted files need to be copied to `outbox` directory on the NFS drive, so that each node is able to access the file. (see also chapter 19 "Server Farm of rvsEVO").
SFS	**S**ecurity **F**eature **S**et; this parameter applies to the format of encryption. Possible values: - `1` (no encryption; default) - `4` (OFTP 2.0 (CMS)) (GUI name: **Security Feature Set**)
SIGN	File signature is activated. Active only in connection with **sfs**=4 (OFTP 2.0). Possible values: - `Y`: File Siganture is active - `N`: None file signature (default) (GUI name: **File Signature**)
SIGNCERTSN	Serial number of the certificate for file signature; active only in connction with **sfs**=`4` and **SIGN**=`Y` (GUI name: **Encryption Cert. Serial Number**)
SIGNCERTIN	Issuer name of the certificate for file signature; active only in connction with **sfs**=`4` and **SIGN**=`Y` (GUI name: **Encryption Cert. Issuer Name**)
SIGNRESP	Request signed EERP/NERP. Active only in connection with **sfs**=`4` (OFTP 2.0). Possible values: - `Y`: signed EERP/NERP is requested - `N`: signed EERP/NERP is not requested (default) (GUI name: **Request Signed EERP/NERP**)
TSTAMP	No Function; for compatibility with rvs® portable. Possible values: - `Y` - `N` (default
XID	Parameter for an external JobID, which can refer to several rvsEVO JobIDs. The external JobID is a string of alphanumeric characters. (GUI name: **External Job ID**)

18.3 Managing of Jobstarts via rvsbat

In this Chapter you get information about managing of jobstarts via `rvsbat`. Jobstarts after receive (`RESENTR` command), after send attempt (`SENDJOB` command) and after processing error (`FAILURE` command) can be created, modified or deleted. Please read chapter 8.1.1 "Create a new Jobstart" for further information about jobstarts.

You can create, update or delete a jobstart by starting `rvsbat` and thereafter entering the appropriate command in the command promt or by reading the command from an input file (see chapter 18.1.2 and 18.1.3).

18.3.1 Jobstart after Receive

You can use the `RESENTR` command of `rvsbat` script for managing Jobstarts after receive (in accordance with resident receive entries in rvs® portable).

The function that is to be executed can be indicated with the `RESENTR` command:

– `RESENTR /CREATE` or `/C`: create a jobstart after receive

– `RESENTR /UPDATE` or `/U`: edit a jobstart after receive

– `RESENTR /DELETE` or `/D`: delete a jobstart after receive

Hint: The commands are not case-sensitive. They can be added in abbreviated or long version and with blank or without. (e.g. `RESENTR /DELETE DSN="*.TENNIS.*" SID="*"` is the same as `RESENTR/D DSN="*.TENNIS.*" SID="*"`).

Syntax:

```
RESENTR /CREATE DSN=<filename> SID=<originator sid>
[CODETABLE=<table name> CODETRANS=<set id>
COMMENT=<text> DSNNEW=<new dsn name> ENABLED=<set id>
EXECSYNC=<set id> EXECTIMEOUT=<set id> JOB=<scrip
name> NEWDIR=<new directory name> NEWNAME=<new file
name> PARAMHANDLING=<set id> REPLACE=<set id>
SHELL=<set id> SIDDEST <receiver sid> TSTAMP=<set id>
TSTAMPFORMAT=<set id> VFTYP=<set id> DISP=<set id>
ACCOUNT=<set id> FLAGCOMP=<set id> FLAGCRYP=N
LUID=<set id> UID=<set id>)
```

RESENTR parameters supported by rvsRVO:

Required parameters: DSN, SID or SIDORIG

Optional parameters: CODETABLE, CODETRANS, COMMENT, DISP, DSNNEW, ENABLED, EXECSYNC, EXECTIMEOUT, JOB, NEWDIR, NEWNAME, PARAMHANDLING, REPLACE, SHELL, SIDDEST, TSTAMP, TSTAMPFORMAT, VFTYP

Parameters without function (for compatibility with rvs® portable): ACCOUNT, FLAGCOMP, FLAGCRYP, LUID, UID

In chapter 18.3.4 "Jobstart Parameters" you can find a description of the jobstart parameters.

CREATE **Example for creating a jobstart after receive:**

```
RESENTR /c SIDORIG=* DSN=*TESTFILE1* SIDDEST=LOC
DSNNEW=C:\Daten-rvsbat-tests\DSNNEW_TEST.txt
ENABLED=Y LUID=* UID=*
```

Files, receiving by local station `LOC` with a virtual filename that includes the character set `"TESTFILE1"` are renamed in `DSNNEW_TEST.txt` and saved in `C:\Daten-rvsbat-tests` directory. The parameters **LUID** and **UID** are without function.

Hint: One of the parameters **SID** (=**SIDORIG**), **DSN** and **SIDDEST** has to distinguish from existing jobstarts after receive.

Update **Example for updating a jobstart after receive:**

```
RESENTR /u SIDORIG=* DSN=*TESTFILE1* SIDDEST=LOC
DSNNEW=C:\Daten-rvsbat-tests\DSNNEW_TEST.txt
REPLACE=Y ENABLED=Y LUID=* UID=*
```

In example above the previously created jobstart is updated as follows: Existing files with the same name like the new one are replaced (`REPLACE=Y`). By default the existing file is kept and a timestamp is added to the name of new file.

Hint: The parameters **SID** (=**SIDORIG**) and **DSN** must be identical to the jobstart that is to be updated.

DELETE **Example for deletion of a jobstart after receive:**

```
RESENTR /DELETE SIDORIG=* DSN=*TESTFILE1*
```

In example obove the previously created jobstart is deleted.

Hint: The parameters **SID** (=**SIDORIG**) and **DSN** are requested.

18.3.2 Jobstart after Send Attampt

Use the `SENDJOB` command to create, edit or delete a jobstart after send attempt. The function that is to be executed can be indicated with the `SENDJOB` command:

Commands:

– `SENDJOB /CREATE or /C:` create a jobstart after send attempt

– `SENDJOB /UPDATE or /U:` edit a jobstart after send attempt

– `SENDJOB /DELETE or /D:` delete a jobstart after send attempt

Syntax:

```
SENDJOB /CREATE VDSN=<filename> SID=<receiver sid>
ATTEMPTS=<set id> COMMENT=<text> ENABLED=<set id>
EXECSYNC=<set id> EXECTIMEOUT=<set id> JOB=<scrip
name> PARAMHANDLING=<set id> SHELL=<set id>
```

```
SIDSENDER=<originator sid>  SIDORIG=<originator sid>
UID=<set id>
```

SENDJOB parameters supported by rvsRVO:

Required parameters: SID (= SIDDEST), VDSN, ATTEMPTS

Optional parameters: COMMENT, ENABLED, EXECSYNC, EXECTIMEOUT, JOB, PARAMHANDLING, SHELL, SIDSENDER = SIDORIG

Parameters without function (for compatibility with rvs® portable):

UID, COMPRESSION; ENCRYPTION

The table in chapter 18.3.4 "Jobstart Parameters" shows the description of the parameters.

CREATE **Example for creating a jobstart after send attempt**

```
SENDJOB /c SIDDEST=EVO54 SIDORIG=LOC VDSN=*TESTF*
ATTEMPTS=0 JOB=C:\rvsTest\scripts\sendjob.bat
ENABLED=Y
```

After successful transmission (**ATTEMPTS**=0) of a file with a virtual filename that includes the character set `"TESTF"` from station `LOC` to station `EVO54` the job `C:\rvsTest\scripts\sendjob.bat` is started.

Hint: One of the parameters **SID** (=**SIDDEST**), **VDSN** and **ATTEMPT** has to distinguish from existing jobstarts after send attempt.

Update **Example for updating a jobstart after send attempt:**

```
SENDJOB /u SIDDEST=EVO54 SIDORIG=LOC VDSN=*TESTF*
ATTEMPTS=0 JOB=C:\rvsTest\scripts\sendjob.bat
ENABLED=N
```

In this example the priviously created jobstart after send attempt is disabled (**ENABLED**=`N`).

Hint: The parameters **SID** (=**SIDDEST**), **VDSN**, **ATTEMPTS** and **SIDORIG** must be identical to the jobstart that is to be updated. (**SIDORIG** only if indicated with creation).

Delete **Example for deletion of a jobstart after send attempt:**

```
SENDJOB /d SID=EVO54 SIDORIG=LOC VDSN=*TESTF*
ATTEMPTS=0
```

The previously created jobstart after send attempt is deleted.

Hint: The parameters **VDSN, ATTEMPTS, SIDDEST** and **SIDORIG** are requested. (**SIDORIG** only if indicated with creation).

18.3.3 Jobstart after Failure

Use the `FAILURE` command to create, edit or delete a jobstart after processing error. The function that is to be executed can be indicated with the `FAILURE` command:

Commands:

– `FAILURE /CREATE or /C:` create a jobstart after processing error

– `FAILURE /UPDATE or /U:` edit a jobstart after processing error

– `FAILURE /DELETE or /D:` delete a jobstart after processing error

Syntax:

```
FAILURE /Create VDSN=<filename> SIDDEST=<receiver sid>
SIDORIG=<originator sid> COMMENT=<text> ENABLED=<set
id> EXECSYNC=<set id> EXECTIMEOUT=<set id> JOB=<scrip
name> PARAMHANDLING=<set id> SHELL=<set id>
```

FAILURE parameters supported by rvsRVO:

Required Parameter: VDSN

Optional parameters: COMMENT, ENABLED, EXECSYNC, EXECTIMEOUT, JOB, PARAMHANDLING, SHELL, SIDDEST, SIDORIG

The table in chapter 18.3.4 "Jobstart Parameters" shows a description of the parameters.

CREATE **Example for creation a jobstart after failure:**

```
FAILURE/C VDSN=Test.txt SIDORIG=LOC SIDDEST=RVS
JOB=C:\rvsTest\scripts\failure.bat
```

After receiving an error massage after send attempt of a file with VDSN `"Test.txt"` from station `LOC` to station `RVS` the job `"C:\rvsTest\scripts\failure.bat"` is to be started.

Hint: The parameter **VDSN** has to distinguish from existing jobstarts after failure.

UPDATE **Example for updating a jobstart after failure:**

```
FAILURE/U VDSN=Test.txt SIDORIG=LOC SIDDEST=RVS
JOB=C:\rvsTest\scripts\failure.bat ENABLED=N
```

In the example above the priviously created jobstart is disabled (**ENABLED**=`Y`)

Hint: The parameters **VDSN**, **SIDORIG** and **SIDDEST** must be identical to the jobstart that is to be updated. (**SIDORIG** and **SIDDEST** only if indicated with creation).

DELETE **Example for deletion a jobstart after failure:**

```
FAILURE/D VDSN=Test.txt SIDORIG=LOC SIDDEST=RVS
```

In the example above the priviously created jobstart after failure is deleted.

Hint: The parameters **VDSN**, **SIDORIG** and **SIDDEST** are requested. (**SIDORIG** and **SIDDEST** only if indicated with creation).

18.3.4 Jobstart Parameters

The following table shows the description of the jobstart parameters.
Required Jobstart Parameters:

Parameter	Description
DSN	(only `RESENTR`) Virtual file name. Allowed: SID string and wildcard '*****' at the end, the beginning or instead of the string. (GUI name: **VDSN**)
SID **or SIDDEST**	(only `SENDJOB`) Station ID of target station. Allowed: SID string and wildcard '*****' at the end. (GUI name: **SID of Destination**)
SID	(only `RESENTR`) Station ID of source station. Allowed: SID string and wildcard '*****' at the end. (GUI name: **SID of Destination**).
VDSN	(`SENDJOB` and `FAILURE`) Path and name of the local file to send. Allowed: SID string and wildcard '*****' at the end, the beginning or instead of the string.

Optional Jobstart Parameters

Parameter	Description
ACCOUNT	without function; for compability with rvs® portable
ATTEMPTS	Number of unsuccessful send attempts. Successful file transmission is indicated by "0". (GUI name: **Send Attempts**)
CODETABLE	Alias of code table that is to be used if parameter **CODETRANS = '**`T`'. See parameter **Conversion table** Example: `RESENTR /C DSN=C:\files\test22.dat CODETRANS=T CODETABLE=ASCII-IBM037 DSNNEW=TEST22`
CODETRANS	Defines whether a code translation has to be executed. Possible values: - no indication: no code translation - `E`: EBCDIC - ASCII conversion - `A`: ASCII - EBCDIC conversion - `T`: your own conversion table **,** that is indicated in **CODETABLE** parameter. (For information about creating own conversion tables, please see the rvsEVO Operator Manual, chapter „Customizing rvsEVO“ / „Conversion Tables“ .)

Optional Jobstart Parameters

Parameter	Description
COMMENT	free text;
COMPRESSION	no function; for compatibility with rvs® portable, possible values: – N - empty string
DISP	no function; for compatibility with rvs® portable, possible value: **K**
DSNNEW	With this parameter you can save the received file with another name and path. - no indication: the file is saved with current name in inbox directory - indication of new path and name: the file is saved in the indicated directory with the new name
ENABLED	With this parameter you decide whether the job is to be started or not. Possible values: - Y (Yes): the job is started (default) - N (No): the job is not started
ENCRYPTION	no function; for compatibility with rvs® portable, possible values: N
EXECSYNC	With this parameter you indicate if the OFTP session is kept alive until the process is finished. Possible values: - y (session is kept alive) - N (Default) (GUI name: **Synchronized**)
EXECTIMEOUT	If setting **EXECSYNC** = Y: Time Out after the connection is closed by the communication program. (GUI name: **Timeout (Sync.)**)
FLAGCOMP	no function; for compatibility with rvs® portable; Possible values: - N - empty string
FLAGCRYP	no function; for compatibility with rvs® portable; Possible value: – N - empty string

Optional Jobstart Parameters

Parameter	Description
JOB	Program to be started if all filter conditions apply. A defined set of parameters is transmitted to the program. (GUI name: **Process**)
LUID	no function; for compatibility with rvs® portable; Possible value: – * - empty string
NEWDIR	With this parameter you can save the received file in another directory than Inbox directory; - no indication: file is saved in Inbox - indication of new path: the file is saved in the indicated directory (GU name: **New Directory**)
NEWNAME	With this parameter you can save the received file with another name. - no indication: VDSN is file name - indication of new name: the file is saved with the new name (GUI name: **New Filename)**
PARAMHANDLING	With this parameter you can decide how to transmit the job data to the process. Possible values: - `ARGS`: job data are passed as arguments (default) - `ENV`: job data are set as environment variables - `REPLACE`: for compatibility with rvs® portable. For further information see the rvsEVO Operator Manual, chapter „Customizing rvsEVO“ / „Pass Parameters to Jobstart Process“.
REPLACE	With **REPLACE** parameter you decide how to handle receiving files which own the same name like existing files. Possible values: - `R` or `Y` (yes): replace the existing file - `N` (no): create new data set with uniquename; timestamp is added (default)
SHELL	Command Shell for executions of the program, inicated with **JOB** parameter e.g. ksh, csh, ... on UNIX systems
SIDDEST	Station ID of target station. (GUI name: **SID of Destination**)

Optional Jobstart Parameters

Parameter	Description
SIDORIG	Station ID of source station (local or virtual station) (GUI name: **SID of Originator**)
TSTAMP	With this parameter you define rules for timestamp creation. Possible values: - `Y` = timestamp is to be generated generally - `N` = timestamp is to be added only if necessary (default) (GUI name: **Timestamp**)
TSTAMPFORMAT	Format of the timestamp. Possible values: - `TIME`: time of creation of the job and counter; format: `hhmmssccc` (default) - `DATETIME`: date and time of creation of the job and counter; format: `YYMMDDhhmmssccc` - `SFID_DATETIME`: Odette timestamp of SFID; date, time and counter in format `YYMMDDhhmmsscccc` - `COUNTER`: `000000 - 999999`. If the counter is not sequential, the gap is filled first before the counter is counted up. (GUI name: **Timestamp format**)
UID	No function; for compatibility with rvs® portable; Possible value: empty string, '*****'
VFTYP	With this parameter you decide, whether each line of a received file in F or V format is terminated by line feed. Possible values: T: Files in F or V format are saved in text mode. Line feed is inserted at the end of each record (default) V: Files in V format are saved in text mode; files in F format are saved in binary mode. (GUI name: **Record Mode**)

19 Server Farm of rvsEVO

On UNIX, Linux and Windows systems rvsEVO provides the Application Cluster functionality with a server farm.

The benefits of a server farm are fail-safe stability, great transmission capacity, scalability and load distribution.

A server farm comprises an array of several computers (rvsEVO servers). In case of a server failure another server can assume the tasks of a failed server.

Hint:

From version 6.02.00 on system requirements, architecture, installation, update, configuration and Server Farm with rvs® OFTP Proxy Implementation is described in the „rvsEVO Operator Manual“ chapter „Server Farm of rvsEVO“.

19.1 Introduction

– The server farm offers significantly higher fail-safety and transmission capacity than single instance.

– A server farm comprises several rvsEVO servers aiming at ensuring high system availability.

– Jobs to be processed are evenly distributed among all rvsEVO servers within the server farm (load balancing).

– Transmission capacity can be increased or decreased by adding or removing rvsEVO servers (scalability).

– To ensure trouble-free rvsEVO operation another server can assume the tasks of a failed server.

– With regard to communication partners and operation (e.g. file transfer), rvsEVO Server Farm behaves like a single rvsEVO.

19.1.1 Fail-Safety

Failure of a node

Failure of a single node causes a restart of the aborted jobs on another server instance. A node failure has not any impact on the overall system.

A failure of the entire system causes a restart of jobs combined with the restart of the rvsEVO systems.

Failure of the database

A temporary failure: the connection is set up automatically such as configured in the Connection Pool (see „rvsEVO Operator Manual“ Chapter „Customizing rvsEVO“ / „Connection Pooling“).

If all connection attempts failed an error message is output, all activities are terminated in a controlled manner and the node or the rvsEVO system is stopped. The activites can be restarted.

Failure of the network file systems

If the read or write access failed the procedure is equivalent to the failure of the database: All activities are terminated in a controlled manner and

the node or the rvsEVO system is stopped. The activities can be restarted.

19.1.2 Load Balancing

As every node in rvsEVO can assume all tasks, it is possible to evenly distribute the load among all nodes. A server with low load will start processing a new job earlier than a server with a high load because it will access the list with upcoming tasks earlier to check for new jobs which are to be processed (first come first serve).

19.1.3 Scalability

You can use the server farm to add new rvsEVO nodes or to remove old ones. This quantitative node scalability allows the server farm to process a significantly greater amount of data and jobs than a single instance. Adding or removing nodes at runtime allows the processing capacity to be dynamically adapted without having to stop rvsEVO.

20 rvsEVO as Docker Container

In order to utilize the advantages of modern container technologies rvsEVO provides the possibility to obtain rvsEVO and rvsProxy as Docker image.

With Docker image you can operate rvsEVO and rvsProxy as container. With different container orchestration tools obtaining, deployment and scaling of containers can be automated.

Please read the rvsEVO Operator Manual, chapter „rvsEVO within Container Environment“ for information about providing and update of rvsEVO within Docker container.

21 Multitenancy Functionality of rvsEVO

From version 6.02 upwards rvsEVO provides the multitenancy functionality via rvsRuntime. rvsRuntime is an rvs system for managing multiple rvsEVO tenants. Creation, customizing and deletion of tenants can be done from a central unit.

The tenants are independent of each other and have no access on the business data of another tenant (client-dependent data).

In addition there is a common data area with technical data for all tenants from runtime environment (cross-client data). The result of this is the advantage of lower disc space and memory requirements and reduced maintenance costs.

Please see the rvsEVO Operator Manual, chapter „Multitenancy Functionality of rvsEVO“ for further information about rvsRuntime.

22 ODETTE Protocol

From version 6.02.00 on „ODETTE Protocol“ is described in rvsEVO Operator Manual.

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23 rvs® OFTP Proxy

This chapter describes the configuration and the functionality of rvs® OFTP Proxy implementation in rvsEVO.

23.1 Basics

Internet

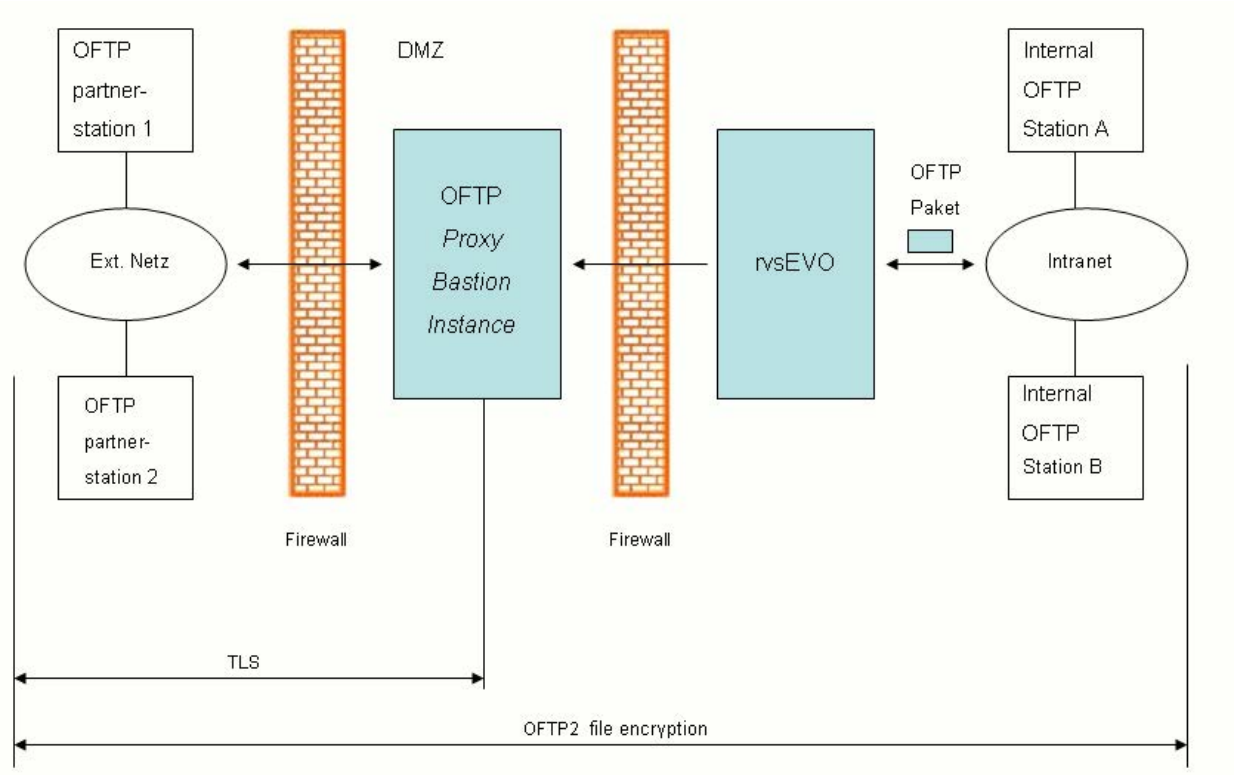
Internet is increasingly used also for transferring the commercial and business-critical data of the content. Such advantages are deciding factors for this, such as its wide distribution, enormous available bandwidths and the reasonable transfer costs.

These advantages are accompanied by the risks to be considered seriously: in internet all data are susceptible to the attacks of their confidentiality and integrity. Their authentication is also not ensured without further ado. This may result into that the data are repudiated by its obvious sender. Such risk is encountered with suitable protocols, such as TLS and HTTPS. For CAD data and also for EDI and any other data, OFTP version 2 has been standardized as a secured internet-protocol.

Any further risk arises through malware, which is spread widely and automatically in internet. Virus, worms, Trojan and other harmful programs can damage the IT-infrastructure of the company considerably. In order to minimize this risk, network areas are developed which are separated from the corporate networks through the firewalls (also referred to as demilitarized Zone - DMZ). For example, http-proxies are set in DMZ, which transport the user data without allowing direct connections between internet and intranet. rvs® OFTP Proxy offers the similar solution, which can be implemented in DMZ in order to transfer the controlled data of OFTP protocol version 1 and 2 between intranet and internet

23.2 rvs® OFTP Proxy Architecture

The rvs® OFTP Proxy is run with two instances: bastion instance and connection instance, which is integrated in rvsEVO. Following figure should illustrate these facts:



The connections are received from the external network using Bastion Instance. These external connections are forwarded to rvsEVO.

The other way round rvsEVO uses the bastion instance for establishing a connection to the OFTP station in the external network.

Communication between rvsEVO and bastion instance is executed exclusively through the connections built by rvsEVO. Communication is restricted to one port, the configurable RMI-port of the bastion instance.

Proxy listener

For being accessible for the external stations one or more Proxy listener are launched by rvsEVO. rvsEVO uses this Proxy listener to establish a connection to the partner station.

Altogether, all functional configurations, keys and certificates required for TLS communication are stored exclusively in rvsEVO.

The Bastion Instance is installed with the program rvs® OFTP Proxy. You can download rvs® OFTP Proxy using the following website:

`https://system-communication-products.t-systems-service.com/de/en/data-communication-suite/rvs/rvs-downloads/software/rvs-oftp-proxy/rvs-oftp-proxy-742610`

If you cannot download the software, please contact your sales partner:

phone from Germany: 0800 664 77 45
phone from other countries: +375 606 19 902
E-email: rvs-service@t-systems.com).

We will send you the software also on a DVD.

Carry out the installation of the Bastion Instance, as explained in „User Manual rvs® OFTP Proxy“.

23.3 Bastion Instance

In order to create a connection with the Bastion Instance, it is required to store the parameter in rvsEVO (Bastion Instances) and to configure a proxy listener.

A list of already created Bastion Instances can be displayed in Stations perspective. Please read Chapter 7.3.2 "Display Proxy Bastions" for more information.

Add a Bastion

You can add a Bastion Instance via Stations perspective. Please read Chapter 7.3.1 "Add Proxy Bastion" to find out how to add a Bastion.

Configure Proxy listener

A receiver/listener must be defined for each Bastion Instance. For this purpose, please proceed, as described in chapter 6.2.1 "Setting up of a Local Station"

You also have to add a neighbour station with Proxy TCP/IP or Proxy TLS network, for the communication via rvs® OFTP Proxy. Please read chapter 6.3.2 "Setting up of a Neighbour Station" for detailed information.

First the Bastion Instance should be started and following the rvsEVO server.

Hints:

Failure of connection attempts

You can start a process after connection problems with rvs® OFTP Proxy. This functionality is to be set via ODETTE parameters **ConnSetupProxyFailCount** and **ConnSetupProxyFailScript** (see chapter 12.6 "ODETTE Parameters").

Please see rvs® OFTP Proxy User Manual, chapter „Setting up of a TLS Connection“ in case of failure due to unvalid certificate for encrypted TLS connection with rvs® OFTP Proxy.

Keep connection

You can configure a time interval for sending heartbeats to prevent the interrupt of connection if there is no transmission. This functionality is to be set up in parameter `listener.controller-thread.sleep-time` and `listener.controller-thread.keep-alive-counter` in `$RVSPROXY_HOME\conf\RemoteListenerProperties.properties` file of rvs® OFTP Proxy program.

Use the `listener.controller-thread.sleep-time` parameter to set up the sleep time (in milliseconds) of the controller thread before it checks next time for status of listener.

Use the `listener.controller-thread.keep-alive-counter` parameter to define the number of checkups before the next heartbeat is sent.

24 File Service Module

24.1 Basics

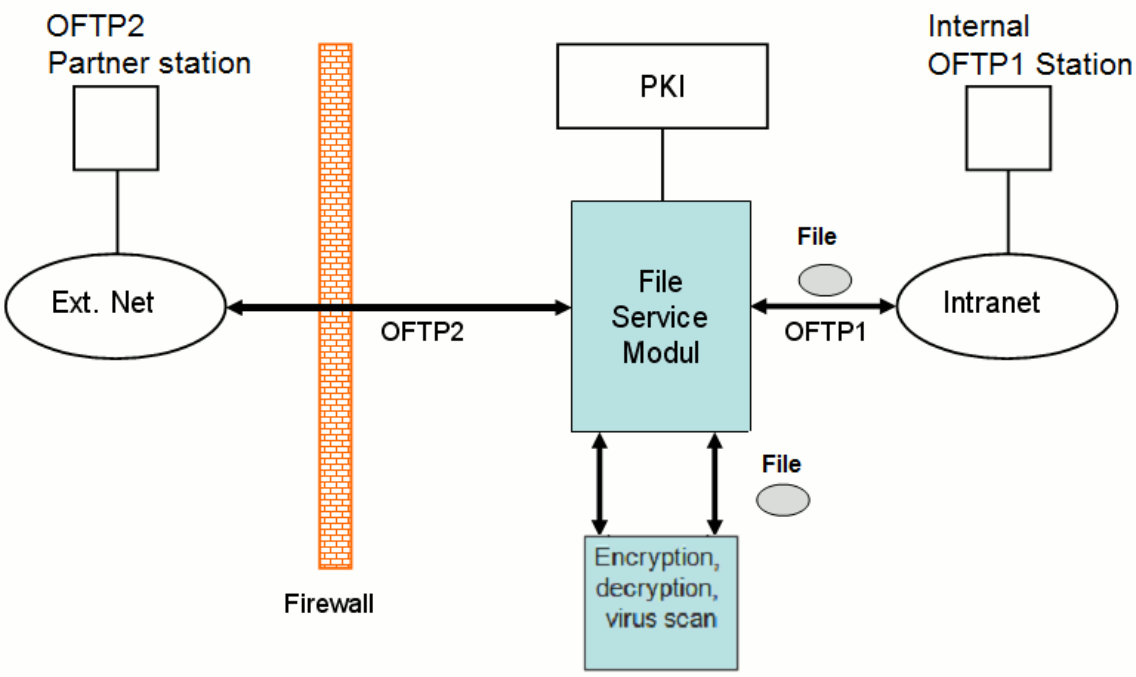
The internet is increasingly used for exchanging commercial and business critical data, which are especially security-relevant. OFTP version 2 is standardized as a secure internet protocol for CAD data, EDI and any other data.

File Service Module works as an OFTP router which de- and encrypts files accordingly OFTP2 for stations which do not support OFTP2 itself.

24.2 Architecture of File Service Module

The File Service Module works as a OFTP router and de- and encrypt files for internal stations substitutional before he forwards them.

The illustration below shows the File Service Module architecture:



Hint: You can combine the File Service Module with rvs® OFTP Proxy (see chapter 23 "rvs® OFTP Proxy").

The following functions are available:

- The File Service Module works as an OFTP router
- Protocol alternation OFTP1 <-> OFTP2, all elements, EERP, even
- En- and decryption of files acting for stations
- Automatc Certificate Exchange acting for stations

– Virus scan of received files
– Connection to the PKI for downloads of certificates, OCSP check, download of CRL

24.3 Setting up of a Neighbour Station for File Service Module

In this chapter is described the adjustment of setting up of a neighbour station for the OFTP File Service Module. Please read chapter 6 "Management of Stations" for detailed description of station configuration.

The relevant parameter for File Service Module is:

Required File Service Module parameter

Parameter	Description
File Service Module	This parameter enables the File Service Module. Possible values: - `NONE`: File Service Module is not needed (default) - `INTERNAL`: this partner station is an internal station (located in the intranet). When receiving an encrypted file which is routed to an internal station the mechanism of encryption at the File Service Module is automatically activated, depending on the encryption component used by the sending partner station. - `EXTERNAL`: this partner station is marked as an external station (located in the internet); for this option all encryption and compression features are available.

The next parameters are only available if FileProxy Service = EXTERN. The table “Security parameters:” on page 39 shows a description of the parameters.

– Security Feature Set
– Offline Compression
– Encryption
– File Signature
– Requested signed EERP/NERP
– Encryption Algorithm

Hint: Both, direct neighbour stations and routing stations can be marked by File Service Module. Therewith the configured File Service Module function is station wide fixed and can not be overwritten while creating a transmission. This corresponds with a forced encryption during activated encryption (Security=`FORCED`).

25 rvsEVO Database

From rvsEVO version 6.00 upwards the following data are written into the database:

– Jobdata

– Data of user management

– Data of rvsEVO configuration

– Data of key management

– Data of jobstarts

– Data of Observer

The following databases are available:

– **Derby** for all editions

– **Microsoft SQL**

– **Oracle Version 12 and higher** for Enterprise Edition.

– **PostgreSQL**

During the installation of rvsEVO, you can specify which database you want to use (see the chapter 2.3 "New Installation of rvsEVO and rvsRuntime").

You can also change the database after installation using the switchDatabase tool. See the chapter 25.7.32 "switchDatabase: Exchange of Database"

25.1 Derby

Derby database is a free, java based relational database from Apache Foundation. If you select Derby Embedded during the installation procedure, a database will be automatically (by rvsEVO) installed in the directory `$RVS_HOME/db`.

25.2 Microsoft SQL Server

Pre-conditions

The following pre-conditions must be fulfilled if you decided for an MS SQL database:

– an MS SQL user was set up with server role PUBLIC and CONNECT permission.

– an MS SQL database was created and the user was defined as database owner.

– rvsEVO was tested with MS SQLServer 2008 R2.

MS SQL connection parameters

During rvsEVO installation process (see chapter 2.4 "Fresh installation of rvsEVO") the following MS SQL connection parameters must be set:

– `jdbc.url`, `jdbc.user`, `jdbc.password`

– `jdbc.url` has the following syntax:

```
jdbc:sqlserver://<host>:<port>;database=<name>
```

Example:

```
jdbc:sqlserver://10.33.72.18:1433;databaseName
=rvsdb
```

Default port for MS SQL is 1433.

25.3 Oracle

Pre-conditions

The following pre-conditions must be fulfilled if you decided yourself for an Oracle database:

– an Oracle database user was set up with the user rights `connect`, `resource`, `create session` and `create table`.

– the quota „unlimited“ has to be assigned to the Oracle database user for the table space used by him**.** This setting must be done **before installation of rvsEVO**.

– your Oracle configuration is accurate. If an Oracle client is installed on your machine, you can test the configuration with the following command:

```
sqlplus ORACLE-user@ORACLE-Network servername/
password
```

Example:

```
sqlplus skk@RVS.TSYSTEMS.DE/skk
```

If this command could be executed successfilly (if a user exists and a database is available), you can start the rvsEVO installation.

Connection parameters

While installation of rvsEVO (see Chapter 2.4 "Fresh installation of rvsEVO") the following Oracle connection parameters have to be set:

`jdbc.url`, `jdbc.user`, `jdbc.password` and `Net Service Name`.

`jdbc.url` has the following Syntax:

```
jdbc:oracle:thin:@<server>:<port>/<service_name>
```

Example:

```
jdbc:oracle:thin:@localhost:1521/rvsORA
```

Default port for Oracle is `1521`.

Note: If Oracle is running on an external computer, this computer must be defined in the Oracle configuration file:

```
$ORACLEHOME/network/admin/tnsnames.ora
```

Example (tnsnames.ora):

```
#TNSNAMES.ORA Network Configuration File: /opt/oracle/
product/8.1.7/network/admin/tnsnames.ora

# Generated by Oracle configuration tools.


RVS.TSYSTEMS.DE =

  (DESCRIPTION =

    (ADDRESS_LIST =

      (ADDRESS = (PROTOCOL = TCP)(HOST = rvsaix3)(PORT
= 1521))

    )

    (CONNECT_DATA =

      (SERVICE_NAME = rvsDB)

    )

  )
```

In this example `RVS.TSYSTEMS.DE` is an Oracle network service name and `rvs` a service name. A network service name is needed to identify an Oracle database in a network.

This name should not be mixed up with the global database name, although it has a similar syntax. The service name identifies a database instance, because several instances can run on the same computer.

`jdbc.user` is the Oracle user and `jdbc.passsword` is his password.

Note: The Oracle connection data from the installation window are stored in the file `$RVS_HOME/conf/jdbc.properties`.

25.4 PostgreSQL

Requirements:

If you want to run rvsEVO with a PostgreSQL database, you will need the following requirements:

– A PostgreSQL user account has been created

– A database has been created and CONNECT rights have been assigned to the user

– The user has been granted full rights for the database

rvsEVO has been tested with PostgreSQL 16.

PostgreSQL connection parameters:

The rvsEVO installation dialog (see Chapter 2.3 "New Installation of rvsEVO and rvsRuntime") requests the following PostgreSQL connection parameters:

– jdbc.url, jdbc.user, jdbc.password

– jdbc.url has the following syntax:

jdbc:postgresql://<host>:<port>/<database-name>

25.5 Connection Pooling

With MS SQL, PostgreSQL or Oracle database rvsEVO offers the functionality **connection pooling**. Please see the rvsEVO Operator Manual, chapter „Customizing rvsEVO“ / „Connection Pooling“ for further information.

25.6 How to Delete and Create the Database Tables

Hints:

The scripts for deletion and creating database tables are located in `$RVS_HOME/system/db` directory.

Stop rvsEVO before executing the database scripts and restart rvsEVO when the new tables are created.

Remove tables

If the data are not required anymore the current database tables can be dropped via the script `$RVS_HOME/system/db/drop`.

Create tables

After removing the database tables must be recreated.

Use the following scripts for recreation of the database tables:

Via `$RVS_HOME/system/db/init_config` script the rvsEVO user table is recreated and the rights are assigned to the user rolls. In addition the configuration table with default values and the values indicated during installation is created

Via `$RVS_HOME/system/db/create_default_user` script the default user is created.

25.7 Database Maintenance and further Administrator Tools

Below you can find a description of the scripts for importing and exporting data into / from the database. The scripts are located in `$RVS_HOME/tools` directory.

Attention:

With Derby database, you can only execute the scripts with started rvsEVO server if the description of parameters refers to this (contains parameter `-a`). Otherwise the rvsEVO server must be stopped before running a script.

With Oralce, PostgreSQL and MS SQL database the scripts can also be executed with a started server. But we highly recommend to stop the rvsEVO server before launching the input scripts.

Exception: the `downloadCRLs` tool must be lauched with started rvsEvo server.

Hints:

The following error messages indicate that another process has access to the database. You have to stop this process before starting the script again.

– java.sql.SQLException: Failed to start database 'C:/rvsEVO/db' with class loader sun.misc.Launcher...

– java.sql.SQLException: Another instance of Derby may have already booted the database `C:\rvsEVO\db`.

Consider that directories and filenames which include blancs are always specified in double quotes.

25.7.1 changeStationId

Use the tool `changeStationId` for setting up a station ID. Via `changeStationId` a SID can also be changed if a certificate or transmission jobs exist for the Station.

Hint: The server must be stopped before starting `changeStationId`. With application cluster system all nodes have to be stopped.

Syntax:

```
changeStationId -s <SID> -n <new SID> [-h] [--verbose]
```

Required changeStationId Parameters:

Parameters	Description
-s <SID> **-sid <SID>**	ID of the station to be set up.
-n <new SID> **-newsid <new SID>**	New station ID.

Optional parameters:

Parameters	Description
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.2 createDefaultQueries

This tool allows you to create and import the default queries for monitor log messages and transmission queries into the database. Existing default queries will be overwritten.

Hints:

When the **MAINTENANCE** parameter `MonlogInDatabase = Y` is set, the MonLog lines must be regularly deleted from the database to prevent it from overflowing.

This can be automated using the `MonlogInDatabaseCleanup` parameter (see Chapter 12.4 "MAINTENANCE") or performed with the command-line tool cleanup (see Chapter 16.7 "cleanup: Deletion of Jobs and Monlog Data from the Database").

Hints:

These functionalities are only available to users with administrator or operator rights.

Syntax:

```
createDefaultQueries -t <Type> [-a] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-t **-type**	Type of query to create and save. Possible values: - all: all types of queries - job: transmission queries - log: Monitor-Log queries

Optional Parameters

Parameter	Description
-a	Use this parameter for executing the script with started rvsEVO server.
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.3 downloadCRLs

Use this tool for importing CRL’s which are needed for certificates with CRL validation into the database. rvsEVO server must be started.

Syntax:
`downloadCRLs [-h] [--verbose]`

Optional Parameters

Parameter	Description
-h, --help, ?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.4 exportAllConfiguration: Export of all Configuration Data from Database

This tool allows you to export data from the database and save the `$RVS_HOME\conf directory`. One XML file is generated for each data group.

One file is generated for each data group.

Job data are not exported. Use the exportJobs tool (see the chapter 25.7.11 "exportJobs: Export of all Send and Receive Jobs") for exporting the job data.

Syntax:
`exportAllConfiguration -d <Directory> [-a] [--include-conf-directory] [-r] [-h] [--verbose]`

Required Parameter

Parameter	Description
-d, --directory	Directory name in which the files are to be saved.

Optional Parameters:

Parameter	Description
-a, --api	Use this parameter for executing the script with started rvsEVO server.
--include-conf-directory	Use this parameter for saving also $RVS_HOME\conf directory.
-r, --replace	To replace files with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.

Optional Parameters:

Parameter	Description
--verbose	Verbose message output.

25.7.5 importAllConfiguration: Import of all rvsEVO Configuration Data

Use this tool to import the rvsEVO configuration data (which were exported with `exportAllConfiguration` tool) into the database and delete the present configuration data.

Syntax:

```
importAllConfiguration -d <directory> -c [-nr] [-h]
[--verbose]
```

Required Parameter:

Parameter	Description
--d **--directory**	Directory name from which the data are to be imported into the database.
-c **--clean**	to delete the old data before new data are imported.

Optional Parameters:

Parameter	Description
-nr **--noRollback**	Import of data, only. Rollback is not possible in case of failure.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, ?	Displays help (Usage) for the command.
--verbose	Verbose message output.

25.7.6 exportCertificateUsageDefinition: Export of the Definition of Certificate Usage

Use this tool to export the definition of certificate usage from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportCertificateUsageDefinition -f <File> [-a] [-r]
[-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the data are to be imported.

Optional Parameters:

Parameter	Description
-a	Use this parameter for executing the script with started rvsEVO server.
-r, --replace	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, ?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.7 importCertificateUsageDefinition: Import of the Definition of Certificate Usage

Use this tool to import the definition of certificate usage into the database and delete the present database entries.

Syntax:

```
importCertificateUsageDefinition -f <File> [-c] [-h]
[--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	to delete the old data before new data are imported.
--add-timestamp	This parameter adds a timestamp to the output

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Optional Parameters:

Parameter	Description
-h, --help, ?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.8 exportConfiguration: Export of rvsEVO Configuration Data

Use this tool to export the rvsEVO configuration data from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportConfiguration -f <File> [-a] [-r] [-h]
[--verbose]
```

Required Parameter

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the configuration data are to be imported.

Optional Parameters:

Parameter	Description
-a	Use this parameter for executing the script with started rvsEVO server.
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, ?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.9 importConfiguration: Import of rvsEVO Configuration Data

Use this tool to import the rvsEVO configuration data into the database and delete the present configuration data.

Syntax:
`importConfiguration -f <File> [-c] [-h] [--verbose]`

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	to delete the old data before new data are imported.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, ?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.10 exportCRLs: export of a CRL

Use this tool to export a CRL (**C**ertificate **R**evocation **L**ist) from the database into a file and replace a file with the same name if necessary.

Syntax:
`exportCRLs -f <File> [-a] [-r] [-h] [--verbose]`

Required Parameter:

Parameter	Beschreibung
-f <File> **-file <File>**	Path and name of the file in which the CRL is to be imported.

Optional Parameters:

Parameter	Beschreibung
-a	Use this parameter for executing the script with started rvsEVO server.
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, ?	Displays a description of the current command.

Optional Parameters:

Parameter	Beschreibung
--verbose	Verbose message output.

25.7.11 exportJobs: Export of all Send and Receive Jobs

Use this tool to export the transmission jobs from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportJobs -f <File> [-r] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the transmission jobs are to be imported.

Optional Parameters:

Parameter	Description
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, ?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.12 importJobs: Import of Send and Receive Jobs

Use this tool to import the transmission jobs into the database and delete the present transmission data.

Syntax:

```
importJobs -f <File> [-c] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	To delete the old data before new transmission data are imported.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.13 exportJobstarters: Export of Jobstart Files

Use this tool to export the configured jobstarts from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportJobstarters -f <File> [-a] [-r] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the jobstart data are to be imported

Optional Parameters:

Parameter	Description
-a	Use this parameter for executing the script with started rvsEVO server.
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.14 importJobstarters: Import of Jobstart Files

Use this tool to import the data of jobstarts into the database and delete the present jobstart data.

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Syntax:

```
importJobstarters -f <File> [-c] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	To delete the old data before new jobstart data are imported.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.15 exportKeys: Export of Keys and Certificates

This tool allows you to export the keys and certificates stored in rvsEVO from the database to a file and, if necessary, replace an existing file with the same name.

In addition to the specified file, a p12 file containing the key pairs (a file with the same name but the .p12 extension) is created, which is required when importing the keys and certificates into the database.

Syntax:

```
exportKeys -f <File> [-a] [-r] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the certificates are to be imported.

Optional Parameters:

Parameter	Description
-a	Use this parameter for executing the script with started rvsEVO server.

Optional Parameters:

Parameter	Description
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.16 importKeys: Import of Keys and Certificates

Use this tool for importing keys and certificates into the database and deleting the present key data.

Syntax:

```
importKeys -f <File> -c [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database. The XML file must be indicated. The same directory has to contain the p12 file from which the certificates can be imported (see chapter 25.7.15).
-c **--clean**	To delete the old data before new keys and certificates data are imported.

Optional Parameters:

Parameter	Description
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.17 importKeystores: Import of a Keystore File

Use this tool to import certificates and private keys from a keystore file into the database.

The keystore file must be saved as `keystore.p12` in `$RVS_HOME/system/data` folder.

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Syntax:

```
importKeystores [-c] [-h] [--verbose]
```

Optional Parameters:

Parameter	Description
-c **--clean**	To delete the old certificates before new data are imported.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.18 importLicense: Import of rvsEVO License Key

Use this tool to import a new license key at runtime. The old license file is renamed in „license.properties.“ plus date pattern and the new one is saved in $RVS_HOME/ conf directory and renamed in „license.properties“.

A Monitor-Log message gives you information about the success of the operation:

LICENSE_IMPORTED: successful import

LICENSE_NOT_IMPORTED: import was not successful

Syntax:

```
importLicense -f <File> [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File>	Path and name of the license file.

Optional Parameters:

Parameter	Description
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.19 exportObservers: Export of the Configured Observers

Use this tool to export the Observer data from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportObserver -f <File> [-a] [-r] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the Observer data are to be imported.

Optional Parameters:

Parameter	Description
-a	Use this parameter for executing the script with started rvsEVO server.
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.20 importObservers: Import Observer Data

Use this tool for importing Observer data into the database and deleting the present Observers.

Syntax:

```
importObserver -f <File> [-c] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	To delete the old Observer data before new data are imported.
--add-timestamp	This parameter adds a timestamp to the output

Optional Parameters:

Parameter	Description
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.21 exportOftpProxies: Export of Proxy Bastion Instances

Use this tool to export the created Proxy Bastion Instances from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportOftpProxies -f <File> [-a] [-r] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the Proxy Bastion Instances are to be imported

Optional Parameters:

Parameter	Description
-a, --api	Use this parameter for executing the script with started rvsEVO server.
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.22 importOftpProxies: Import of Proxy Bastion Instances

Use this tool for importing Proxy Bastion Instances into the database and deleting the present key data.

Syntax:

```
importOftpProxies -f <File> [-c] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	To delete the old Proxy Bastions before new data are imported.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.23 exportOftpRoutingRules: Export of OFTP Routing Rules

Use this tool to export the OFTP routing rules from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportOftpRoutingRules -f <File> [-a] [-r] [-h]
[--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the routing rules are to be imported.

Optional Parameters:

Parameter	Description
-a	Use this parameter for executing the script with started rvsEVO server.
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output

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Optional Parameters:

Parameter	Description
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.24 importOftpRoutingRules: Import of OFTP Routing Rules

Use this tool for importing the previously exported OFTP routing rules into the database and deleting the present OFTP routing rules.

Syntax:

```
importOftpRoutingRules -f <File> [-c] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	To delete the old data before the new data are imported.
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.25 exportQueries: Export of Queries

Use this tool to export the queries from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportQueries -f <File> [-a] [-r] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the queries are to be imported.

Optional Parameters:

Parameter	Description
-a	Use this parameter for executing the script with started rvsEVO server.
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.26 importQueries: Import of Queries

Use this tool for importing the previously exported queries into the database and deleting the present queries.

Syntax:

```
importQueries -f <File> [-c] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	to delete the old queries before new data are imported.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.27 exportStationlist: Export of the Stationlist

Use this tool to export the stationlist from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportStationlist -f <File> [-a] [-r] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File>	Path and name of the file in which the stationlist is to be imported.

Optional Parameters:

Parameter	Description
-a, --api	Use this parameter for executing the script with started rvsEVO server.
-r **--replace**	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.28 importStationlist: Import of a Stationlist

Use this tool for importing a stationlist into the database and deleting the present station settings.

Syntax:

```
importStationlist -f <File> -c [-n] [-nr] [-nv] [-h]
[--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file from which the data are to be imported into the database.
-c **--clean**	to delete the old station data before new data are imported.

Optional Parameters:

Parameter	Description
-nr **---noRollback**	Only import of data. No export of data and no roll-back in case of failure.
-n **--nd** **--nodeps**	No restoration of dependencies to certificates and certificate-usages
-nv **---novalidation**	Import without checking XML file by validation schema.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.29 importTrustedServiceList: Import of a TSL

Use this tool to import a TSL (**T**rust **S**ervice Status **L**ist) into the rvsEVO database.

Syntax:

```
importTrustedServiceList -f <TSL filename> -u <TLS
URL> [--remote] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <TSL filename>	Path of the file which includes the TSL **Hint**: Only one parameter can be set: -f or -u
-u <URL>	URL for CRL download **Hint**: Only one parameter can be set: -f or -u

Optional Parameters:

Parameter	Description
--remote	Availbles to start the tool during rvsEVO is running.
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.30 exportUsers: Export of User Data

Use this tool to export the user data from the database into a file and replace a file with the same name if necessary.

Syntax:

```
exportUsers -f <file> [-a] [-r] [-arf <File> -srf
<File>] [-s] [-h]  [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **-file <File>**	Path and name of the file in which the user data are to be imported.

Optional Parameters:

Parameter	Description
-a, --api	Use this parameter for executing the script with started rvsEVO server.
-r, --replace	To replace a file with the same name if necessary. If a file with the same name is present and this parameter was not set the process is failed with an error message.
--arf <CSV File>	Exports user accounts to file in CSV format. Further requested parameters: **--srf** and **-a**. Not usable together with **-f**.
--srf <CSV File>	Exports system roles of users to file in CSV format. Further requested parameters: **--arf** and **-a**. Not usable together with **-f**.
-s <Seperator>	Only with **--arf** and **--srf**: Defines separator in CSV file, default is ’,’.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.31 importUsers: Import of User Data

Use this tool for importing user data into the database and deleting the present users.

Syntax:

```
importUsers -f <File> [-c] [-h] [--verbose]
```

Required Parameter:

Parameter	Description
-f <File> **--file <File>**	Path and name of the file from which the data are to be imported into the database.

Optional Parameters:

Parameter	Description
-c **--clean**	to delete the old user data before new data are imported.
--add-timestamp	This parameter adds a timestamp to the output
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

25.7.32 switchDatabase: Exchange of Database

Use this tool to exchange the rvsEVO database. The data of original database are preserved.

Syntax with Parameter dbtype:

```
switchDatabase -dbtype <type> -url <url> -username
<name> -password <password> [-deleteorigdb -skiptest
switchonly -testonly] [-h] [--verbose]
```

Syntax with Parameter driver:

```
switchDatabase -driver <classname> -url <url>
username <name> -password <password> [-deleteorigdb
skiptest -switchonly -testonly] [-h] [--verbose]
```

Syntax with Parameter jdbcfile:

```
switchDatabase -jdbcfile <file> [-deleteorigdb
skiptest -switchonly -testonly] [-h] [--verbose]
```

Required Parameters:

Parameter	Description
-dbtype **or** **-driver**	Required. Type of database. Possible values: - derby - oracle - mssql - Postgres - Name and path of the jdbc driver
-url	Connection URL; see chapter Kapiteln 25.2 (MS SQL) and 25.3 (Oracle).
-username	Login name of database user.
-password	Password of database user.

Optional Parameters:

Parameter	Description
-deleteorigdb	Delete origin database after export of data. **Attention:** In case of using **-deleteorigdb** together with parameter **-switchonly** the data are not imported into the new database but deleted from the old one. We would urgently recommend to backup your rvsEVO system first.
-jdbcfile	If the parameters are imported via imputfile: path and name of the file. Replaces ALL other required parameters. Optional if **dbtype**=derby.
-skiptest	Skip test connection to database
-switchonly	Switch database without export and import of data. **Hint:** Change back to the former database via `switchdatabase` tool with **-switchonly** parameter to make the data available again.
-testonly	Make only test connection to database by using new parameters.
-h, --help, -?	Displays a description of the current command.
--verbose	Verbose message output.

Examples:

Test connection to Oracle database determined by dbtype:

```
switchdatabase -dbtype oracle -url
jdbc:oracle:thin:@rdde9art.p1.sl-si.com:1521:mydb -
username name -password password -testonly
```

Test connection to database by values in jdbc.file:

```
switchdatabase -jdbcfile jdbc.properties  -testonly
```

Swtitch to Oracle database determined by driver:

```
switchdatabase -driver oracle.jdbc.OracleDriver -url
jdbc:oracle:thin:@rdde9art.p1.sl-si.com:1521:mydb -
username name -password password
```

Switch to database by values in jdbc file:

```
switchdatabase -jdbcfile myJdbcFile
```

Switch to database by values in file without export/import:

```
switchdatabase -jdbcfile myJdbcFile -switchonly
```

Switch to database by values in jdbc file and delete origin database after export of data:

```
switchdatabase -jdbcfile myJdbcFile -deleteorigdb
```

Example of jdbc.properties file:

```
jdbc.driverClassName=oracle.jdbc.OracleDriver

jdbc.url=jdbc:oracle:thin:@rdde9art.p1.sl-
si.com:1521:mydb

jdbc.username=name

jdbc.password=qassword

#--------- c3p0 parameters are only for Oracle + MSSQL
databases ---------

c3p0.minPoolSize=5

c3p0.maxPoolSize=25

c3p0.checkoutTimeout=0

c3p0.maxIdleTime=0

c3p0.idleConnectionTestPeriod=60

c3p0.automaticTestTable=POOLING_TEST

c3p0.acquireRetryAttempts=30

c3p0.acquireRetryDelay=1000

c3p0.breakAfterAcquireFailure=false
```


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25.7.33 keytool: Key Management for RMI Communication over TLS

The tool **keytool** is used for managing and generating keys and certificates in **.p12** files.

Syntax:

```
keytool [-l <file> | -g | -i <file> | -u <file> |
-e <string>] [-k <file>] [-t <file>] [-p <password>]
[-c <file>] [-vt <date>] [-lp <password>] [-ip
<password>] [-ia <name>] [-h] [--verbose]
```

Action Parameters:

Parameter	Description
-l, --list <file>	List keypairs with expiration date of specified PKCS#12 file. Further requested parameter: **-lp**
-g, --generate	Generate keypair and export self signed certificate into trustkeystore. Further requested parameters: **-k**, **-t**, **-p, -vt** Further optional parameters: **-c**
-i, --import <file>	Import keypair from specified PKCS#12 file into keystore and export the public certificate into the trustkeystore. Further requested parameters: **-ip, -k**, **-t**, **-p** Further optional parameter: **-ia**
-u, --update <File>	Update certificate in keystore and trustkeystore (e.g. if the certificate is expired) from the source file. Further requested parameters: **-k**, **-t**, **-p**
-e, --encrypt <data string>	Encrypt given data string (e. g. password for keystore)

Additional Parameters:

Parameter	Description
-k, --keystore <file>	Path and name of the target keystore file.
-p, --password <password>	Password for the target keystore file.
-t, --trustore <file>	Path and name of the target trustkeystore file for the X.509 certificate.
-c, --csr <file>	Path and name of the target certificate sign request file. The file is only created if parameter is specified.

Additional Parameters:

Parameter	Description
-vt, --valid-to <date>	Date until certificate is valid. Default: 2 years. Format: yyyy-MM-dd
-lp, --list-password <password>	Password of PKCS#12 file in which keypairs should be listed. (See parameter **-l)**
-ia, --import-alias <name>	The key alias to import. **Hint:** The alias is displayed with `keytool -l <Schlüsseldatei> -lp <Passwort>` (See parameter **-i)**
-ip, --import-password <password>	Password of the source keystore (See parameter **-i)**
-h, --help, -?	Requests help information
--verbose	Verbose message output

Hint:

You get the alias via `keytool -l -lp`.

Examples:

Generate keypair:

```
keytool -g -k <KeyStoreFile> -p <password> -t <Trust-
StoreFile> -c <csr-File> -vt <date>
```

Update certificate in key- and truststore:

```
keytool -u <CertFile> -k <KeyStoreFile> -p <password>
-t <TrustStoreFile>
```

Import keypair into a keystore:

```
C:\rvsProxy\bin>keytool -i "c:\rvsPROXY\system\
NEUrmiTlsKeyStore.p12" -k "c:\rvsPROXY\system\
rmiTlsKeyStore.p12" -t "c:\rvsPROXY\system\
rmiTlsTrustStore.p12" -ip tls1 -p tls --ia LOC.2
```

In the example above a keypair with alias `LOC.2` is imported from `NEUr-miTlsKeyStore.p12` keystore with password `tls1` into `c:\rvsPROXY\system\rmiTlsKeyStore.p12` keystore with passsword `tls`.

The x.509 certificate is safed in `c:\rvsPROXY\system\rmiTlsTrustStore.p12` truststore

25.7.34 writeTlsCipherConfig

This tool writes the available ciphers for each TLS listeners or for HTTPS REST Server into a configuration file or the console.

The lists are subdivided in `deniedCiphers` and `availableCiphers`.

Via configuration file you may adjust the settings to your requirements. That means that the algorithms can be shifted between the sectors `deniedCiphers` and `availableCiphers`.

To be effective for rvsEVO the files must be saved into `$RVS_HOME/conf` directory as

– `tlsCipherConfig.yaml` (TLS configuration file) or

– `tlsHttpsRestCipherConfig.yaml` (HTTPS REST Server configuration file)

After changing a configuration file it is necessary to restart the adjusted listeners or the rvsEVO Server.

Syntax:

```
writeTlsCipherConfig [-c | -f <File>] [--https] [-h]
[--verbose]
```

Mandatory Parameters:

Parameter	Description
-c, --console	Writes the output into console (alternativ to -f)
-f, --file <Dateiname>	Writes the output into a file (alternativ to -c)

Optional Parameters

Parameter	Description
--https	Output for HTTPS REST Server (By default the algorithm for TLS listeners are written)
-h, --help, -?	Requests help information
--verbose	Verbose message output

25.7.35 Command Line Tools for Backup and Recovery

The description of the following programs, you can find in chapter 28 "Backing Up and Recovering rvsEVO Data"

– `backup`: Export of the whole database

– `restore`: Import of the whole database

25.7.36 Command Line Tools for Internal Use

The following programs are only for internal use:

- `importCRLs`

25.7.37 Command Line Tools for Migration from rvs® portable

The description of the following programs, you can find in rvsEVO Operator Manual, chapter „Migration from rvs® portable to rvsEVO“:

- `portable2codetable`
- `portable2jobstart`
- `portable2stationlist`

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26 Archiving the rvsEVO Data

The archiving process takes place automatically after transmission jobs are terminated or failed. During the process the job data are written into the `RevisionLog.xml` and `RevisionLog.txt` files in the `$RVS_HOME\archive` directory by default. Name and path of the files can be set up in **PersistenceArchive** and **PersistenceArchiveText** parameters. (Please see chapter 12.4 "MAINTENANCE").

The following settings of RevisionLog file can be done via parameter group `MAINTENANCE` of `Parameter` perspective:

The maximum size and maximum number of revision files can be indicated via **MaxRevisionLogSize** and **MaxRevisionLogCount** parameters. If the value in **MaxRevisionLogSize** parameter is reached a new RevisionLog file is created and the old file is extended by a sequential counter (Format: `Filename.xml.c` etc `Filename.txt.c`) .

Deletion of job data from database

Any terminated and failed jobs older than 7 days will be deleted every 24 hours, by default. This configuration can be set up via the parameters **CleanupDays**, **CleanupInterval** and **CleanupTime** (please see chapter 12.4 "MAINTENANCE" for getting the description of the parameters).

Hint: You can also start the deletion of job data by launching the script `archiveJobs` at the command prompt as an alternative. (See chapter 16.8 "archiveJobs: Deletion of Job Data from Database").

Deletion of an Individual Job

You also have the option to delete an individual job. This can be reached via the script `archiveJob` (see chapter 16.9 "archiveJob: Deletion of an Individual Job from the Database") or via the GUI (see chapter 5.6 "Operations on Jobs").

27 PKI Connection

In this chapter the configuration and the functionality of the rvsEVO PKI binding will be described. You also get information about certificate validation via GUI.

27.1 Introduction

In a PKI (**P**ublic **K**ey **I**nfrastructure) public keys with their certificates are stored.

Following features are available for rvsEVO PKI binding:

– obtaining of public keys and certificates from a PKI for a partner station.

– certificate validation with OCSP (**O**nline **C**ertificate **S**tatus **P**rotocol), CRL (**C**ertificate **R**evocation **L**ist) or CertPath.

– Transfer of information from a OCSP server by HTTP

Hint: From rvsEVO version 6.00 upwards certificate validation via OCSP and CRL is also available for TLS certificates.

27.2 Configuration of PKI Binding

The PKI binding should be configured in the `Stations` perspective via the OFTP parameters **Certificate Validation Type** and **PKI**.

Hint: Usually the PKI binding should be configured for a partner station. For the local station it can be interesting, only if you want to validate your own certificate.

Parameters for PKI Binding

Parameter	Description
Certificate Validation Type	This parameter describes the type of the certificate validation. The following three types are supported: - `OCSP` (Online Certificate Status Protocol) - `CRL` (Certificate Revocation List) - `CertPath`. `None` in this field means: No validation will be done.
PKI	If the check box is activated the PKI functionality is used.

27.3 Validation Methods

Which validation method will be used depends on security standards in the company.

OCSP If you use OCSP as validation method, the OCSP service must be available for the rvsEVO nodes.

CRL The validation with CRL can be done offline. The rvsEVO user/administrator can provide the list of certificates (CRL) and store it manually or download the list via the key management of rvsEVO (see 27.4 "Certificate Validation via Key Management Perspective"). The disadvantage of this way is, that certificate can expire between two updates.

CertPath For CertPath validation you only need the certificate to be validated and the certificates of the issuer up to the root certificate. Normaly these certificates change rarely. So not much service is necessary. With CertPath validation the formal correctness of the certificate, the certificate of the issuer ... up to the root certificate is checked. It is not checked whether a certificate was revocated.

Default Procedure with Encryption via TLS

You need to set up parameter **Certificate Validation Type** to `CRL` or `OCSP` for CRL or OCSP validation. Else CertPath validation takes place. With all outgoing calls certificate validation takes place. To validate certificates of incoming calls parameter **Client Authentification** is to be set up to `NEEDED` or `WANTED` (see table “Network Parameters” on page 60).

Default Procedure with File Encryption

There is no certificate validation by default.Validation with OCSP or CRL always includes the validation of the chain. For using OCSP and CRL validation you need to set up parameter **Certificate Validation Type**.

Hint: rvsEVO don’t accepets expired certificates.

27.4 Certificate Validation via Key Management Perspective

A certificate can also be validated via Key Management perspective in the detailed view of the key, section `Certificate validation`. The detailed view can be opened by a double-click on the certificate. Precondition is that the parameter **Certificate Validation Type** was set up (see chapter 27.2).



Validate certificate

To validate a certificate:

- Specify the OCSP address or download address of CRL in input field **Validation URI** and select the validation type (OCSP or CRL) from the combo box. Press to specify more than one address and to open a further input field for validation URI or press to delete an entry.

  Hint: Via mouse click on **Reset from Certificate** button the validation URI is read from the certificate.
- Press the **Validate Certificate** button to validate the certificate.

Import / update CRL

Hint: A current CRL can be downloaded by pressing the **Import CRL** button. The CRL is imported into the rvsEVO system. Set the security parameter **CRLUseHttpProxy**=`Y` for downloading the CRL via Http Proxy server. Via security parameter **CRLAutoDownload** you can configure that the CRL is downloaded and imported during the next establishment of connection.

Via security parameters **CRLAutoUpdate** and **CRLAutoUpdateInterval** you can configure that automatic updates should take place (See chapter 12.8 "Security Parameters" for the description of the parameters).

Please use the following format to imort a CRL from your system:

"file:/PATH/FILENAME.crl"

Example for Windows systems:

```
file:/C:/certs/rvsEVO_Dev_Intermediate_CA_1.crl
```

Example for UNIX systems:

```
file://home/certs/rvsEVO_Dev_Intermediate_CA_1.crl
```

28 Backing Up and Recovering rvsEVO Data

You can back up all relevant rvsEVO data and recover them if necessary. This is particularly important when an error has occurred in rvsEVO and the user wishes to revert to the old status prior to the error.

Important: The rvsEVO version of backup must be identical to the version where the backup is to convey to.

28.1 Backup

Pre-condition

The rvsEVO server has to be stopped before the backup function can be launched.

Start the `$RVS_HOME/tools/backup` script to perform the backup functionality.

Syntax:

```
backup [-u] [-s] [-verbose][-help][-?]
```

All parameters are optional.

Backup Parameters:

Parameter	Description
-file <file> **-f**	Without function
-userfiles **-u**	Use this parameter to save the user data from `Files` directory (subdirectories `inbox`, `outbox` and `temp`)
-skipjobs **-s**	Use this parameter for desabling backup of job-data.
-?	Requests help information.
-help **-h**	Displays a description of the current command.
-verbose	Verbose message output

By default, the data are saved in a jar file in `$RVS_HOME/archive` directory. The jar file name constists of “EVO6B-“, date, “-“, time and the extension “.jar“. (Format: `EVO6B-YYYYMMTT-hhmmss.jar`).

What is backed up?

The following directories are backed up:

- `$RVS_HOME/conf`
- `$RVS_HOME/files`
- `$RVS_HOME/system`
- `$RVS_HOME/jobs` (if used)

The following data from the database are backed up:

– configuration data
– stationlist
– OFTP-Proxy Bastions
– jobdata
– jobstarts
– certificates and their assignments
– Observer
– user data

28.2 Recovering the rvsEVO Data

Use the `$RVS_HOME/tools/restore.bat` script to recover the rvsEVO data.

Important:

Syntax:

```
restore -backup|-b <file> [-noinitdb|-n] [-t <file>]
[-u] [-skipjobs|-s] [-verbose] [-help][-?]
```

Required Parameter:

Parameter	Description
-backup <file> **-b**	Name of the backup file (jar file). See chapter 28.1.

Optional Parameters:

Parameter	Description
-noinitdb **-n**	With this parameter the database will not be initialized before restore.
-temp **-t**	Directory for temporary data
-userfiles **-u**	Parameter for importing the user data (see also the same parameter of backup script)
-skipjobs **-s**	Parameter for disabling the import of job data.
-?	Requests help information.
-help **-h**	Displays a description of the current command.

Optional Parameters:

Parameter	Description
-verbose	Verbose message output

29 Abbreviations

3DES	Triple Data Encryption Algorithm (deprecated)
AES	Advanced Encryption Standard
CA	Certificate Authority
CMS	Cryptographic Message Syntax
CRL	Certificate Revocation List
EERP	End-to-End-Response
EFID	End File ID
EFNA	End File Negative Answer
ESID	End Session ID
GUI	Graphical User Interface
JRE	Java Runtime Environment
JS	Jobstart
JVM	Java Virtual Machine
LCI	Local Configuration Instance
LDAP	Ligthweight Directory Access Protocol
NAS	Network-Attached Storage
NERP	Negative-End-to-End-Response
OCSP	Online Certificate Status Protocol
OFTP	Odette File Transfer Protocol
PKI	Public Key Infrastructure
RCI	Remote Controlled Instance
RSA	Asymmetrical cryptographic technique (initials of the developers second name: Rivest, Shamir, Adleman)
SAN	Storage Area Network
SFID	Start File ID
SFPA	Start File Positive Answer
SID	Station ID
SNMP	Simple Network Management Protocol
SP	Service Provider
TLS	Transport Layer Security

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VDSN	Virtual Dataset Name